Introduction to Tensors

Introduction

The origin of tensors…
Why we need tensors?
What tensors are actually?

The origin of tensors

- The word **tensor** has its root tensus in Latin, meaning stretch or tension.
- J. W. Gibbs in (1884) developed the concept of dyadics (object with two parts or elements) and polyadics (an object comprising of many elements) which are actually tensors in modern sense but only different by names.

The origin of tensors

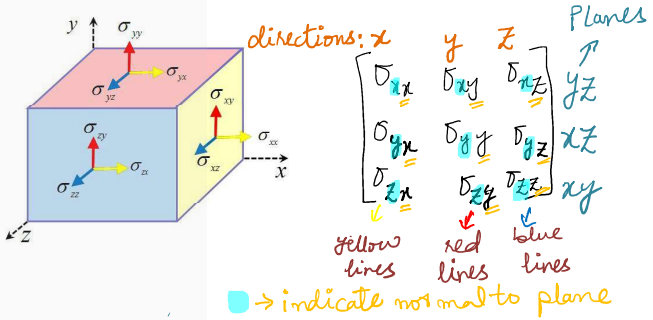
- W. Voigt in (1898) introduced the term tensor in his study of stress and strain of crystals.
- Later on many physicists including Albert Einstein and Marcel Grossmann in (1913) used the Ricci's concepts with the name tensors instead of systems.

Why we need tensors?

- A **scalar** is a physical quantity that is represented by **a number** at a particular point in space. For example, mass, density and temperature.
- A **vector** is a tool to keep track of **two pieces of information,** typically magnitude and direction, for a physical quantity. For example position, force and velocity.

Why we need tensors?

What happens when we need to keep track of three pieces of information for a given physical quantity? Say, three pieces of information are magnitude, direction and plane. For example stress, strain and inertia tensors.



Why we need tensors?

In short:

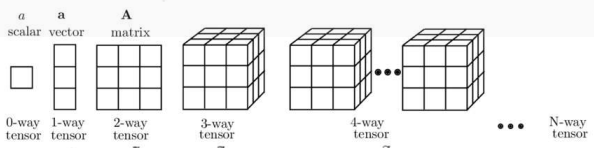
A scalar is a quantity whose specification requires just one number.

On the other hand, a vector is a quantity whose specification requires three numbers, namely its components with respect to some basis.

Scalars and vectors are both special cases of a more general term called a tensor.

What is a tensor?

Tensor can be regarded as a general name of multi-way array data.



Tensor Definition:

In an m dimensional space, a tensor of rank n is a mathematical object that has n indices, m^n components and obey certain transformation rule.

Rank of a tensor

The rank of a tensor also known as order or degree of tensor is the total number of free indices in a tensor.

For example,

1) m (scalar) tensor of rank zero

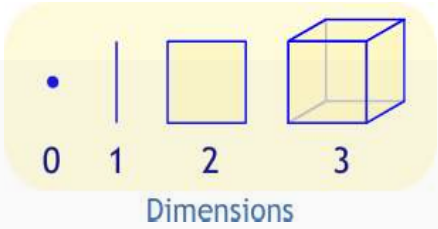
2) v_α (vector) tensor of rank 1,
   v^β (vector) tensor of rank 1

3) $T_{\mu\delta}$ (tensor) tensor of rank 2

Higher Dimensional Spaces

Dimensions:

The **dimensions** of a mathematical space (or object) is informally defined as the **minimum number of coordinates** needed to specify any point within it.



2-Dimensional space

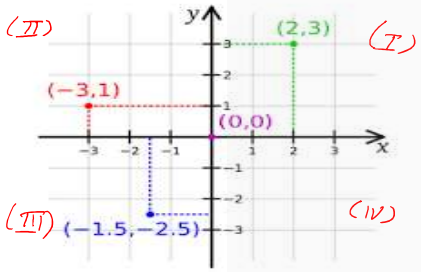
A coordinate system in which the exact position of a point is determined with two numerical values, is called the two dimensional space.

In Euclidean geometry, two perpendicular number-lines are intersected at their middle points for splitting the whole space into four equal parts. The number lines are used as base line to measure the distance of every point in two perpendicular directions from their bisection point.

2-Dimensional Space

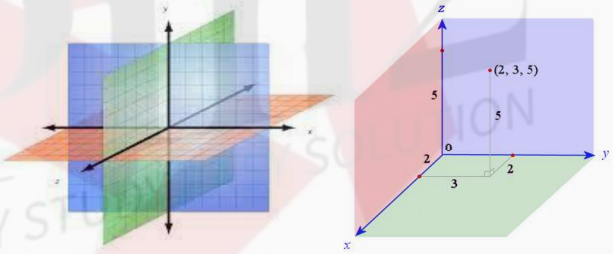
There are four fundamental concepts related to two dimensional Cartesian coordinate system:

Coordinate axes, origin, coordinates of a point and quadrants.



3-Dimensional space

We can expand our 2-dimensional (xy) coordinate system into a **3-dimensional coordinate system**, using x, y, and z-axes. In 3-dimensional space the exact position of a point is determined with three numerical values, called the coordinates.

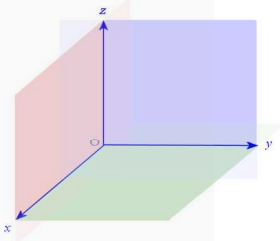


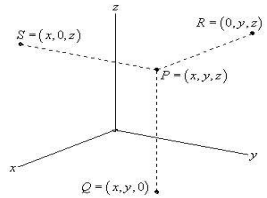
3-Dimensional space

The ***xy*-plane** is horizontal in diagram below and shaded green. It can also be described using the equation $z = 0$,

The ***xz*-plane** is vertical and shaded pink above. This plane can be described using the equation $y=0$.

The ***yz*-plane** is also vertical and shaded blue. The yz-plane can be described using the equation $x=0$.





Higher Dimensional Spaces

More than three dimensional spaces are treated as higher dimensional spaces.

For example, we can think of dimensions as ways in which we can move up and down, left and right, forward and backwards, which leads us to believe we live in a three-dimensional world. Every point in this world is defined by three coordinates. We can't imagine living in a four-dimensional world because we simply can't conceive of another direction to move in.

Higher Dimensional Spaces

However, now imagine you are arranging to meet someone. Not only do you need to give them the three coordinates of the place you are meeting, you also need to tell them the time you are meeting. Thus, time and space together form a four-dimensional space: you need four numbers to specify a point.

Generally, a point in an n-dimensional space is given by n numbers, or coordinates. Even though we can't visualize higher dimensional spaces and objects within them as geometric objects. Algebra gives the tools to deal with these objects.

Applications of Tensors

Applications of tensors

Tensors have a wide range of applications in different fields. Following are different fields of Mathematics and Physics where tensors are used.

- Mechanics

  Stress tensor, Strain tensor, Elasticity tensor, Inertia tensor
- Electromagnetics

  Maxwell equations and related tensors
- Special theory of relativity

  Metric tensor, four vector formalism
- General theory of relativity

  Riemannian tensor, Ricci tensor, Einstein tensor

Applications of tensors

- Following are some applications of tensors in **Engineering**
- To represent mechanical stress as the stress tensor in continuum mechanics
- To measure deformations through deformation tensor and strain via strain tensor in continuum mechanics
- To represent diffusion as a tensor in diffusion tensor imaging.
- To describe moment of inertia as inertia tensor.

Applications of tensors

Tensor Flow

Tensor flow is an open-source end-to-end platform for creating machine learning applications. It is a symbolic math library that uses dataflow and differentiable programming to perform various tasks focused on training and inference of deep neural networks.

How Tensor Flow Works

Tensor Flow enables you to build dataflow graphs and structures to define how data moves through a graph by taking inputs as a multi-dimensional array called Tensor. It allows you to construct a flowchart of operations that can be performed on these inputs, which goes at one end and comes at the other end as output.

Applications of tensor flow

Currently, tensor flow is used in a variety of Google applications like Google photos, Google translate, Gmail, search engine, translation, image captioning or recommendations.

Free and Dummy Indices

Tensor Notation:

Tensor notation provides a convenient and unified system for describing physical quantities.

In the most general representation, a tensor is denoted by a symbol followed by a collection of subscripts (or superscripts). For example

$$X_i,\ \sigma_{ij}, U_l,\ B_{klm}$$

X^i f^{ij}

Subscript and Superscript:

A subscript is a character (such as a number or letter) that is written slightly below and a superscript is written above the normal line of type. It is usually smaller than the rest of the text.

Subscripts appear **below** the baseline, while **superscripts** are **above**.

For example: $A_i,\ u_j$ → **subscripts**

$B^k,\ C^{ij}$ → **superscripts**

These subscripts and superscripts are also called **indices**.

Indices:

Indices assume values from an index set. They successively assume each value from the given index set.

For example, if i takes the values from 1 to 4, then a_i denotes the set (a_1, a_2, a_3, a_4) and we say that the index i runs from 1 to 4.

Example:

$a_j; j = 1,2,3$ a_1, a_2, a_3

$b_k; k = 0,1,2,3$

↳ b_0, b_1, b_2, b_3

Free and dummy Indices:

In the index notation, indices are categorized into two groups: free indices and dummy indices.

- An index which may take on any particular value from the index set is called a free index. For example

$A_i; i = 1,2,3$ $\quad A_1, A_2, A_3$

$B_{ij}; i = 1,2,3 \,\&\, j = 1,2,3$

- A dummy index means summation. Dummy indices are repeated indices in an expression. For example

$\sum_i A_{ii}; i = 1,2,3 \longrightarrow A_{11} + A_{22} + A_{33}$

$\sum_i A_i B_{ij}; i = 1,2,3 \,\&\, j = 1,2,3$

$\hookrightarrow A_1 B_{1j} + A_2 B_{2j} + A_3 B_{3j}$

Example 1:

1) Write down explicitly the equations represented by the expression $y_i = \sum_{j=1}^{3} a_{ij} x_j$ where $i = 1,2,3$ and $j = 1,2,3$

Solution: i) First differentiate the free and dummy indices in expression.

Here i is free index and j is dummy index.

ii) Then apply the sum over the dummy index.

$y_i = a_{i1}x_1 + a_{i2}x_2 + a_{i3}x_3$

iii) After that, put the values of free index one by one which yield a set of equations.

$y_1 = a_{11}x_1 + a_{12}x_2 + a_{13}x_3$

$y_2 = a_{21}x_1 + a_{22}x_2 + a_{23}x_3$

$y_3 = a_{31}x_1 + a_{32}x_2 + a_{33}x_3$

Example 2:

Show that if $i = 1,2,3,\ldots,n$ and $j = 1,2,3,\ldots,n$ then $\sum_{i=1}^{n} a_i x^i = \sum_{j=1}^{n} a_j x^j$.

Solution: First consider

L.H.S.$= \sum_{i=1}^{n} a_i x^i = a_1x^1 + a_2x^2 + a_3x^3 + \ldots + a_nx^n$

R.H.S.$= \sum_{j=1}^{n} a_j x^j = a_1x^1 + a_2x^2 + a_3x^3 + \ldots + a_nx^n$

Note: So any dummy index can be replaced by other index ranging the same numbers.

Example 3: Consider the expression $T_{ij} = \sum_m A_{ijm}x_m + \sum_p B_{ip}C_{jp}$

1) Which are the free indices?

2) Which are the dummy indices?

3) How many terms are on the right hand side when m and p has variation 1,2.

4) How many equations are described by the given expression when i and j has variation 1,2?

1) $T_{ij} = \sum_m A_{ijm}x_m + \sum_p B_{ip}C_{jp}$ ← i and j are free indices

2) $T_{ij} = \left(\sum_m\right) A_{ijm}x_m + \left(\sum_p\right) B_{ip}C_{jp}$

m, p → dummy indices

Example 3: Consider the expression $T_{ij} = \sum_m A_{ijm}x_m + \sum_p B_{ip}C_{jp}$

3) How many terms are on the right hand side when m and p has variation 1,2.

4) How many equations are described by the given expression when i and j has variation 1,2?

3) $T_{ij} = \sum_m A_{ijm}x_m + \sum_p B_{ip}C_{jp}$

$= A_{ij1}x_1 + A_{ij2}x_2 + B_{i1}C_{j1} + B_{i2}C_{j2}$

So, there are four terms on R.H.S.

Example 3: Consider the expression $T_{ij} = \sum_m A_{ijm}x_m + \sum_p B_{ip}C_{jp}$

3) How many terms are on the right hand side when m and p has variation 1,2.

4) How many equations are described by the given expression when i and j has variation 1,2?

4) $T_{ij} = \sum_m A_{ijm}x_m + \sum_p B_{ip}C_{jp}$

here $i = 1,2$ and $j = 1,2$ implies

$T_{ij} \Rightarrow T_{11}, T_{12} \Rightarrow$ 2 equations

$T_{2j} \Rightarrow T_{21}, T_{22} \Rightarrow$ 2 equations

$\Rightarrow$ The given expression describe a set of four equations.

Tensor Definition:

In simple words, tensors are simply arrays of numbers or functions that transform according to certain rules under a change of coordinates.

OR

In an m dimensional space, a tensor of rank n is a mathematical object that has n indices, m^n components and obey certain transformation rule.

Rank/order of a tensor

The rank of a tensor is the number of indices required to uniquely select each element of the tensor. Rank is also known as order, degree. In simple words, It is the total number of free indices in a tensor.

For example,

1) m (scalar) tensor of rank zero

2) v_α (vector) tensor of rank 1,
v^β (vector) tensor of rank 1

3) $T_{\mu\delta}$ (tensor) tensor of rank 2

4) $A^{\mu}_{\nu\mu}$ (tensor) tensor of rank 1

Einstein Summation Convention

Einstein Summation Convention:

This convention was introduced with Einstein's words:

"I have made a great discovery in mathematics; I have suppressed the summation sign every time that the summation must be made over an index which occurs twice..."

Einstein summation is a notational convention for simplifying expressions including summations of vectors, matrices and tensors. Following are the rules of Einstein summation notation.

1. Repeated indices are summed over.

$$a_i a_i = \sum_{i=1}^{n} a_i a_i$$

$$a_{ik} a_{ij} = \sum_{i=1}^{n} a_{ik} a_{ij}$$

2. Each index can appear at most twice in any term.

3. Each term must contain identical non-repeated indices.

So,

$$M_{ij} v_j = \sum_j M_{ij} v_j$$

is a valid expression whereas

$$M_{ij} u_j v_j + w_i \quad and \quad T_{ijk} u_k + M_{ip}$$

are not valid expressions.

Question: Check which of the following is a valid expression.

1) $a_{ms} = b_m(c_r - d_r)$
not valid
2) $a_{ms} = b_m(c_s - d_s)$
valid
3) $t_i = a_{ji} n_j$
valid
4) $t_i = a_{ji} n_i$
not valid
5) $\sigma_{ij} = x_i x_{ij}$
not valid
6) $b_{ij} c_j = 3$
not valid

Example 1: Expand the summation, $c_i(x_i + y_i)$ for $n = 4$

$$\begin{aligned} c_i(x_i + y_i) = {} & c_1(x_1 + y_1) \\ & + c_2(x_2 + y_2) \\ & + c_3(x_3 + y_3) \\ & + c_4(x_4 + y_4) \end{aligned}$$

Double Sums:

- An expression can involve more than one summation index.
- For example $a_{ij} x_i y_j$ indicates a summation taking place on both i and j simultaneously.
- If an expression has two summation indices varying from 1 to n, there will be a total of n^2 terms in the sum. If there are three summation indices, there will be n^3 terms in the sum and so on.

Example 2

Simplify $a_{ij} x_i y_j$, where $i = 1,2,3$ and $j = 1,2,3$.

Solution: First sum over i

$$a_{ij} x_i y_j = a_{1j} x_1 y_j + a_{2j} x_2 y_j + a_{3j} x_3 y_j$$

Then sum over j

$$\begin{aligned} a_{ij} x_i y_j = {} & a_{11} x_1 y_1 + a_{12} x_1 y_2 + a_{13} x_1 y_3 \\ & + a_{21} x_2 y_1 + a_{22} x_2 y_2 + a_{23} x_2 y_3 \\ & + a_{31} x_3 y_1 + a_{32} x_3 y_2 + a_{33} x_3 y_3 \end{aligned}$$

Example 3: Write down the equations for n=2 corresponding to the expression: $y_i = c_i^r a_{rs} x_s$

Solution:

$y_i = c_i^r a_{rs} x_s$; $n=2$

r and s are dummy, i is free and all i, r, s have values 1 and 2.

$y_1 = c_1^r a_{rs} x_s = c_1^r a_{r1} x_1 + c_1^r a_{r2} x_2$

$y_1 = c_1^1 a_{11} x_1 + c_1^2 a_{21} x_1 + c_1^1 a_{12} x_2 + c_1^2 a_{22} x_2$

Similarly:

$y_2 = c_2^r a_{rs} x_s = c_2^1 a_{11} x_1 + c_2^2 a_{21} x_1 + c_2^1 a_{12} x_2 + c^2 a_{22} x_2$

Substitutions:

Suppose it is required to substitute $y_i = a_{ij} x_j$ in the equation $Q = b_{ij} y_i x_j$ it would lead to $Q = b_{ij} a_{ij} x_j x_j$ which is an invalid indicial expression.

The correct method is that first we identify any dummy indices in the expression to be substituted that coincide with indices occurring in the main expression. Changing these dummy indices to characters not found in the main expression one can proceed substitution in the usual fashion.

Example 4

If $y_i = a_{ij}x_j$, express the quadratic form $Q = g_{ij}y_iy_j$ in terms of x-variable.

Solution:

First replace j with k in $y_i = a_{ij}x_j$ for y_i : $y_i = a_{ik}x_k$

and i with j replace j with s in $y_i = a_{ij}x_j$ for y_j : $y_j = a_{js}x_s$

Then by substitution it becomes

$$Q = g_{ij}a_{ik}x_ka_{js}x_s = g_{ij}a_{ik}a_{js}x_kx_s$$

$$Q = h_{rs}x_kx_s; \qquad h_{rs} = g_{ij}a_{ik}a_{js}$$

Example 5: Express $b^{ij}y_iy_j$ in terms of x variable, if $y_i = c_{ij}x_j$ and $b^{ij}c_{ik} = \delta^j_k$.

Solution:

Consider $b^{ij}y_iy_j$

$y_i = c_{ir}x_r$ and $y_j = c_{js}x_s$

$$\Rightarrow b^{ij}y_iy_j = b^{ij}c_{ir}x_r\,c_{js}x_s$$
$$= b^{ij}c_{ir}\,x_r\,c_{js}x_s$$
$$= \delta^j_r\,x_r\,c_{js}x_s = x_j\,c_{js}\,x_s$$
$$= c_{js}x_j\,x_s = c_{ij}\,x_i\,x_j$$
$$\Rightarrow b^{ij}y_iy_j = c_{ij}\,x_i\,x_j$$

Kronecker Delta Symbol

Kronecker Delta

The Kronecker delta is defined as

$$\delta_{ij} = \delta^j_i = \delta^{ij} = \begin{cases} 1; & i = j \\ 0; & i \neq j \end{cases}$$

Clearly $\delta_{ij} = \delta_{ji}$ for all i, j.

The Kronecker-δ can be thought of as the identity matrix, for example for $i = 1,2,3$ and $j = 1,2,3$

$$\delta_{ij} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Example: If $i = 1,2,3$ and $j = 1,2,3$ simplify $\delta_{ij}x_ix_j$.

Solution:

$$\delta_{ij}x_ix_j = \delta_{1j}x_1x_j + \delta_{2j}x_2x_j + \delta_{3j}x_3x_j$$
$$= \delta_{11}x_1x_1 + \delta_{12}x_1x_2 + \delta_{13}x_1x_3$$
$$+ \delta_{21}x_2x_1 + \delta_{22}x_2x_2 + \delta_{23}x_2x_3$$
$$+ \delta_{31}x_3x_1 + \delta_{32}x_3x_2 + \delta_{33}x_3x_3$$
$$= (1)x_1x_1 + (0)x_1x_2 + (0)x_1x_3$$
$$+ (0)x_2x_1 + (1)x_2x_2 + (0)x_2x_3$$
$$+ (0)x_3x_1 + (0)x_3x_2 + (1)x_3x_3$$
$$\delta_{ij}x_ix_j = x_1x_1 + x_2x_2 + x_3x_3 = x_ix_i$$

So, $\delta_{ij}x_ix_j = x_ix_i$ OR $\delta_{ij}x_ix_j = x_jx_j$

Hence in general

$$\delta_{ij}x_ix_j = x_ix_i$$
$$\delta^r_j a_{ir}x_i = a_{ij}x_i$$

Kronecker delta is a much used symbol in tensor calculus. It has the effect of annihilating the off diagonal terms.

Also, $\delta_{ik}\delta_{kj} = \delta_{ij}$

$\delta_{ik}\delta_{kj} = \delta_{i1}\delta_{1j} + \delta_{i2}\delta_{2j} + \delta_{i3}\delta_{3j}$

$\delta_{1k}\delta_{1k} = \delta_{11}\delta_{11} + 0 + 0 = \delta_{11} = 1$

$\delta_{2k}\delta_{2k} = \delta_{22} = 1$

$\delta_{3k}\delta_{3k} = \delta_{33} = 1$

Example: For arbitrary n, evaluate δ_{ii}.

Solution:

Consider δ_{ii} ; $i = 1,2,\ldots,n$

i is dummy.

$$\delta_{ii} = \delta_{11} + \delta_{22} + \delta_{33} + \cdots + \delta_{nn}$$
$$= 1 + 1 + 1 + \cdots + 1$$

(n times)

$$\delta_{ii} = n$$

Example: For $n = 3$, evaluate $\delta_{ij}\delta_{ij}$.

Solution:

Consider $\delta_{ij}\delta_{ij}$ → { i and j both are dummy indices.

$$\delta_{ij}\delta_{ij} = \delta_{1j}\delta_{1j} + \delta_{2j}\delta_{2j} + \delta_{3j}\delta_{3j}$$
$$= \delta_{11}\delta_{11} + \delta_{12}\delta_{12} + \delta_{13}\delta_{13}$$
$$+ \delta_{21}\delta_{21} + \delta_{22}\delta_{22} + \delta_{23}\delta_{23}$$
$$+ \delta_{31}\delta_{31} + \delta_{32}\delta_{32} + \delta_{33}\delta_{33}$$
$$= 1 + 1 + 1 = 3$$
$$\Rightarrow \delta_{ij}\delta_{ij} = 3$$

Alternatively:

$\delta_{ij}\delta_{ij} = \delta_{ii}$

and $\delta_{ii} = 3$ if $n = 3$ and for arbitrary n, $\delta_{ii} = n$

$\delta_{ik}\delta_{kj} = \delta_{ij}$

$\delta_{ik}\delta_{jk} = \delta_{ij}$

$\delta_{ik}\delta_{ik} = \delta_{ii}$

$\delta_{ij}\delta_{ij} = \delta_{ii}$

Example: For arbitrary n, evaluate $\delta_{ij}\delta^{j}_{k}c_{ik}$.

Solution: In $\delta_{ij}\delta^{j}_{k}C_{ik}$ all i,j,k are dummy.

Recall first example: $\delta_{ij}x_i x_j = x_i x_i$

$\delta_{ij}\delta^{j}_{k}C_{ik} = \delta_{ij}C_{ij}$

Again keeping in mind property of Kronecker delta:

$\delta_{ij}\delta^{j}_{k}C_{ik} = C_{ii}$

Permutation symbol

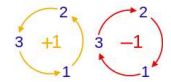
The Permutation Symbol or Levi-Civita Symbol

In three dimensions, the permutation symbol is a three index object also called the Levi-Civita symbol and alternating tensor. It is defined by

$$\varepsilon_{ijk} = \begin{cases} 0 & for\ i=j, j=k\ or\ k=i \\ 1 & for\ (i,j,k)\in\{(1,2,3),(2,3,1),(3,1,2)\} \\ -1 & for\ (i,j,k)\in\{(1,3,2),(3,2,1),(2,1,3)\} \end{cases}$$

OR $\varepsilon_{ijk} = \frac{1}{2}(i-j)(j-k)(k-i)$

For the indices (i, j, k) in ε_{ijk}, the values 1, 2, 3 occurring in the cyclic order (1, 2, 3) correspond to $\varepsilon = +1$, while occurring in the reverse cyclic order correspond to $\varepsilon = -1$, otherwise $\varepsilon = 0$.



Levi-Civita Symbol

Example: The permutation symbol satisfies :

i) $\delta_{ij}\varepsilon_{ijk} = 0$

ii) $\varepsilon_{ipq}\varepsilon_{jpq} = 2\delta_{ij}$

iii) $\varepsilon_{ijk}\varepsilon_{ijk} = 6$

where δ_{ij} is the Kronecker delta.

i) $\delta_{ij}\varepsilon_{ijk} = 0$

$\delta_{ij}\varepsilon_{ijk} = \varepsilon_{iik} = 0$ (by definition of ε_{ijk})

Alternatively: let $n=3$, i.e. $i=1,2,3$; $j=1,2,3$ and $k=1,2,3$.

$$\delta_{ij}\varepsilon_{ijk} = \delta_{1j}\varepsilon_{1jk} + \delta_{2j}\varepsilon_{2jk} + \delta_{3j}\varepsilon_{3jk}$$
$$= \delta_{11}\varepsilon_{11k} + \delta_{12}\varepsilon_{12k} + \delta_{13}\varepsilon_{13k} + \delta_{21}\varepsilon_{21k} + \delta_{22}\varepsilon_{22k} + \delta_{23}\varepsilon_{23k} + \delta_{31}\varepsilon_{31k} + \delta_{32}\varepsilon_{32k} + \delta_{33}\varepsilon_{33k}$$
$$= 0$$

ii) $\varepsilon_{ipq}\varepsilon_{jpq} = 2\delta_{ij}$

$\varepsilon_{ipq}\varepsilon_{jpq}$ → i and j are free indices; p and q are dummy indices.

Here, $n=3$. Then

$$\varepsilon_{ipq}\varepsilon_{jpq} = \varepsilon_{i1q}\varepsilon_{j1q} + \varepsilon_{i2q}\varepsilon_{j2q} + \varepsilon_{i3q}\varepsilon_{j3q}$$
$$= \varepsilon_{i11}\varepsilon_{j11} + \varepsilon_{i12}\varepsilon_{j12} + \varepsilon_{i13}\varepsilon_{j13} + \varepsilon_{i21}\varepsilon_{j21} + \varepsilon_{i22}\varepsilon_{j22} + \varepsilon_{i23}\varepsilon_{j23} + \varepsilon_{i31}\varepsilon_{j31} + \varepsilon_{i32}\varepsilon_{j32} + \varepsilon_{i33}\varepsilon_{j33}$$
$$= \varepsilon_{i12}\varepsilon_{j12} + \varepsilon_{i13}\varepsilon_{j13} + \varepsilon_{i21}\varepsilon_{j21} + \varepsilon_{i23}\varepsilon_{j23} + \varepsilon_{i31}\varepsilon_{j31} + \varepsilon_{i32}\varepsilon_{j32}$$

($i=j=3$; $i=j=2$; $i=j=1$; $i=j=1$; $i=j=2$)

$\varepsilon_{ipq}\varepsilon_{jpq} =$

$$= \varepsilon_{i12}\varepsilon_{j12} + \varepsilon_{i13}\varepsilon_{j13} + \varepsilon_{i21}\varepsilon_{j21} + \varepsilon_{i23}\varepsilon_{j23} + \varepsilon_{i31}\varepsilon_{j31} + \varepsilon_{i32}\varepsilon_{j32}$$

But i and j are free indices, so in one equation they can assume one value:

$\varepsilon_{1pq}\varepsilon_{1pq} = 2$	$\varepsilon_{1pq}\varepsilon_{2pq} = 0$	$\varepsilon_{1pq}\varepsilon_{3pq} = 0$
$\varepsilon_{2pq}\varepsilon_{1pq} = 0$	$\varepsilon_{2pq}\varepsilon_{2pq} = 2$	$\varepsilon_{2pq}\varepsilon_{3pq} = 0$
$\varepsilon_{3pq}\varepsilon_{1pq} = 0$	$\varepsilon_{3pq}\varepsilon_{2pq} = 0$	$\varepsilon_{3pq}\varepsilon_{3pq} = 2$

iii) $\varepsilon_{ijk}\varepsilon_{ijk} = 6$

all indices are dummy.

$$\varepsilon_{ijk}\varepsilon_{ijk} = \varepsilon_{1jk}\varepsilon_{1jk} + \varepsilon_{2jk}\varepsilon_{2jk} + \varepsilon_{3jk}\varepsilon_{3jk}$$
$$= \varepsilon_{12k}\varepsilon_{12k} + \varepsilon_{13k}\varepsilon_{13k} + \varepsilon_{21k}\varepsilon_{21k} + \varepsilon_{23k}\varepsilon_{23k} + \varepsilon_{31k}\varepsilon_{31k} + \varepsilon_{32k}\varepsilon_{32k}$$
$$= \varepsilon_{123}\varepsilon_{123} + \varepsilon_{132}\varepsilon_{132} + \varepsilon_{213}\varepsilon_{213} + \varepsilon_{231}\varepsilon_{231} + \varepsilon_{312}\varepsilon_{312} + \varepsilon_{321}\varepsilon_{321}$$
$$= (1)^2 + (-1)^2 + (-1)^2 + (1)^2 + (1)^2 + (-1)^2$$
$$= 1+1+1+1+1+1$$

$\varepsilon_{ijk}\varepsilon_{ijk} = 6$

Tensor Notation for Matrices

Tensor Notation for Matrices

In the matrix notation $[a_{ij}]_{mn}$, the first subscript *i,* tells what row the number a_{ij} lies in and the second *j* designates the column, *m* denotes the number of rows and *n* the number of columns.

Examples:

$$C = [c_{ij}]_{33} = \begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{bmatrix}$$

$$D = [d_{ij}]_{24} = \begin{bmatrix} d_{11} & d_{12} & d_{13} & d_{14} \\ d_{21} & d_{22} & d_{23} & d_{24} \end{bmatrix}$$

$A = [a_{ij}]_{mn}$, a_{11}, a_{32}

Basic Formulas:

The essential formulas involving matrices, vectors and determinants will be written in terms of summation convention.

- **Matrix Multiplication**

If $A = [a_{ij}]_{mn}$ and $B = [b_{ij}]_{nk}$, then

$$AB = [a_{ir}b_{rj}]_{mk}$$

- **Identity Matrix**

In terms of the kronecker delta, the identity matrix of order *n* is

$$I = [\delta_{ij}]_{nn}$$

Which has the property

$$A = AI = IA = [a_{ij}\delta_{ji}]_{nn}$$

For any square matrix of order n.

Recall that

$$\delta_{ij} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- **Inverse of a square matrix**

A square matrix $A = [a_{ij}]_{nn}$ is invertible if there exists a unique matrix $B = [b_{ij}]_{nn}$ called the inverse of A, such that

$$AB = BA = I$$

In terms of components:

$$a_{ir}b_{rj} = b_{ir}a_{rj} = \delta_{ij}$$

- **Transpose of a Matrix**

The transpose of a matrix in index notation is defined as

$$A^T = [a_{ij}]^T_{mn} = [a^t_{ij}]_{nm}, \text{ where } a^t_{ij} = a_{ji}$$

A matrix A is orthogonal if $A^T = A^{-1}$

Example: Display explicitly the matrices

(a) $[b_{ij}]_{42}$ (b) $[h_{ij}]_{33}$

a) $[b_{ij}]_{42} = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \\ b_{41} & b_{42} \end{bmatrix}$ (rows ↙, columns ↘)

b) $[h_{ij}]_{33} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix}$

Example: Prove by using tensor notation and product rule for matrices that $(AB)^T = B^T A^T$.

Let $A = [a_{ij}]_{mn}$, $B = [b_{ij}]_{nk}$

$AB = [c_{ij}]_{mk}$ and for all *i* and *j*

$a^t_{ij} = a_{ji}$, $b^t_{ij} = b_{ji}$, $c^t_{ij} = c_{ji}$

Now, $A^T = [a^t_{ij}]_{nm}$, $B^T = [b^t_{ij}]_{kn}$ and

$(AB)^T = [c^t_{ij}]_{km}$

By definition of matrix product:

$B^T A^T = [b^t_{ir}a^t_{rj}]_{km} = [b_{ri}a_{jr}]_{km} = [c^t_{ij}]_{km}$

$b_{ri}a_{jr} = a_{jr}b_{ri} = c_{ji} = c^t_{ij} = (AB)^T$

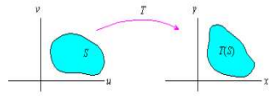
Coordinate Transformations

2-Dimensional Coordinate Transformations

A two dimensional coordinate transformation is a mapping of the form

$$T(u, v) = (x(u, v), y(u, v))$$

The transformation *T* maps a set *S* in the *uv*-plane to a set *T(S)* in the *xy*-plane.



Example:

If P has the coordinates $(1,2)$ in uv-plane then find the coordinates in xy-plane if: $x = u + v$ and $y = 2u - v$.

Solution: $P:(u,v) = (1,2)$

using the transformation:

$x = u + v$ and $y = 2u - v$

$\Rightarrow x = 1 + 2 = 3$, $y = 2(1) - 2$

$y = 2 - 2 = 0$

So, coordinates of P in xy-plane are: $(3,0)$

Coordinate Transformations

Example:

Transform the line PQ where P has coordinates $(1,-1)$ and Q has coordinates $(0,1)$ from uv-plane to xy-plane if: $x = u + v$ and $y = 2u - v$.

Solution: To transform the line we find the new coordinates of both its end points.

$P(u,v) = (1,-1)$, $Q(u,v) = (0,1)$

$P(x,y) = (u+v, 2u-v)$ $Q(x,y) = (u+v, 2u-v)$

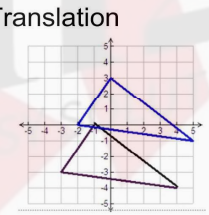
$= (0, 2+1)$ $Q(x,y) = (1,-1)$

$P(x,y) = (0,3)$

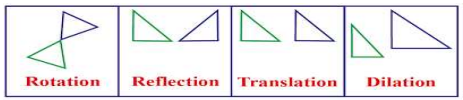
$\Rightarrow$ The line PQ is transformed to a line joining P(0,3) and Q(1,-1) in xy-plane.

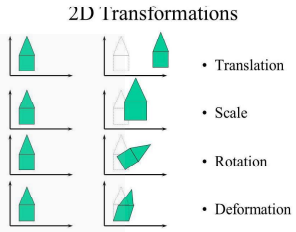
Translation

A translation slides an object a fixed distance in a given direction. The original object and its translation have the same shape and size, and they face in the same direction. To translate a shape every point of the shape must move the same distance and in the same direction.



Some Well-Known Transformations

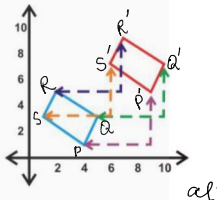




For example the transformation

$$(x, y) \to (x + 5, y + 3)$$

moves the parallelogram to the right 5 units and up 3 units.



$P(x,y) = (4,1)$

$Q(x,y) = (5,3)$

$R(x,y) = (2,5)$

$S(x,y) = (1,3)$

after translation:

$P' = (9,4)$

$Q' = (10,6)$

$R' = (7,8)$

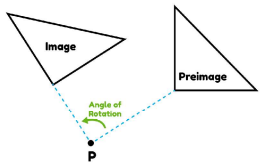
$S' = (6,6)$

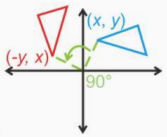
Rotation

A rotation is a transformation where a figure is turned around a fixed point to create an image. The two dimensions rotation matrix is

$$R = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

To perform the rotation on a plane point with coordinates (x, y), it should be written as a column vector, and multiplied by the matrix R:





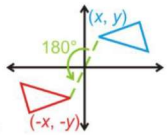
Rotation of $90°: (x, y) \to (-y, x)$

here $\theta = 90°$, So

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$= \begin{bmatrix} 0-y \\ x+0 \end{bmatrix} = \begin{bmatrix} -y \\ x \end{bmatrix}$$

$\Rightarrow$ under 90° rotation a point (x,y) is transformed to $(-y,x)$



Rotation of $180°: (x, y) \to (-x, -y)$

for $\theta = 180°$, rotation matrix becomes:

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -x+0 \\ 0-y \end{bmatrix} = \begin{bmatrix} -x \\ -y \end{bmatrix}$$

implies under 180° rotation a point

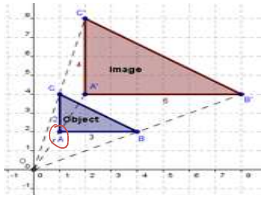
$(x, y) \longrightarrow (-x, -y)$

Scaling

To change the size of an object, scaling transformation is used. In the scaling process, you either expand or compress the dimensions of the object. Scaling can be achieved by multiplying the original coordinates of the object with the scaling factor to get the desired result.

$$\begin{bmatrix} X_{new} \\ Y_{new} \end{bmatrix} = \begin{bmatrix} S_x & 0 \\ 0 & S_y \end{bmatrix} \times \begin{bmatrix} X_{old} \\ Y_{old} \end{bmatrix}$$

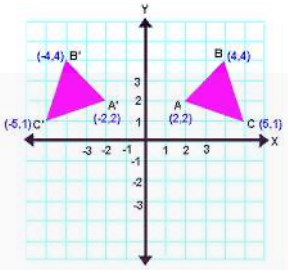
Scaling Matrix



consider point A:
A: (1,2)
and Scaling matrix:
$\begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}$
Then:
$\begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}\begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2+0 \\ 0+8 \end{bmatrix} = \begin{bmatrix} 2 \\ 8 \end{bmatrix}$
$\Rightarrow A_{new}: (2,8)$

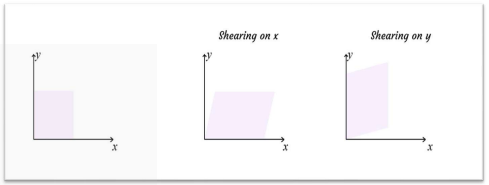
Reflection

A reflection is a transformation that turns a figure into its mirror image by flipping it over a line. Reflection is the mirror image of original object. In other words, we can say that it is a rotation operation with 180°. In reflection transformation, the size of the object does not change.



Shear

A transformation that slants the shape of an object is called the shear transformation. There are two shear transformations X-Shear and Y-Shear. One shifts X coordinate values and other shifts Y coordinate values. However; in both the cases only one coordinate changes its coordinates and other preserves its values. Shearing is also termed as Skewing.



A shear parallel to x-axis results in $x' = x + \alpha y$ and $y' = y$

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{bmatrix} 1 & \alpha \\ 0 & 1 \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

Similarly a shear parallel to y-axis results in $x' = x$ and $y' = y + \alpha x$

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{bmatrix} 1 & 0 \\ \alpha & 1 \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

If A: (1,2) and $x' = x+2y$, $y' = y$
then $x' = 1+2(2) = 5$ and $y' = 2$
$\Rightarrow$ (1, 2) $\rightarrow$ (5, 2)

Applications of Coordinate Transformations

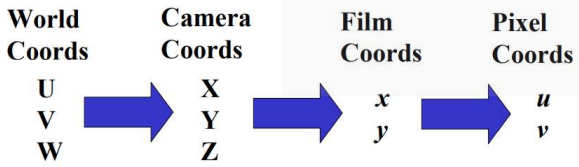
Applications of coordinate transformations

Transformations can represent geometric operations, which are caused by movement or action, as well as changes of coordinates, which are caused by changes of interpretation. Many common spatial transformations, including translations, rotations, and scaling are represented by matrix operations.

Here we will discuss some applications of transformations.

- Camera image processing
- Robotics
- Computer graphics

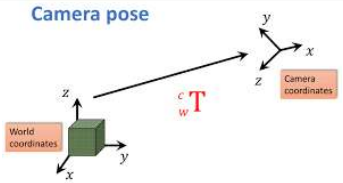
Camera image processing



Consider the three coordinate frames:

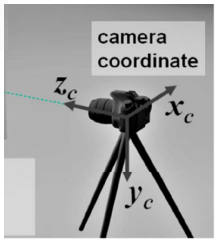
- **World coordinate frame:**

These are 3D coordinates fixed in the world, say with respect to one corner of the room.

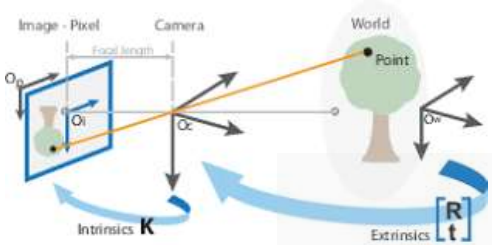


- **Camera coordinate frame:**

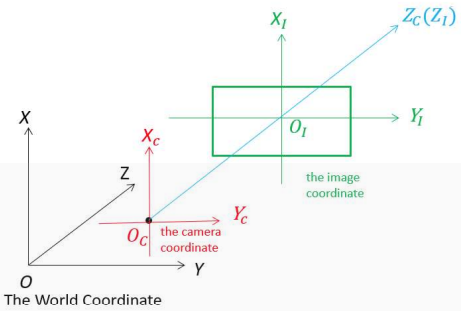
These are 3D coordinates fixed in the camera. The origin of the camera coordinates is at the center of projection of the camera. The z-axis is taken to be the optical axis of the camera (with points in front of the camera in the positive z direction).



- **Image coordinate frame:**

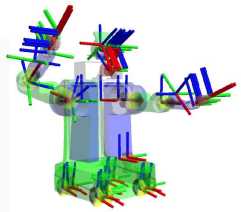


- **Coordinate transformations:**



Robotics:

For the purpose of controlling a robot, it is important to know the kinematics where transformations between the coordinate frames are performed. These frames are attached to different robot links of the robot. Vectors and matrices are used in robotics to represent 2D and 3D positions, directions, rigid body motion, and coordinate transformations.



Computer Graphics:

Computer graphics deals with generating images with the aid of computers. Today, computer graphics is a core technology in digital photography, film, video games, cell phone and computer displays, and many specialized applications.

A transformation is used to change the object's

- Position (translation)
- Size (scaling)
- Orientation (rotation)
- Shapes (shear)

Coordinate transformations

Position vector and point coordinates:

We will discuss the coordinates and position vector of a point in a three dimensional Cartesian coordinate system. The unit vector $\hat{\imath}, \hat{\jmath}, \hat{k}$ along x, y, z axes are represented by $\hat{e}_1, \hat{e}_2, \hat{e}_3$ in tensor notation or $\hat{\tilde{e}}_i, i = 1,2,3$.

Similarly, the coordinates of a point (x, y, z) are represented as:

Coordinates of P: $(x, y, z) = (x_1, x_2, x_3) = x_i \; ; \; i = 1, 2, 3$

and the position vector is represented as:

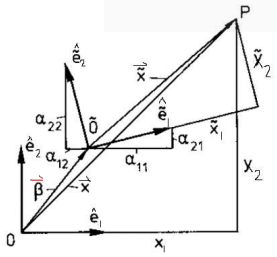
$$\vec{x} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$\vec{x} = x_1\hat{e}_1 + x_2\hat{e}_2 + x_3\hat{e}_3$$

$$\vec{x} = x_i\hat{e}_i$$

Transformation of Cartesian Coordinate Systems

Consider two Cartesian coordinate systems, in an arbitrary orientation relative to each other. We label one system by a tilde, to distinguish one from another, and call it the tilde coordinate system. The relative position of two coordinate systems in two dimensions is given in figure.



The origin of the tilde coordinate system has the coordinates w.r.t non-tilde system

$$\tilde{O} : (\beta_x, \beta_y, \beta_z) \qquad (\beta_1, \beta_2, \beta_3)$$

So, its position vector can be written as

$$\vec{\beta} = \beta_x\hat{e}_x + \beta_y\hat{e}_y + \beta_z\hat{e}_z = \beta_i\hat{e}_i$$
$$= \beta_1\hat{e}_1 + \beta_2\hat{e}_2 + \beta_3\hat{e}_3 = \beta_i\hat{e}_i$$

The basis vectors $\hat{\tilde{e}}_i$ of the tilde coordinate system can be written in terms of the no-tilde system as

$$\hat{\tilde{e}}_x = \alpha_{11}\hat{e}_x + \alpha_{21}\hat{e}_y + \alpha_{31}\hat{e}_z = \alpha_{i1}\hat{e}_i,$$

$$\hat{\tilde{e}}_y = \alpha_{12}\hat{e}_x + \alpha_{22}\hat{e}_y + \alpha_{32}\hat{e}_z = \alpha_{i2}\hat{e}_i,$$

$$\hat{\tilde{e}}_z = \alpha_{13}\hat{e}_x + \alpha_{23}\hat{e}_y + \alpha_{33}\hat{e}_z = \alpha_{i3}\hat{e}_i,$$

Implies $\hat{\tilde{e}}_j = \alpha_{ij}\hat{e}_i$.

j is free index; i is dummy index

- The transformation between no-tilde and tilde coordinate systems is completely described by β_i and α_{ij}.
- The α_{ij} are called the transformation coefficients or the transformation matrix.

Properties of the Transformation Coefficients

- Let $\tilde{\gamma}_i$ be the coordinates of the vector $\hat{\tilde{e}}_j$, i.e.,

$$\hat{\tilde{e}}_j = \tilde{\gamma}_i \hat{e}_i$$

Comparing this expression with

$$\hat{\tilde{e}}_j = \alpha_{ij} \hat{e}_i$$

we have

$$\alpha_{ij} = \tilde{\gamma}_i$$

Implies that α_{ij} is the *ith*-component of the vector $\hat{\tilde{e}}_j$ with respect to basis $\hat{e}_i$.

- Another interpretation of α_{ij} is as follows. As

$$\hat{\tilde{e}}_j = \alpha_{ij} \hat{e}_i,$$

$$\hat{\tilde{e}}_j . \hat{e}_k = \alpha_{ij} \hat{e}_i . \hat{e}_k,$$

$$\hat{\tilde{e}}_j . \hat{e}_k = \alpha_{ij} \delta_{ik} = \alpha_{kj}.$$

Since $\hat{\tilde{e}}_j . \hat{e}_k$ is the cosine of the angle between $\hat{\tilde{e}}_j$ and $\hat{e}_k$, we get

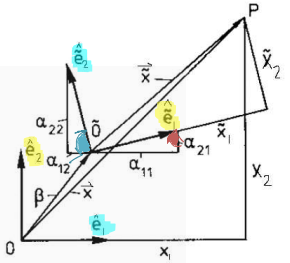
$$\alpha_{ij} = \cos(\hat{e}_i, \hat{\tilde{e}}_j).$$

Where the α_{ij} are the so called direction cosines.

- α_{ij} is not the same as α_{ji}.

For example

$$\alpha_{ij} = \cos(\hat{e}_i, \hat{\tilde{e}}_j) \quad and \quad \alpha_{ji} = \cos(\hat{e}_j, \hat{\tilde{e}}_i)$$



For example:

$\alpha_{12} = \cos(\hat{e}_1, \hat{\tilde{e}}_2)$

$\alpha_{21} = \cos(\hat{e}_2, \hat{\tilde{e}}_1)$

- Since both the $\hat{e}_i$ and the $\hat{\tilde{e}}_i$ are pairwise perpendicular unit vectors, they satisfy the relations

$$\hat{e}_i . \hat{e}_j = \delta_{ij} \quad and \quad \hat{\tilde{e}}_i . \hat{\tilde{e}}_j = \delta_{ij}$$

This gives

$$\begin{aligned} \delta_{ij} = \hat{\tilde{e}}_i . \hat{\tilde{e}}_j &= \alpha_{mi} \hat{e}_m . \alpha_{nj} \hat{e}_n, \\ &= \alpha_{mi} \alpha_{nj} \hat{e}_m . \hat{e}_n, \\ &= \alpha_{mi} \alpha_{nj} \delta_{mn}, \\ \delta_{ij} &= \alpha_{mi} \alpha_{mj}. \end{aligned}$$

Hence the transformation coefficients form an orthogonal matrix and

$$\alpha_{ik} \alpha_{jk} = \delta_{ij}, \qquad \alpha_{ki} \alpha_{kj} = \delta_{ij}$$

These are called orthogonality relations.
Where orthogonal matrices satisfy the relation

$$\det\left[\alpha_{ij}\right] = \pm 1$$

Transformation Law for Basis Vectors

Since we have:

$$\hat{\tilde{e}}_j = \alpha_{ij} \hat{e}_i.$$

Multiply the above equation with α_{kj}:

$$\alpha_{kj} \hat{\tilde{e}}_j = \alpha_{kj} \alpha_{ij} \hat{e}_i$$

$$\alpha_{kj} \hat{\tilde{e}}_j = \delta_{ki} \hat{e}_i = \hat{e}_k \Rightarrow \hat{e}_k = \alpha_{kj} \hat{\tilde{e}}_j$$

After renaming the indices, we obtain the following transformation law for basis vectors:

$$\hat{e}_i = \alpha_{ij} \hat{\tilde{e}}_j,$$

$$\hat{\tilde{e}}_i = \alpha_{ji} \hat{e}_j$$

Example: let $\hat{e}_1 = (1,0,0)$, $\hat{e}_2 = (0,1,0)$, $\hat{e}_3 = (0,0,1)$ and $a_{ij} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is the transformation matrix then find the new basis vectors.

Solution: Assume the transformation equation for basis: $\hat{\tilde{e}}_j = \alpha_{ji} \hat{e}_i$

$\hat{\tilde{e}}_1 = \alpha_{1i} \hat{e}_i = \alpha_{11} \hat{e}_1 + \alpha_{12} \hat{e}_2 + \alpha_{13} \hat{e}_3$

$\hat{\tilde{e}}_1 = (-2)(1,0,0) + 0 + 0 = (-2,0,0) = \hat{\tilde{e}}_1$

$\hat{\tilde{e}}_2 = \alpha_{2i} \hat{e}_i = \alpha_{21} \hat{e}_1 + \alpha_{22} \hat{e}_2 + \alpha_{23} \hat{e}_3$

$= 0 + (-2)(0,1,0) + 0 = (0,-2,0) = \hat{\tilde{e}}_2$

$\hat{\tilde{e}}_3 = \alpha_{3i} \hat{e}_i = \alpha_{31} \hat{e}_1 + \alpha_{32} \hat{e}_2 + \alpha_{33} \hat{e}_3 = (0,0,1) = \hat{\tilde{e}}_3$

Coordinate Transformations

Transformation law for point coordinates

The position vector to a point P can be expressed as

$\vec{x} = x_i \hat{e}_i \rightarrow$ (in no-tilde system)

$\vec{x} = \tilde{x}_i \hat{\tilde{e}}_i \rightarrow$ (in tilde system)

And following is the relation between them

$$\vec{x} = \vec{\beta} + \vec{\tilde{x}}$$

Implies

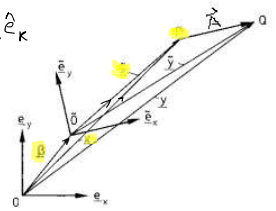
$$\vec{\beta} = \beta_i \hat{e}_i, \quad \hat{\tilde{e}}_i = \alpha_{ki} \hat{e}_k$$

$$\vec{x} = \beta_i \hat{e}_i + \tilde{x}_i \alpha_{ki} \hat{e}_k$$

$$x_i \hat{e}_i = \beta_i \hat{e}_i + \tilde{x}_i \alpha_{ki} \hat{e}_k$$

$$x_i \hat{e}_i = \beta_i \hat{e}_i + \tilde{x}_j \alpha_{ij} \hat{e}_i$$

$$x_i \hat{e}_i = (\beta_i + \tilde{x}_j \alpha_{ij}) \hat{e}_i$$



Since the representation of a position vector with respect to a basis is unique, we have

$$x_i = \beta_i + \tilde{x}_j \alpha_{ij}$$

Similarly, the inverse relation is

$$\tilde{x}_i = \alpha_{ji} x_j - \alpha_{ji}\beta_j$$

Consider $x_i = \beta_i + \tilde{x}_j \alpha_{ij}$ (multiply by α_{ik})

$\alpha_{ik} x_i = \alpha_{ik}\beta_i + \alpha_{ik}\tilde{x}_j \alpha_{ij}$

$\Rightarrow \alpha_{ik} x_i = \alpha_{ik}\beta_i + \alpha_{ik}\alpha_{ij}\tilde{x}_j$ ($\because \alpha_{ik}\alpha_{ij} = \delta_{kj}$)

$\Rightarrow \alpha_{ik} x_i = \alpha_{ik}\beta_i + \delta_{kj}\tilde{x}_j$ ($\delta_{kj} = 1$ if $k = j$)

$\Rightarrow \alpha_{ik} x_i - \alpha_{ik}\beta_i = \tilde{x}_k$

OR $\tilde{x}_i = \alpha_{ji} x_j - \alpha_{ji}\beta_j$ (just replacing k with i and dummy i with j)

2 marks

Thus we say that a vector is a quantity which in a coordinate system can be represented by

$$\vec{A} = A_1\hat{e}_1 + A_2\hat{e}_2 + A_3\hat{e}_3 = A_i\hat{e}_i$$

And whose coordinates transform according to following rule

$$A_i = \alpha_{ij}\tilde{A}_j$$
$$and\ \tilde{A}_i = \alpha_{ji}A_j$$

The Transformation Law for Vector Coordinates

A vector $\vec{A}$, in a Cartesian coordinate system with basis vectors $\hat{e}_i$, can be written as

$$\vec{A} = A_1\hat{e}_1 + A_2\hat{e}_2 + A_3\hat{e}_3 = A_i\hat{e}_i$$

We can obtain the transformation law for the coordinates of a vector from the transformation law for point coordinates. From the figure

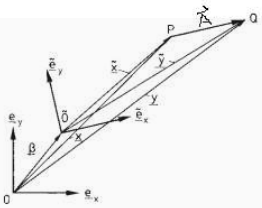
$\vec{A} = \vec{y} - \vec{x}$ or $\vec{\tilde{A}} = \vec{\tilde{y}} - \vec{\tilde{x}}$

so $\vec{A} = \vec{y} - \vec{x}$ implies

$A_i\hat{e}_i = (y_i - x_i)\hat{e}_i$

and $\vec{\tilde{A}} = \vec{\tilde{y}} - \vec{\tilde{x}}$

$\tilde{A}_i\hat{\tilde{e}}_i = (\tilde{y}_i - \tilde{x}_i)\hat{\tilde{e}}_i$



So we obtain

$A_i\hat{e}_i = (y_i - x_i)\hat{e}_i$

$implies\ A_i = y_i - x_i$

$and\ similarly$

$\tilde{A}_i = \tilde{y}_i - \tilde{x}_i$

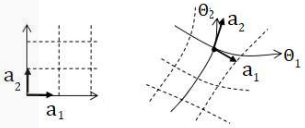
After using the transformation law for coordinates of points we obtain

$$A_i = \alpha_{ij}\tilde{A}_j$$
$$and\ \tilde{A}_i = \alpha_{ji}A_j$$

Curvilinear Coordinate Systems

Curvilinear Coordinate System

- In Curvilinear Coordinate system, the coordinate lines may be curved and this is not allowed in Cartesian Coordinate system.
- The basis set in Curvilinear coordinate system may not be orthogonal everywhere in space i.e. the basis vectors don't have to be at 90 degrees to each other, and they don't have a magnitude of 1.



Cartesian **Curvilinear**

Orthogonal Curvilinear Coordinate System

A curvilinear coordinate system is called orthogonal when the basis vectors are always orthogonal.

Coordinate Lines	Basis Vectors	Example
Straight	Orthogonal	Cartesian coordinate systems
Straight	Not Orthogonal	Skew coordinate systems
Curved	Orthogonal	Plane polar coordinate system, Cylindrical coordinate system, Spherical coordinate system
Curved	Not orthogonal	Elliptical cylindrical coordinates

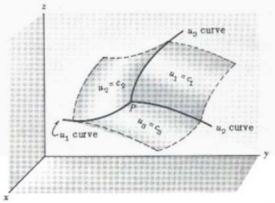
Transformation of Coordinates

Let the rectangular coordinates (x, y, z) of any point can be expressed as the function of (u_1, u_2, u_3) so that

$$x = x(u_1, u_2, u_3)$$
$$y = y(u_1, u_2, u_3)$$
$$z = z(u_1, u_2, u_3)$$

and

$$u_1 = u_1(x, y, z)$$
$$u_2 = u_2(x, y, z)$$
$$u_3 = u_3(x, y, z)$$



Given a point P with rectangular coordinates (x, y, z), we can associate a unique set of coordinates (u_1, u_2, u_3) called the curvilinear coordinates of the point P. Hence any point P can then be defined in space not only by rectangular coordinates (x, y, z), but also by curvilinear coordinates (u_1, u_2, u_3).

Unit vectors

Let $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

be the position vector of a point P. Then using the transformation between cartesian and curvilinear coordinates we have

$\vec{r} = \vec{r}(u_1, u_2, u_3)$

A tangent vector to the u_1 curve at P (for which u_2 and u_3 are constants) is $\frac{\partial \vec{r}}{\partial u_1}$. Then a unit tangent vector in this direction is

$$\hat{e}_1 = \frac{\frac{\partial \vec{r}}{\partial u_1}}{\left|\frac{\partial \vec{r}}{\partial u_1}\right|}$$

Similarly if $\hat{e}_2$ and $\hat{e}_3$ are unit tangent vectors to the u_2 and u_3 curves at P, respectively, then

$$\hat{e}_2 = \frac{\frac{\partial \vec{r}}{\partial u_2}}{\left|\frac{\partial \vec{r}}{\partial u_2}\right|},\quad \hat{e}_3 = \frac{\frac{\partial \vec{r}}{\partial u_3}}{\left|\frac{\partial \vec{r}}{\partial u_3}\right|}$$

Curvilinear Coordinate Systems

Scale factors

From the unit tangents along u_1, u_2 and u_3 curve we have

$\frac{\partial \vec{r}}{\partial u_1} = \left|\frac{\partial \vec{r}}{\partial u_1}\right| \hat{e}_1 = h_1 \hat{e}_1$ where $h_1 = \left|\frac{\partial \vec{r}}{\partial u_1}\right|$

And similarly

$\frac{\partial \vec{r}}{\partial u_2} = \left|\frac{\partial \vec{r}}{\partial u_2}\right| \hat{e}_2 = h_2 \hat{e}_2$ where $h_2 = \left|\frac{\partial \vec{r}}{\partial u_2}\right|$

$\frac{\partial \vec{r}}{\partial u_3} = \left|\frac{\partial \vec{r}}{\partial u_3}\right| \hat{e}_3 = h_3 \hat{e}_3$ where $h_3 = \left|\frac{\partial \vec{r}}{\partial u_3}\right|$

The quantities h_1, h_2 and h_3 are called scale factors. The scale factor gives a measure of how a change in the coordinate, changes the position of a point.

Unit normal vectors:

∇u_1 is a vector at P normal to the surface $u_1 = c_1$, a unit vector in this direction is given by

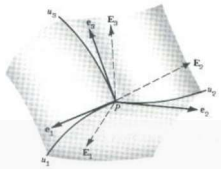
$\hat{E}_1 = \frac{\nabla u_1}{|\nabla u_1|}$

Similarly, the unit vectors

$\hat{E}_2 = \frac{\nabla u_2}{|\nabla u_2|}$ and

$\hat{E}_3 = \frac{\nabla u_3}{|\nabla u_3|}$

at P are normal to the surfaces $u_2 = c_2$ and $u_3 = c_3$, respectively.



- Thus at each point P of a curvilinear coordinate system there exist two sets of unit vectors $\hat{e}_1, \hat{e}_2, \hat{e}_3$ tangent to the coordinate curves and $\hat{E}_1, \hat{E}_2, \hat{E}_3$ normal to the coordinate surface.
- The two sets become identical if and only if the curvilinear system is orthogonal.
- Both sets are analogous to $\hat{i}, \hat{j}, \hat{k}$ unit vectors in rectangular Cartesian coordinates but are unlike in the way that they may change directions from point to point.
- These tangent vector and normal vectors constitute reciprocal system of vectors. (A vector having the same direction as that of a given vector but magnitude equal to the reciprocal of the given vector is known as the reciprocal of vector.)

Orthogonal curvilinear coordinate system

If the coordinate curves intersect at right angles, the curvilinear coordinate system is called **orthogonal.** The u_1, u_2 *and* u_3 coordinate curves of an orthogonal curvilinear system are similar to the x, y *and* z coordinate axes of a rectangular Cartesian system. In an orthogonal curvilinear coordinate system, the unit vectors $\hat{e}_1, \hat{e}_2, \hat{e}_3$ are mutually orthogonal.

$$\hat{e}_1.\hat{e}_1 = \hat{e}_2.\hat{e}_2 = \hat{e}_3.\hat{e}_3 = 1, \text{ and } \quad \hat{e}_1.\hat{e}_2 = \hat{e}_2.\hat{e}_3 = \hat{e}_3.\hat{e}_1 = 0$$

If the system is **right-handed** then

$$\hat{e}_1 \times \hat{e}_2 = -\hat{e}_2 \times \hat{e}_1 = \hat{e}_3, \quad \hat{e}_2 \times \hat{e}_3 = \hat{e}_3 \times \hat{e}_2 = \hat{e}_1, \quad \hat{e}_3 \times \hat{e}_1 = -\hat{e}_1 \times \hat{e}_3 = \hat{e}_2$$

If the system is **left-handed** then

$$\hat{e}_1 \times \hat{e}_2 = -\hat{e}_2 \times \hat{e}_1 = -\hat{e}_3, \quad \hat{e}_2 \times \hat{e}_3 = \hat{e}_3 \times \hat{e}_2 = -\hat{e}_1, \quad \hat{e}_3 \times \hat{e}_1 = -\hat{e}_1 \times \hat{e}_3 = -\hat{e}_2$$

Left-handed system

Right-handed system

A vector in curvilinear coordinate system:

A vector $\vec{A}$ can be represented in terms of the unit basis vectors $\hat{e}_1, \hat{e}_2, \hat{e}_3$ or $\hat{E}_1$ $\hat{E}_2, \hat{E}_3$ in the form

$$\vec{A} = A_1\hat{e}_1 + A_2\hat{e}_2 + A_3\hat{e}_3$$
$$\vec{A} = a_1\hat{E}_1 + a_2\hat{E}_2 + a_3\hat{E}_3$$

Where A_1, A_2, A_3 and a_1, a_2, a_3 are the respective components of vector $\vec{A}$ in each system.

We can also represent $\vec{A}$ in terms of $\frac{\partial \vec{r}}{\partial u_1}, \frac{\partial \vec{r}}{\partial u_2}, \frac{\partial \vec{r}}{\partial u_3}$ or $\nabla u_1, \nabla u_2, \nabla u_3$ as

$$\vec{A} = C_1\frac{\partial \vec{r}}{\partial u_1} + C_2\frac{\partial \vec{r}}{\partial u_2} + C_3\frac{\partial \vec{r}}{\partial u_3}$$
$$\vec{A} = c_1\nabla u_1 + c_2\nabla u_2 + c_3\nabla u_3$$

Where C_1, C_2, C_3 are called the contravariant components of the vector $\vec{A}$ and c_1, c_2, c_3 are called the covariant components of $\vec{A}$.

Note: If u_1, u_2, u_3 are general coordinates, then $\frac{\partial \vec{r}}{\partial u_1}, \frac{\partial \vec{r}}{\partial u_2}, \frac{\partial \vec{r}}{\partial u_3}$ and $\nabla u_1, \nabla u_2, \nabla u_3$ are reciprocal system of vectors.

i.e. $\frac{\partial \vec{r}}{\partial u_i} \cdot \nabla u_j = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$

where $i = 1, 2, 3$

$j = 1, 2, 3$

Arc length in general curvilinear coordinates

The square of the element of arc length in general curvilinear coordinates can be expressed by

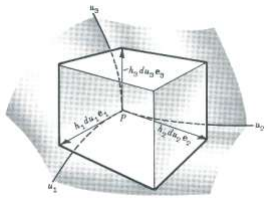
$$ds^2 = \sum_{i=1}^{3}\sum_{j=1}^{3} h_i h_j (\hat{e}_i.\hat{e}_j) du_i du_j$$

$$\vec{r} = \vec{r}(u_1, u_2, u_3)$$

$$d\vec{r} = \frac{\partial \vec{r}}{\partial u_1}du_1 + \frac{\partial \vec{r}}{\partial u_2}du_2 + \frac{\partial \vec{r}}{\partial u_3}du_3$$

$$d\vec{r} = h_1 du_1 \hat{e}_1 + h_2 du_2 \hat{e}_2 + h_3 du_3 \hat{e}_3$$

$$ds^2 = d\vec{r}.d\vec{r}$$



Arc length element in orthogonal curvilinear coordinates

The arc length ds is the length of the infinitesimal vector $d\vec{r}$:

$$(ds)^2 = d\vec{r}.d\vec{r}$$

In curvilinear coordinates, we have

$$\vec{r} = \vec{r}(u_1, u_2, u_3)$$

$$d\vec{r} = \frac{\partial \vec{r}}{\partial u_1} du_1 + \frac{\partial \vec{r}}{\partial u_2} du_2 + \frac{\partial \vec{r}}{\partial u_3} du_3$$

$$d\vec{r} = h_1 du_1 \hat{e}_1 + h_2 du_2 \hat{e}_2 + h_3 du_3 \hat{e}_3$$

$$ds^2 = h_1^2 du_1^2 + h_2^2 du_2^2 + h_3^2 du_3^2$$

Area element in an orthogonal curvilinear system:

Along a u_1 curve, u_2 *and* u_3 are constants so that $d\vec{r} = h_1 du_1 \hat{e}_1$. Then the differential of arc length ds_1 along u_1 at P is $h_1 du_1$.
The area elements are given by

$$dA_1 = \left|(h_2 du_2 \hat{e}_2) \times (h_3 du_3 \hat{e}_3)\right|$$

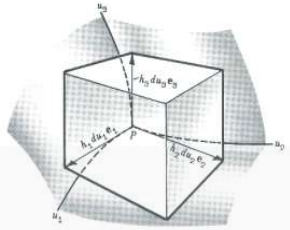
$$dA_1 = h_2 h_3 \left|\hat{e}_2 \times \hat{e}_3\right| du_2 du_3 = h_2 h_3 du_2 du_3$$

$$As \ \left|\hat{e}_2 \times \hat{e}_3\right| = \left|\hat{e}_1\right| = 1$$

similarly

$$dA_2 = \left|(h_1 du_1 \hat{e}_1) \times (h_3 du_3 \hat{e}_3)\right| = h_1 h_3 du_1 du_3$$

$$dA_3 = \left|(h_1 du_1 \hat{e}_1) \times (h_2 du_2 \hat{e}_2)\right| = h_1 h_2 du_1 du_2$$



Volume element:

We know that the absolute value of the scalar triple product gives the **volume of the parallelepiped**.

$$dv = \left|(h_1 du_1 \hat{e}_1).(h_2 du_2 \hat{e}_2) \times (h_3 du_3 \hat{e}_3)\right|$$

$$dv = h_1 h_2 h_3 du_1 du_2 du_3 \left|\hat{e}_1.\hat{e}_2 \times \hat{e}_3\right| = h_1 h_2 h_3 du_1 du_2 du_3$$

$$As \left|\hat{e}_1.\hat{e}_2 \times \hat{e}_3\right| = 1$$

Jacobian of transformation

If u_1, u_2 and u_3 are the orthogonal curvilinear coordinates and

$$x = x(u_1, u_2, u_3), \ y = y(u_1, u_2, u_3), \ z = z(u_1, u_2, u_3)$$

then the Jacobian of x, y, z is denoted by J or $J(\frac{x,y,z}{u_1,u_2,u_3})$ is defined as

$$J = J(\frac{x,y,z}{u_1,u_2,u_3}) = \frac{\partial(x,y,z)}{\partial(u_1,u_2,u_3)} = \begin{vmatrix} \frac{\partial x}{\partial u_1} & \frac{\partial y}{\partial u_1} & \frac{\partial z}{\partial u_1} \\ \frac{\partial x}{\partial u_2} & \frac{\partial y}{\partial u_2} & \frac{\partial z}{\partial u_2} \\ \frac{\partial x}{\partial u_3} & \frac{\partial y}{\partial u_3} & \frac{\partial z}{\partial u_3} \end{vmatrix}$$

From the definition of a scalar triple product, the above determinant can be written as

$$J(\frac{x,y,z}{u_1,u_2,u_3}) = \left(\frac{\partial x}{\partial u_1}\hat{i} + \frac{\partial y}{\partial u_1} j + \frac{\partial z}{\partial u_1}k\right).\left(\frac{\partial x}{\partial u_2}\hat{i} + \frac{\partial y}{\partial u_2} j + \frac{\partial z}{\partial u_2}k\right) \times \left(\frac{\partial x}{\partial u_3}\hat{i} + \frac{\partial y}{\partial u_3} j + \frac{\partial z}{\partial u_3}k\right)$$

$$J(\frac{x,y,z}{u_1,u_2,u_3}) = \frac{\partial \vec{r}}{\partial u_1}.\frac{\partial \vec{r}}{\partial u_2} \times \frac{\partial \vec{r}}{\partial u_3} = h_1 \hat{e}_1 . h_2 \hat{e}_2 \times h_3 \hat{e}_3$$

$$J(\frac{x,y,z}{u_1,u_2,u_3}) = h_1 h_2 h_3 (\hat{e}_1 . \hat{e}_2 \times \hat{e}_3)$$

$$J(\frac{x,y,z}{u_1,u_2,u_3}) = h_1 h_2 h_3 \quad (\text{since} \quad \hat{e}_1 . \hat{e}_2 \times \hat{e}_3 = 1)$$

If the Jacobian equals to zero identically then $\frac{\partial \vec{r}}{\partial u_1}, \frac{\partial \vec{r}}{\partial u_2}, \frac{\partial \vec{r}}{\partial u_3}$ are coplanar vectors and the linear coordinate transformation breaks down, i.e., there is a relation between x, y, z having the form $F(x, y, z) = 0$. We shall therefore require the Jacobian to be different from zero.

Curvilinear Coordinate Systems

Example: Find the scale factors, arc length and Jacobian in rectangular Cartesian coordinates.

Solution: If $\vec{r}$ is the position vector of any point P in a coordinate system then the scale factors are defined by:

$$h_1 = \left|\frac{\partial \vec{r}}{\partial u_1}\right|, \ h_2 = \left|\frac{\partial \vec{r}}{\partial u_2}\right|, \ h_3 = \left|\frac{\partial \vec{r}}{\partial u_3}\right|$$

also,

$$\hat{e}_1 = \frac{\left(\frac{\partial \vec{r}}{\partial u_1}\right)}{h_1}, \ \hat{e}_2 = \frac{\left(\frac{\partial \vec{r}}{\partial u_2}\right)}{h_2}, \ \hat{e}_3 = \frac{\left(\frac{\partial \vec{r}}{\partial u_3}\right)}{h_3}$$

Now, the position vector of a point 'P' in rectangular Cartesian Coordinate is: $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

and $(u_1, u_2, u_3) \to (x, y, z)$

Implies:

$$\frac{\partial \vec{r}}{\partial x} = \hat{i}, \ \frac{\partial \vec{r}}{\partial y} = \hat{j}, \ \frac{\partial \vec{r}}{\partial z} = \hat{k}$$

$$\left|\frac{\partial \vec{r}}{\partial x}\right| = 1, \ \left|\frac{\partial \vec{r}}{\partial y}\right| = 1, \ \left|\frac{\partial \vec{r}}{\partial z}\right| = 1$$

$$\Rightarrow h_1 = 1, \ h_2 = 1, \ h_3 = 1$$

Also, $\hat{e}_1 = \hat{i}, \ \hat{e}_2 = \hat{j}, \ \hat{e}_3 = \hat{k}$

$$ds^2 = d\vec{r} . d\vec{r}$$

$$d\vec{r} = \frac{\partial \vec{r}}{\partial x} dx + \frac{\partial \vec{r}}{\partial y} dy + \frac{\partial \vec{r}}{\partial z} dz$$

$$d\vec{r} = dx\,\hat{i} + dy\,\hat{j} + dz\,\hat{k}$$

$$\Rightarrow ds^2 = (dx\hat{i} + dy\hat{j} + dz\hat{k}).(dx\hat{i} + dy\hat{j} + dz\hat{k})$$

$$ds^2 = (dx)^2 + (dy)^2 + (dz)^2$$

Similarly: $J = h_1 h_2 h_3 = 1$

Example: Consider the curvilinear coordinates system defined by

$x = u_3, \qquad y = e^{u_2}\cos u_1, \qquad z = e^{u_2}\sin u_1$

Find the unit vectors $\hat{e}_1, \hat{e}_2, \hat{e}_3$ and show that the system is orthogonal. Compute the scales factors $h_1, h_2, and\ h_3$.

5 marks

Solution: Consider:

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$\Rightarrow \vec{r} = u_3\hat{i} + e^{u_2}\cos u_1\,\hat{j} + e^{u_2}\sin u_1\,\hat{k}$$

Then $\dfrac{\partial \vec{r}}{\partial u_1} = -e^{u_2}\sin u_1\,\hat{j} + e^{u_2}\cos u_1\,\hat{k}$

$\dfrac{\partial \vec{r}}{\partial u_2} = e^{u_2}\cos u_1\,\hat{j} + e^{u_2}\sin u_1\,\hat{k}$

and $\partial\vec{r}/\partial u_3 = \hat{i}$

Now, we show that the system is orthogonal. So consider:

$$\hat{e}_1 \cdot \hat{e}_2 = (-\sin u_1\,\hat{j} + \cos u_1\,\hat{k}) \cdot (\cos u_1\,\hat{j} + \sin u_1\,\hat{k})$$

$$= -\sin u_1 \cos u_1 + \cos u_1 \sin u_1 = 0$$

$$\Rightarrow \hat{e}_1 \cdot \hat{e}_2 = 0$$

$$\hat{e}_2 \cdot \hat{e}_3 = (\cos u_1\,\hat{j} + \sin u_1\,\hat{k}) \cdot \hat{i} = 0$$

$$\Rightarrow \hat{e}_2 \cdot \hat{e}_3 = 0$$

and $\hat{e}_3 \cdot \hat{e}_1 = \hat{i} \cdot (-\sin u_1\,\hat{j} + \cos u_1\,\hat{k})$

$$\hat{e}_3 \cdot \hat{e}_1 = 0$$

$$\hat{e}_3 = \frac{\partial\vec{r}/\partial z}{|\partial\vec{r}/\partial z|} = \hat{k}$$

Now:

$$\hat{e}_1 \cdot \hat{e}_2 = \frac{-a^2\sin\theta\cos\theta + b^2\sin\theta\cos\theta}{\sqrt{a^2\cos^2\theta + b^2\sin^2\theta}\,\sqrt{a^2\sin^2\theta + b^2\cos^2\theta}} \neq 0 \quad b/c\ a \neq b$$

$\Rightarrow$ The elliptical cylindrical coordinate system is not orthogonal.

Now,

$$\left|\frac{\partial\vec{r}}{\partial u_1}\right| = \sqrt{(-e^{u_2}\sin u_1)^2 + (e^{u_2}\cos u_1)^2}$$

$$= \sqrt{e^{2u_2}(\sin^2 u_1 + \cos^2 u_1)} = \sqrt{e^{2u_2}}$$

$$\Rightarrow \left|\frac{\partial\vec{r}}{\partial u_1}\right| = e^{u_2}$$

$$\left|\frac{\partial\vec{r}}{\partial u_2}\right| = \sqrt{(e^{u_2}\cos u_1)^2 + (e^{u_2}\sin u_1)^2}$$

$$= e^{u_2}$$

and $\left|\dfrac{\partial\vec{r}}{\partial u_3}\right| = |\hat{i}| = 1$

Scale factors:

$$h_1 = \left|\frac{\partial\vec{r}}{\partial u_1}\right| = e^{u_2}$$

$$h_2 = \left|\frac{\partial\vec{r}}{\partial u_2}\right| = e^{u_2}$$

$$h_3 = \left|\frac{\partial\vec{r}}{\partial u_3}\right| = 1$$

Differential Operators in Orthogonal Curvilinear Coordinate Systems

So, $\hat{e}_1 = \dfrac{\partial\vec{r}/\partial u_1}{|\partial\vec{r}/\partial u_1|} = \dfrac{-e^{u_2}\sin u_1\,\hat{j} + e^{u_2}\cos u_1\,\hat{k}}{e^{u_2}}$

$$\hat{e}_1 = -\sin u_1\,\hat{j} + \cos u_1\,\hat{k}$$

Similarly:

$$\hat{e}_2 = \cos u_1\,\hat{j} + \sin u_1\,\hat{k}$$

and $\hat{e}_3 = \hat{i}$

Example: Check that the following system of elliptical cylindrical coordinates are orthogonal or not?

3marks

$\vec{r} = a\rho\cos\theta\,\hat{i} + b\rho\sin\theta\,\hat{j} + z\hat{k}\ ;\ a \neq b$

Solution: First we have to find $\hat{e}_1, \hat{e}_2, \hat{e}_3$ for the given systems

$$\hat{e}_1 = \frac{\partial\vec{r}/\partial\rho}{|\partial\vec{r}/\partial\rho|} = \frac{(a\cos\theta\,\hat{i} + b\sin\theta\,\hat{j})}{\sqrt{a^2\cos^2\theta + b^2\sin^2\theta}}$$

$$\hat{e}_2 = \frac{\partial\vec{r}/\partial\theta}{|\partial\vec{r}/\partial\theta|} = \frac{(-a\rho\sin\theta\,\hat{i} + b\rho\cos\theta\,\hat{j})}{\rho\sqrt{a^2\sin^2\theta + b^2\cos^2\theta}}$$

$$= \frac{-a\sin\theta\,\hat{i} + b\cos\theta\,\hat{j}}{\sqrt{a^2\sin^2\theta + b^2\cos^2\theta}}$$

Nabla operator

The differential operator nabla often appear in vector analysis. In the space of three variables it is defined as

$$\nabla = \frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}j + \frac{\partial}{\partial z}k$$

- Gradient $\rightarrow \nabla\psi$
- Divergence $\rightarrow \nabla \cdot \vec{A}$
- Curl $\longrightarrow \nabla \times A$
- Laplacian $\rightarrow \nabla^2\psi$

Gradient of a scalar field in curvilinear coordinates

Let u_1,u_2 and u_3 are the curvilinear coordinates.

Let $\nabla\psi = f_1\hat{e}_1 + f_2\hat{e}_2 + f_3\hat{e}_3$

Where f_1, f_2 *and* f_3 are to be determined.

Since $d\vec{r} = \frac{\partial\vec{r}}{\partial u_1}du_1 + \frac{\partial\vec{r}}{\partial u_2}du_2 + \frac{\partial\vec{r}}{\partial u_3}du_3$

$d\vec{r} = h_1du_1\hat{e}_1 + h_2du_2\hat{e}_2 + h_3du_3\hat{e}_3$

we have

$$d\psi = \nabla\psi.d\vec{r} = h_1f_1du_1 + h_2f_2du_2 + h_3f_3du_3$$

$$d\psi = \frac{\partial\psi}{\partial u_1}du_1 + \frac{\partial\psi}{\partial u_2}du_2 + \frac{\partial\psi}{\partial u_3}du_3$$

$$f_1 = \frac{1}{h_1}\frac{\partial\psi}{\partial u_1}, \quad f_2 = \frac{1}{h_2}\frac{\partial\psi}{\partial u_2}, \quad f_3 = \frac{1}{h_3}\frac{\partial\psi}{\partial u_3}$$

Then

$$\nabla\psi = \frac{1}{h_1}\frac{\partial\psi}{\partial u_1}\hat{e}_1 + \frac{1}{h_2}\frac{\partial\psi}{\partial u_2}\hat{e}_2 + \frac{1}{h_3}\frac{\partial\psi}{\partial u_3}\hat{e}_3$$

On comparing:

$$\nabla = \frac{1}{h_1}\frac{\partial}{\partial u_1}\hat{e}_1 + \frac{1}{h_2}\frac{\partial}{\partial u_2}\hat{e}_2 + \frac{1}{h_3}\frac{\partial}{\partial u_3}\hat{e}_3$$

Expression of curl

Consider

$$\nabla\times\vec{A} = \nabla\times(A_1\hat{e}_1 + A_2\hat{e}_2 + A_3\hat{e}_3)$$

$$= \nabla\times(A_1\hat{e}_1) + \nabla\times(A_2\hat{e}_2) + \nabla\times(A_3\hat{e}_3)$$

$$\nabla\times(A_1\hat{e}_1) = \nabla\times(A_1h_1\nabla u_1) \qquad (\text{As, } \hat{e}_1 = h_1\nabla u_1)$$

$$= \nabla(A_1h_1)\times\nabla u_1 + A_1h_1\nabla\times\nabla u_1$$

$$= \nabla(A_1h_1)\times\frac{\hat{e}_1}{h_1} + 0 \qquad (\text{As}, \nabla\times\nabla u_1 = 0)$$

$$= \left[\frac{1}{h_1}\frac{\partial}{\partial u_1}(A_1h_1)\hat{e}_1 + \frac{1}{h_2}\frac{\partial}{\partial u_2}(A_1h_1)\hat{e}_2 + \frac{1}{h_3}\frac{\partial}{\partial u_3}(A_1h_1)\hat{e}_3\right]\times\frac{\hat{e}_1}{h_1}$$

$$\nabla\times(A_1\hat{e}_1) = \frac{1}{h_3h_1}\frac{\partial}{\partial u_3}(A_1h_1)\hat{e}_2 - \frac{1}{h_1h_2}\frac{\partial}{\partial u_2}(A_1h_1)\hat{e}_3$$

similarly

$$\nabla\times(A_2\hat{e}_2) = \frac{1}{h_1h_2}\frac{\partial}{\partial u_1}(A_2h_2)\hat{e}_3 - \frac{1}{h_2h_3}\frac{\partial}{\partial u_3}(A_2h_2)\hat{e}_1$$

$$\nabla\times(A_3\hat{e}_3) = \frac{1}{h_2h_3}\frac{\partial}{\partial u_2}(A_3h_3)\hat{e}_1 - \frac{1}{h_3h_1}\frac{\partial}{\partial u_1}(A_3h_3)\hat{e}_2$$

Hence,

$$\nabla\times\vec{A} = \frac{\hat{e}_1}{h_2h_3}\left[\frac{\partial}{\partial u_2}(A_3h_3) - \frac{\partial}{\partial u_3}(A_2h_2)\right] + \frac{\hat{e}_2}{h_3h_1}\left[\frac{\partial}{\partial u_3}(A_1h_1) - \frac{\partial}{\partial u_1}(A_3h_3)\right]$$

$$+ \frac{\hat{e}_3}{h_1h_2}\left[\frac{\partial}{\partial u_1}(A_2h_2) - \frac{\partial}{\partial u_2}(A_1h_1)\right]$$

This can be write as

$$\nabla\times\vec{A} = \frac{1}{h_1h_2h_3}\begin{vmatrix} h_1\hat{e}_1 & h_2\hat{e}_2 & h_3\hat{e}_3 \\ \frac{\partial}{\partial u_1} & \frac{\partial}{\partial u_2} & \frac{\partial}{\partial u_3} \\ A_1h_1 & A_2h_2 & A_3h_3 \end{vmatrix}$$

Expression for Divergence of a vector field

Consider

$\nabla.\vec{A} = \nabla.(A_1\hat{e}_1 + A_2\hat{e}_2 + A_3\hat{e}_3)$

$\nabla.\vec{A} = \nabla.(A_1\hat{e}_1) + \nabla.(A_2\hat{e}_2) + \nabla.(A_3\hat{e}_3)$

since $\hat{e}_1, \hat{e}_2, and\ \hat{e}_3$ form a right handed system, therefore

$\hat{e}_1 = \hat{e}_2\times\hat{e}_3 = h_2h_3\nabla u_2\times\nabla u_3$

$\hat{e}_2 = \hat{e}_3\times\hat{e}_1 = h_3h_1\nabla u_3\times\nabla u_1$

$\hat{e}_3 = \hat{e}_1\times\hat{e}_2 = h_1h_2\nabla u_1\times\nabla u_2$

Now $\nabla.(A_1\hat{e}_1) = \nabla.(A_1h_2h_3\nabla u_2\times\nabla u_3)$

$= \nabla(A_1h_2h_3).\nabla u_2\times\nabla u_3 + A_1h_2h_3\nabla.(\nabla u_2\times\nabla u_3)$

$\therefore \nabla.(\vec{A}\times\vec{B}) = \vec{B}.(\nabla\times\vec{A}) - \vec{A}.(\nabla\times\vec{B})$ *and* $\nabla\times\nabla\phi = 0$

$\Rightarrow \nabla.(\nabla u_2\times\nabla u_3) = \nabla u_3.(\nabla\times\nabla u_2) - \nabla u_2.(\nabla\times\nabla u_3) = 0$

$\Rightarrow \nabla.(A_1\hat{e}_1) = \nabla(A_1h_2h_3).\frac{\hat{e}_2}{h_2}\times\frac{\hat{e}_3}{h_3} + 0$

$= \nabla(A_1h_2h_3).\frac{\hat{e}_1}{h_2h_3}$

$= \left[\frac{1}{h_1}\frac{\partial}{\partial u_1}(A_1h_2h_3)\hat{e}_1 + \frac{1}{h_2}\frac{\partial}{\partial u_2}(A_1h_2h_3)\hat{e}_2 + \frac{1}{h_3}\frac{\partial}{\partial u_3}(A_1h_2h_3)\hat{e}_3\right].\frac{\hat{e}_1}{h_2h_3}$

$$\nabla.(A_1\hat{e}_1) = \frac{1}{h_1h_2h_3}\frac{\partial}{\partial u_1}(A_1h_2h_3)$$

similarly

$$\nabla.(A_2\hat{e}_2) = \frac{1}{h_1h_2h_3}\frac{\partial}{\partial u_2}(A_2h_3h_1)$$

and $$\nabla.(A_3\hat{e}_3) = \frac{1}{h_1h_2h_3}\frac{\partial}{\partial u_3}(A_3h_1h_2)$$

Hence,

$$\nabla.\vec{A} = \frac{1}{h_1h_2h_3}\left[\frac{\partial}{\partial u_1}(A_1h_2h_3) + \frac{\partial}{\partial u_2}(A_2h_3h_1) + \frac{\partial}{\partial u_3}(A_3h_1h_2)\right]$$

Expression of Laplacian

Consider

$$\nabla\psi = \frac{1}{h_1}\frac{\partial\psi}{\partial u_1}\hat{e}_1 + \frac{1}{h_2}\frac{\partial\psi}{\partial u_2}\hat{e}_2 + \frac{1}{h_3}\frac{\partial\psi}{\partial u_3}\hat{e}_3.\ and$$

$$\nabla.\vec{A} = \frac{1}{h_1h_2h_3}\left[\frac{\partial}{\partial u_1}(A_1h_2h_3) + \frac{1}{h_1h_2h_3}\frac{\partial}{\partial u_2}(A_2h_3h_1) + \frac{1}{h_1h_2h_3}\frac{\partial}{\partial u_3}(A_3h_1h_2)\right]$$

Let, $\vec{A} = \nabla\psi$ *then* $A_1 = \frac{1}{h_1}\frac{\partial\psi}{\partial u_1}, \quad A_2 = \frac{1}{h_2}\frac{\partial\psi}{\partial u_2}, \quad A_3 = \frac{1}{h_3}\frac{\partial\psi}{\partial u_3}$

and $\nabla.\vec{A} = \nabla.\nabla\psi = \nabla^2\psi$

$$\nabla^2\psi = \frac{1}{h_1h_2h_3}\left[\frac{\partial}{\partial u_1}\left(\frac{h_2h_3}{h_1}\frac{\partial\psi}{\partial u_1}\right) + \frac{\partial}{\partial u_2}\left(\frac{h_3h_1}{h_2}\frac{\partial\psi}{\partial u_2}\right) + \frac{\partial}{\partial u_3}\left(\frac{h_1h_2}{h_3}\frac{\partial\psi}{\partial u_3}\right)\right]$$

Differential Operators in Orthogonal Curvilinear Coordinate Systems

Example: Consider the curvilinear coordinate system defined by

$$x = u_3, \quad y = e^{u_2}\cos u_1, \quad z = e^{u_2}\sin u_1$$

Find $\nabla\psi$ and $\nabla^2\psi$, If $\psi = u_1^2 + u_2^2 + u_3^2$.

Sol:

$$\frac{\partial\psi}{\partial u_1} = 2u_1, \qquad \frac{\partial\psi}{\partial u_2} = 2u_2, \qquad \frac{\partial\psi}{\partial u_3} = 2u_3$$

In an orthogonal coordinate system

$$\nabla\psi = \frac{1}{h_1}\frac{\partial\psi}{\partial u_1}\hat{e}_1 + \frac{1}{h_2}\frac{\partial\psi}{\partial u_2}\hat{e}_2 + \frac{1}{h_3}\frac{\partial\psi}{\partial u_3}\hat{e}_3$$

$$= e^{-u_2}(2u_1)\hat{e}_1 + e^{-u_2}(2u_2)\hat{e}_2 + (2u_3)\hat{e}_3$$

$$= 2e^{-u_2}(u_1\hat{e}_1 + u_2\hat{e}_2) + (2u_3)\hat{e}_3$$

$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

$h_1 = \left|\frac{\partial\vec{r}}{\partial u_1}\right| = e^{u_2}$

$h_2 = \left|\frac{\partial\vec{r}}{\partial u_2}\right| = e^{u_2}$

$h_3 = \left|\frac{\partial\vec{r}}{\partial u_3}\right| = 1$

and

$$\nabla^2\psi = \frac{1}{h_1h_2h_3}\left[\frac{\partial}{\partial u_1}\left(\frac{h_2h_3}{h_1}\frac{\partial\psi}{\partial u_1}\right)+\frac{\partial}{\partial u_2}\left(\frac{h_3h_1}{h_2}\frac{\partial\psi}{\partial u_2}\right)+\frac{\partial}{\partial u_3}\left(\frac{h_1h_2}{h_3}\frac{\partial\psi}{\partial u_3}\right)\right]$$

$$= \frac{1}{(e^{u_2})(e^{u_2})(1)}\left[\frac{\partial}{\partial u_1}\left(\frac{e^{u_2}}{e^{u_2}}(2u_1)\right)+\frac{\partial}{\partial u_2}\left(\frac{e^{u_2}}{e^{u_2}}(2u_2)\right)+\frac{\partial}{\partial u_3}\left(e^{u_2}e^{u_2}(2u_3)\right)\right]$$

$$= e^{-2u_2}\left[\frac{\partial}{\partial u_1}(2u_1)+\frac{\partial}{\partial u_2}(2u_2)+\frac{\partial}{\partial u_3}\left(e^{2u_2}(2u_3)\right)\right]$$

$$= e^{-2u_2}\left[2+2+2e^{2u_2}\right] = e^{-2u_2}\left[4+2e^{2u_2}\right]$$

Example. Find $\nabla.\vec{A}$ and $\nabla\times\vec{A}$ in the system defined in previous example for the vector field

$\vec{A} = u_3\hat{e}_1 - u_2\hat{e}_3$.

$$\nabla.\vec{A} = \frac{1}{h_1h_2h_3}\left[\frac{\partial}{\partial u_1}(h_2h_3A_1)+\frac{\partial}{\partial u_2}(h_1h_3A_2)+\frac{\partial}{\partial u_3}(h_1h_2A_3)\right]$$

$$= \frac{1}{(e^{u_2})(e^{u_2})(1)}\left[\frac{\partial}{\partial u_1}(u_3(1)e^{u_2})+\frac{\partial}{\partial u_2}((0)(1)e^{u_2})+\frac{\partial}{\partial u_3}(-u_2(e^{u_2})(e^{u_2}))\right]$$

$$= \frac{1}{e^{2u_2}}\left[\frac{\partial}{\partial u_1}(u_3e^{u_2})+\frac{\partial}{\partial u_2}(0)-\frac{\partial}{\partial u_3}(e^{3u_2})\right] = \frac{1}{e^{2u_2}}\left[\frac{\partial}{\partial u_1}(0)+\frac{\partial}{\partial u_2}(0)-\frac{\partial}{\partial u_3}(0)\right]$$

$$\nabla.\vec{A} = \frac{1}{e^{2u_2}}\left[(0)+(0)-(0)\right]$$

$$\nabla.\vec{A} = 0$$

and

$$\nabla\times\vec{A} = \frac{1}{h_1h_2h_3}\begin{vmatrix} h_1\hat{e}_1 & h_2\hat{e}_2 & h_3\hat{e}_3 \\ \frac{\partial}{\partial u_1} & \frac{\partial}{\partial u_2} & \frac{\partial}{\partial u_3} \\ A_1h_1 & A_2h_2 & A_3h_3 \end{vmatrix}$$

$$= \frac{1}{(e^{u_2})(e^{u_2})(1)}\begin{vmatrix} e^{u_2}\hat{e}_1 & e^{u_2}\hat{e}_2 & \hat{e}_3 \\ \frac{\partial}{\partial u_1} & \frac{\partial}{\partial u_2} & \frac{\partial}{\partial u_3} \\ u_3e^{u_2} & 0 & -e^{u_2} \end{vmatrix}$$

$$= \frac{1}{e^{2u_2}}\left[e^{u_2}\hat{e}_1\frac{\partial}{\partial u_2}(-e^{u_2}-0)+e^{u_2}\hat{e}_2\frac{\partial}{\partial u_3}(u_3e^{u_2}-0)+\hat{e}_3\left(0-\frac{\partial}{\partial u_2}(u_3e^{u_2})\right)\right]$$

$$= \frac{1}{e^{2u_2}}\left[e^{u_2}\hat{e}_1(-e^{u_2})+e^{u_2}\hat{e}_2(e^{u_2})-(u_3\hat{e}_3)\right]$$

$$\nabla\times\vec{A} = \frac{1}{e^{u_2}}\left[-e^{u_2}\hat{e}_1+e^{u_2}\hat{e}_2-u_3\hat{e}_3\right]$$

Example: Find the expressions of gradient, divergence and curl in parabolic cylindrical coordinates defined by

$x = \frac{1}{2}(u^2-v^2)$, $y = uv$, $z = z$

Solution: First we find scale factors

here $u_1 = u$, $u_2 = v$, $u_3 = z$

$\vec{r} = \frac{1}{2}(u^2-v^2)\hat{i} + uv\hat{j} + z\hat{k}$

$\Rightarrow \frac{\partial\vec{r}}{\partial u_1} = u\hat{i}+v\hat{j}$, $\frac{\partial\vec{r}}{\partial u_2} = -v\hat{i}+u\hat{j}$, $\frac{\partial\vec{r}}{\partial u_3} = \hat{k}$

$\Rightarrow h_1 = \sqrt{u^2+v^2}$, $h_2 = \sqrt{u^2+v^2}$, $h_3 = 1$

$$\nabla\psi = \frac{1}{h_1}\frac{\partial\psi}{\partial u_1}\hat{e}_1+\frac{1}{h_2}\frac{\partial\psi}{\partial u_2}\hat{e}_2+\frac{1}{h_3}\frac{\partial\psi}{\partial u_3}\hat{e}_3$$

$$\nabla\psi = \frac{1}{\sqrt{u^2+v^2}}\frac{\partial\psi}{\partial u}\hat{e}_1+\frac{1}{\sqrt{u^2+v^2}}\frac{\partial\psi}{\partial v}\hat{e}_2+\frac{\partial\psi}{\partial u_3}\hat{e}_3$$

$$\nabla.\vec{A} = \frac{1}{h_1h_2h_3}\left[\frac{\partial}{\partial u_1}(h_2h_3A_1)+\frac{\partial}{\partial u_2}(h_3h_1A_2)+\frac{\partial}{\partial u_3}(h_1h_2A_3)\right]$$

$$= \frac{1}{(u^2+v^2)}\left[\frac{\partial}{\partial u_1}\left(A_1\sqrt{u^2+v^2}\right)+\frac{\partial}{\partial u_2}\left(\sqrt{u^2+v^2}\,A_2\right)+\frac{\partial}{\partial u_3}\left(A_3(u^2+v^2)\right)\right]$$

$$\nabla\times\vec{A} = \frac{1}{h_1h_2h_3}\begin{vmatrix} h_1\hat{e}_1 & h_2\hat{e}_2 & h_3\hat{e}_3 \\ \partial/\partial u_1 & \partial/\partial u_2 & \partial/\partial u_3 \\ h_1A_1 & h_2A_2 & h_3A_3 \end{vmatrix}$$

$$= \frac{1}{(u^2+v^2)}\begin{vmatrix} \sqrt{u^2+v^2}\,\hat{e}_1 & \sqrt{u^2+v^2}\,\hat{e}_2 & \hat{e}_3 \\ \partial/\partial u & \partial/\partial v & \partial/\partial z \\ \sqrt{u^2+v^2}\,A_1 & \sqrt{u^2+v^2}\,A_2 & A_3 \end{vmatrix}$$

Example: Using the expressions of the last example evaluate the gradient of scalar field

$\psi = u^2+v^2+z^2$

$$\nabla\psi = \frac{1}{\sqrt{u^2+v^2}}\frac{\partial\psi}{\partial u}\hat{e}_1+\frac{1}{u^2+v^2}\frac{\partial\psi}{\partial v}\hat{e}_2+\frac{\partial\psi}{\partial z}\hat{e}_3$$

$$\nabla\psi = \frac{2u}{\sqrt{u^2+v^2}}\hat{e}_1+\frac{2v}{\sqrt{u^2+v^2}}\hat{e}_2+2z\hat{e}_3$$

Example. Find $\nabla.\vec{A}$ and $\nabla\times\vec{A}$ in parabolic cylindrical coordinates system for the vector field

$\vec{A} = u_3\hat{e}_1 - u_2\hat{e}_3$.

Solution: $\vec{A} = z\hat{e}_1 - v\hat{e}_3$

$$\nabla.\vec{A} = \frac{1}{u^2+v^2}\left[\frac{\partial}{\partial u}\left(\sqrt{u^2+v^2}\,z\right)+0+\frac{\partial}{\partial z}\left(-(u^2+v^2)v\right)\right]$$

$$= \frac{1}{u^2+v^2}z\left(\frac{1}{2}\right)\frac{1}{\sqrt{u^2+v^2}}(2u)$$

$$\nabla.\vec{A} = \frac{zu}{(u^2+v^2)^{3/2}}$$

$$\nabla\times\vec{A} = \frac{1}{(u^2+v^2)}\begin{vmatrix} \sqrt{u^2+v^2}\,\hat{e}_1 & \sqrt{u^2+v^2}\,\hat{e}_2 & \hat{e}_3 \\ \partial/\partial u & \partial/\partial v & \partial/\partial z \\ \sqrt{u^2+v^2}\,z & 0 & -v \end{vmatrix}$$

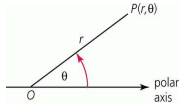
$$= \frac{1}{u^2+v^2}\left[\sqrt{u^2+v^2}\,\hat{e}_1(-1)-\sqrt{u^2+v^2}\,\hat{e}_2\left(-\sqrt{u^2+v^2}\right)+\hat{e}_3\left(\frac{-zv}{\sqrt{u^2+v^2}}\right)\right]$$

$$= \frac{-1}{\sqrt{u^2+v^2}}\hat{e}_1+\hat{e}_2-\frac{zv}{(u^2+v^2)^{3/2}}\hat{e}_3$$

Polar Coordinate System

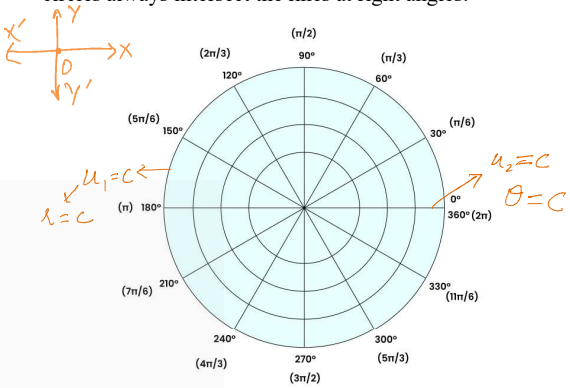
Polar coordinate system

The **polar coordinate** system is a two-dimensional coordinate system in which each point on a plane is determined by a distance "r" from the pole or reference point and an angle "θ".



- The reference point is called **pole.**
- The distance r from the pole is called the **radial coordinate** and the angle is called the **angular coordinate.**

The coordinate lines for r are circles, those for theta are lines. The circles always intersect the lines at right angles.



Polar coordinates to rectangular coordinates

Following are the conversion formulas from polar to rectangular coordinates.

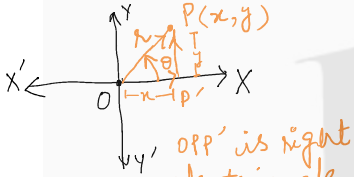
$x = r\cos\theta$

$y = r\sin\theta$

and

$r^2 = x^2 + y^2 \Rightarrow r = \sqrt{x^2 + y^2}$

$\text{Tan}\theta = y/x \Rightarrow \theta = \text{Tan}^{-1} y/x$



opp' is right angle triangle.

$\Rightarrow x = r\cos\theta$

$y = r\sin\theta$

Example No 1:
Convert the polar coordinates $(3, \frac{\pi}{2})$ as rectangular coordinates.

$$x = r\cos\theta$$
$$= 3\cos\pi/2$$
$$= 3(0) = 0$$
$$\boxed{x = 0}$$

$$y = r\sin\theta$$
$$= 3\sin\pi/2$$
$$= 3(1) = 3$$
$$\boxed{y = 3}$$

The rectangular coordinates are $(0,3)$.

Example No 2:
Convert the rectangular coordinates $(3,3)$ to polar coordinate.

$$r^2 = x^2 + y^2$$
$$r^2 = (3)^2 + (3)^2 \quad \Rightarrow r^2 = 9 + 9$$
$$r = \sqrt{18} \quad \Rightarrow \boxed{r = 3\sqrt{2}}$$

$$\text{Tan}\,\theta = \frac{y}{x}$$
$$\text{Tan}\,\theta = \frac{3}{3} \quad \Rightarrow \text{Tan}\,\theta = 1$$
$$\theta = \text{Tan}^{-1}(1) \quad \Rightarrow \theta = \frac{\pi}{4}$$

Hence the polar coordinates are $(3\sqrt{22}, \frac{\pi}{4})$.

Example No 3

Convert the polar equation $4r\cos\theta + r\sin\theta = 8$ into a rectangular equation that expresses y in terms of x.
Solution: Using

$x = r\cos\theta$
$y = r\sin\theta$

$4r\cos\theta + r\sin\theta = 8$
$4(x) + (y) = 8$
$4x + y = 8$
$y = 8 - 4x$

Unit Vectors in Polar Coordinates:

The position vector in plane polar coordinates is:

$$\vec{r} = x\hat{\imath} + y\hat{\jmath}$$

Using the transformation $x = r\cos\theta$, $y = r\sin\theta$

$\vec{r} = r\cos\theta\,\hat{\imath} + r\sin\theta\,\hat{\jmath}$

$\frac{\partial\vec{r}}{\partial r} = \cos\theta\,\hat{\imath} + \sin\theta\,\hat{\jmath}$

$\frac{\partial\vec{r}}{\partial\theta} = -r\sin\theta\,\hat{\imath} + r\cos\theta\,\hat{\jmath}$

$\hat{e}_1 = \frac{\frac{\partial\vec{r}}{\partial r}}{\left|\frac{\partial\vec{r}}{\partial r}\right|} = \cos\theta\,\hat{\imath} + \sin\theta\,\hat{\jmath}$

$\hat{e}_2 = \frac{\frac{\partial\vec{r}}{\partial\theta}}{\left|\frac{\partial\vec{r}}{\partial\theta}\right|} = \frac{-r\sin\theta\hat{\imath} + r\cos\theta\hat{\jmath}}{r} = -\sin\theta\,\hat{\imath} + \cos\theta\,\hat{\jmath}$

The scale factors are:

$h_1 = \left|\frac{\partial\vec{r}}{\partial r}\right| = 1$ and $h_2 = \left|\frac{\partial\vec{r}}{\partial\theta}\right| = r$

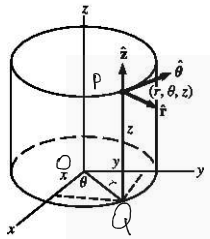
$ds^2 = dr^2 + r^2 d\theta^2$

Cylindrical Polar Coordinates

Cylindrical Polar Coordinates

Cylindrical coordinates simply combine the polar coordinates in the xy-plane with the usual z coordinate of Cartesian coordinate system. Hence cylindrical polar coordinates are represented by (r, θ, z).

Let $P(x, y, z)$ be any point whose projection on the $xy-plane$ is $Q(x, y)$. Then its cylindrical coordinates are (r, θ, z).



Example. Convert the cylindrical coordinates $(4, \pi, 1)$ in rectangular coordinates.

Solution: Using the transformation:

$x = r\cos\theta = 4\cos\pi = -4$

$y = r\sin\theta = 4\sin\pi = 0$

$z = z = 1$

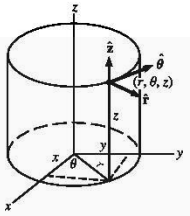
$\Rightarrow (4, \pi, 1) \longrightarrow (-4, 0, 1)$

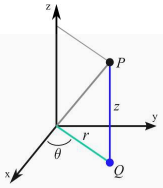
Transformation Equations from Cylindrical coordinates to Cartesian coordinates:

From this figure the transformation equations expressing the rectangular coordinates in terms of cylindrical coordinate:

$$x = r\cos\theta$$
$$y = r\sin\theta$$
$$z = z$$

Where $r \geq 0, 0 \leq \theta \leq 2\pi$, and $-\infty < z < \infty$.





Example: Convert $(1, 1, -1)$ in cylindrical coordinates.

using the inverse transformation:

$r = \sqrt{x^2+y^2} = \sqrt{1+1} = \sqrt{2}$

$\theta = \tan^{-1}\frac{y}{x} = \tan^{-1}(1) = \pi/4$

$z = -1$

$(1, 1, -1) \rightarrow (\sqrt{2}, \pi/4, -1)$

Transformation Equations from Cartesian coordinates to Cylindrical coordinates:

squaring and adding,

$x = r\cos\theta$

$y = r\sin\theta$

we have

$x^2 + y^2 = r^2(\cos^2\theta + \sin^2\theta) = r^2$

or $r = \sqrt{x^2+y^2}$

Moreover:

$\frac{y}{x} = \frac{r\cos\theta}{r\sin\theta} = \tan\theta$

or $\theta = \tan^{-1}\frac{y}{x}$

and $z = z$

Tangent vectors in cylindrical coordinate system:

The position vector of any point P in cylindrical polar coordinates is

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$
$$\vec{r} = r\cos\theta\hat{i} + r\sin\theta\,\hat{j} + z\hat{k}$$

The tangent vectors in the direction of r, θ *and* z respectively, are given by

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$
$$\frac{\partial \vec{r}}{\partial r} = \cos\theta i + \sin\theta j$$
$$\frac{\partial \vec{r}}{\partial \theta} = -r\sin\theta i + r\cos\theta j$$
$$\frac{\partial \vec{r}}{\partial z} = \hat{k}$$

Unit tangent vectors in cylindrical coordinate system:

The unit vectors in these direction of r, θ *and* z *are* given by

$$\hat{e}_r = \frac{\frac{\partial \vec{r}}{\partial r}}{\left|\frac{\partial \vec{r}}{\partial r}\right|} = \frac{\cos\theta\hat{i} + \sin\theta\, j}{\sqrt{\cos^2\theta + \sin^2\theta}} = \cos\theta\hat{i} + \sin\theta\, j$$

$$\hat{e}_\theta = \frac{\frac{\partial \vec{r}}{\partial \theta}}{\left|\frac{\partial \vec{r}}{\partial \theta}\right|} = \frac{-r\sin\theta\hat{i} + r\cos\theta\, j}{\sqrt{r^2\sin^2\theta + r^2\cos^2\theta}} = -\sin\theta\hat{i} + \cos\theta\, j$$

$$\hat{e}_z = \frac{\frac{\partial \vec{r}}{\partial z}}{\left|\frac{\partial \vec{r}}{\partial z}\right|} = k$$

In matrix notaion, we have

$$\begin{bmatrix} \hat{e}_r \\ \hat{e}_\theta \\ \hat{e}_z \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \hat{i} \\ j \\ k \end{bmatrix}$$

Scale factors:

The scale factor for the cylindrical coordinates are given by

$$h_1 = h_r = \left|\frac{\partial \vec{r}}{\partial r}\right| = \sqrt{\cos^2\theta + \sin^2\theta} = 1$$

$$h_2 = h_\theta = \left|\frac{\partial \vec{r}}{\partial \theta}\right| = \sqrt{(-r\sin\theta)^2 + (r\cos\theta)^2}$$
$$= r\sqrt{\cos^2\theta + \sin^2\theta} = r$$

$$h_3 = h_z = \left|\frac{\partial \vec{r}}{\partial z}\right| = |k| = 1$$

hence the scale factor are:

$\boxed{h_1 = h_r = 1}$ $\boxed{h_2 = h_\theta = r}$ $\boxed{h_3 = h_z = 1}$

Question: In cylindrical polar coordinates, show that.

$$\frac{\partial \hat{e}_r}{\partial r} = 0, \frac{\partial \hat{e}_r}{\partial \theta} = \hat{e}_\theta, \frac{\partial \hat{e}_r}{\partial z} = 0$$
$$\frac{\partial \hat{e}_\theta}{\partial r} = 0, \frac{\partial \hat{e}_\theta}{\partial \theta} = -\hat{e}_r, \frac{\partial \hat{e}_\theta}{\partial z} = 0$$
$$\frac{\partial \hat{e}_z}{\partial r} = 0, \frac{\partial \hat{e}_z}{\partial \theta} = 0, \frac{\partial \hat{e}_z}{\partial z} = 0$$

Solution: In cylindrical coordinates

$\hat{e}_r = \cos\theta\hat{i} + \sin\theta\,\hat{j}$

$\Rightarrow \frac{\partial \hat{e}_r}{\partial r} = 0, \frac{\partial \hat{e}_r}{\partial \theta} = -\sin\theta\hat{i} + \cos\theta\hat{j} = \hat{e}_\theta$

and $\frac{\partial \hat{e}_r}{\partial z} = 0$

Question: Prove that in cylindrical polar coordinates

$$\frac{d}{dt}\hat{e}_r = \dot{\theta}\hat{e}_\theta,\quad \frac{d}{dt}\hat{e}_\theta = -\dot{\theta}\hat{e}_r,\quad \frac{d}{dt}\hat{e}_z = \bar{0}$$

Where dot denote differentiation w.r.t time t.

$$\hat{e}_r = \cos\theta\hat{i} + \sin\theta j,\ \hat{e}_\theta = -\sin\theta\hat{i} + \cos\theta j,\ \hat{e}_z = k$$

$$\frac{d}{dt}\hat{e}_r = -\sin\theta\frac{d\theta}{dt}\hat{i} + \cos\theta\frac{d\theta}{dt}j = (-\sin\theta\hat{i} + \cos\theta j)\frac{d\theta}{dt} = \dot{\theta}\hat{e}_\theta,$$

$$\frac{d}{dt}\hat{e}_\theta = -\cos\theta\frac{d\theta}{dt}\hat{i} - \sin\theta\frac{d\theta}{dt}j = -(\cos\theta\hat{i} + \sin\theta j)\frac{d\theta}{dt} = -\dot{\theta}\hat{e}_r$$

$$\frac{d}{dt}\hat{e}_z = \frac{d}{dt}k = \bar{0}$$

$$\frac{d\hat{e}_r}{dt} = \frac{\partial\hat{e}_r}{\partial r}\frac{dr}{dt} + \frac{\partial\hat{e}_r}{\partial\theta}\frac{d\theta}{dt} + \frac{\partial\hat{e}_r}{\partial z}\frac{dz}{dt}$$

Expression for Jacobian in cylindrical polar coordinates:

We know that the Jacobian in orthogonal curvilinear coordinates

u_1, u_2, u_3 is given by

$$J = J\left(\frac{x,y,z}{u_1,u_2,u_3}\right) = h_1h_2h_3$$

$$h_1 = 1,\ h_2 = r,\ h_3 = 1 \quad and \quad u_1 = r,\ u_2 = \theta,\ u_3 = z$$

$$J\left(\frac{x,y,z}{r,\theta,z}\right) = (1)(r)(1) = r$$

Cylindrical Coordinate System

Arc length element:

In cylindrical polar coordinates, the elements of arc length are determined from

$$u_1 = r,\ u_2 = \theta,\ u_3 = z,\ h_1 = 1,\ h_2 = r,\ h_3 = 1$$

$$(ds)^2 = h_1^2(du_1)^2 + h_2^2(du_2)^2 + h_3^2(du_3)^2$$

In cylindrical polar coorrdinates, these becomes

$$(ds)^2 = (1)^2(dr)^2 + (r)^2(d\theta)^2 + (1)^2(dz)^2$$

$$(ds)^2 = (dr)^2 + r^2(d\theta)^2 + (dz)^2$$

Example: Represent the vector

$$\vec{A} = z\hat{i} - 2x\hat{j} + y\hat{k}$$

in terms of cylindrical coordinates.

Solution: Consider the transformation

$$x = r\cos\theta,\ y = r\sin\theta,\ z = z$$

and

$$\hat{e}_r = \cos\theta\hat{i} + \sin\theta\hat{j},\ \hat{e}_\theta = -\sin\theta\hat{i} + \cos\theta\hat{j},\ \hat{e}_z = \hat{k}$$

Implies:

$$\cos\theta\hat{e}_r - \sin\theta\hat{e}_\theta = \hat{i}$$

and

$$\sin\theta\hat{e}_r + \cos\theta\hat{e}_\theta = \hat{j}$$

Example: Find the expressions of $\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}$ in cylindrical coordinate system.

$$x = r\cos\theta,\quad y = r\sin\theta,\quad z = z \quad and$$

$$r = \sqrt{x^2 + y^2},\quad \theta = \tan^{-1}\left(\frac{y}{x}\right),\quad z = z$$

$$\frac{\partial r}{\partial x} = \frac{x}{\sqrt{x^2+y^2}} = \frac{r\cos\theta}{r} = \cos\theta$$

$$\frac{\partial r}{\partial y} = \frac{y}{\sqrt{x^2+y^2}} = \frac{r\sin\theta}{r} = \sin\theta$$

$$\frac{\partial r}{\partial z} = 0$$

$$\frac{\partial\theta}{\partial x} = -\frac{y}{x^2+y^2} = \frac{r\sin\theta}{r^2} = -\frac{\sin\theta}{r}$$

$$\frac{\partial\theta}{\partial y} = \frac{x}{x^2+y^2} = \frac{r\cos\theta}{r^2} = \frac{\cos\theta}{r}$$

$$\frac{\partial\theta}{\partial z} = 0$$

$$\frac{\partial z}{\partial x} = 0,\ \frac{\partial z}{\partial y} = 0,\ \frac{\partial z}{\partial z} = 1$$

Area elements

we know that the elements of area in an orthogonal curvilinear coordinates are:

$dA_1 = h_2h_3du_2du_3,\quad dA_2 = h_1h_3du_1du_3,\quad dA_3 = h_1h_2du_1du_2$

In cylindrical polar coordinates, we have

$dA_1 = (r)(1)d\theta dz = rd\theta dz$

$dA_2 = (1)(1)drdz = drdz$

$dA_3 = (1)(r)drd\theta = rdrd\theta$

Volume element

$$dv = h_1h_2h_3du_1du_2du_3$$

In cylindrical polar coordinate, we *have*

$$dv = (1)(r)(1)drd\theta dz = rdrd\theta dz$$

Replacing the values of x, y, z and $\hat{i}, \hat{j}, \hat{k}$ in terms of cylindrical coordinates we have

$$\vec{A} = z(\cos\theta\hat{e}_r - \sin\theta\hat{e}_\theta) - 2r\cos\theta(\sin\theta\hat{e}_r + \cos\theta\hat{e}_\theta) + r\sin\theta\hat{e}_z$$

$$\vec{A} = (z\cos\theta - 2r\sin\theta\cos\theta)\hat{e}_r - (z\sin\theta - 2r\cos^2\theta)\hat{e}_\theta + (r\sin\theta)\hat{e}_z$$

Now by the chain rule, we have

$$\frac{\partial}{\partial x} = \frac{\partial}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial}{\partial\theta}\frac{\partial\theta}{\partial x} + \frac{\partial}{\partial z}\frac{\partial z}{\partial x} = \cos\theta\frac{\partial}{\partial r} - \frac{\sin\theta}{r}\frac{\partial}{\partial\theta} + (0)\frac{\partial}{\partial z}$$

$$= \cos\theta\frac{\partial}{\partial r} - \frac{\sin\theta}{r}\frac{\partial}{\partial\theta}$$

$$\frac{\partial}{\partial y} = \frac{\partial}{\partial r}\frac{\partial r}{\partial y} + \frac{\partial}{\partial\theta}\frac{\partial\theta}{\partial y} + \frac{\partial}{\partial z}\frac{\partial z}{\partial y} = \sin\theta\frac{\partial}{\partial r} + \frac{\cos\theta}{r}\frac{\partial}{\partial\theta} + (0)\frac{\partial}{\partial z}$$

$$= \sin\theta\frac{\partial}{\partial r} + \frac{\cos\theta}{r}\frac{\partial}{\partial\theta}$$

$$\frac{\partial}{\partial z} = \frac{\partial}{\partial r}\frac{\partial r}{\partial z} + \frac{\partial}{\partial\theta}\frac{\partial\theta}{\partial z} + \frac{\partial}{\partial z}\frac{\partial z}{\partial z} = (0)\frac{\partial}{\partial r} + (0)\frac{\partial}{\partial\theta} + (1)\frac{\partial}{\partial z}$$

$$= \frac{\partial}{\partial z}$$

Expression for gradient, divergence, curl, and Laplacian in cylindrical polar coordinates:

Cylindrical polar coordinates are (r, θ, z).

$h_1 = 1, \quad h_2 = r, \quad h_3 = 1,$

Gradient:

In orthogonal curvilinear coordinates, we have

$$\nabla\psi = \frac{1}{h_1}\frac{\partial\psi}{\partial u_1}\hat{e}_1 + \frac{1}{h_2}\frac{\partial\psi}{\partial u_2}\hat{e}_2 + \frac{1}{h_3}\frac{\partial\psi}{\partial u_3}\hat{e}_3$$

In cylindrical polar coordinates, this becomes

$$\nabla\psi = \frac{1}{1}\frac{\partial\psi}{\partial r}\hat{e}_r + \frac{1}{r}\frac{\partial\psi}{\partial\theta}\hat{e}_\theta + \frac{1}{1}\frac{\partial\psi}{\partial z}\hat{e}_z$$

$$\nabla\psi = \frac{\partial\psi}{\partial r}\hat{e}_r + \frac{1}{r}\frac{\partial\psi}{\partial\theta}\hat{e}_\theta + \frac{\partial\psi}{\partial z}\hat{e}_z$$

Laplacian:

In orthogonal curvilinear coordinates, we have

$$\nabla^2\psi = \frac{1}{h_1h_2h_3}\left[\frac{\partial}{\partial u_1}\left(\frac{h_2h_3}{h_1}\frac{\partial\psi}{\partial u_1}\right) + \frac{\partial}{\partial u_2}\left(\frac{h_3h_1}{h_2}\frac{\partial\psi}{\partial u_2}\right) + \frac{\partial}{\partial u_3}\left(\frac{h_1h_2}{h_3}\frac{\partial\psi}{\partial u_3}\right)\right]$$

In cylindrical polar coordinates, it becomes

$$\nabla^2\psi = \frac{1}{(1)(r)(1)}\left[\frac{\partial}{\partial r}\left(\frac{(r)(1)h_3}{(1)}\frac{\partial\psi}{\partial r}\right) + \frac{\partial}{\partial\theta}\left(\frac{(1)(1)}{(r)}\frac{\partial\psi}{\partial\theta}\right) + \frac{\partial}{\partial z}\left(\frac{(1)(r)}{(1)}\frac{\partial\psi}{\partial z}\right)\right]$$

$$\nabla^2\psi = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial\psi}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2\psi}{\partial\theta^2} + \frac{\partial^2\psi}{\partial z^2}$$

Example: Express heat equation in cylindrical coordinates.

$$\frac{\partial u}{\partial t} = \alpha\nabla^2 u$$

Solution:

Replacing ∇^2 in cylindrical coordinates we have:

$$\frac{\partial u}{\partial t} = \alpha\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 u}{\partial\theta^2} + \frac{\partial^2 u}{\partial z^2}\right]$$

Divergence:

In orthogonal curvilinear coordinates, we have

$$\nabla.\vec{A} = \frac{1}{h_1h_2h_3}\left[\frac{\partial}{\partial u_1}(A_1h_2h_3) + \frac{\partial}{\partial u_2}(A_2h_3h_1) + \frac{\partial}{\partial u_3}(A_3h_1h_2)\right]$$

In cylindrical polar coordinates, we have

$$\nabla.\vec{A} = \frac{1}{(1)(r)(1)}\left[\frac{\partial}{\partial r}((r)(1)A_r) + \frac{\partial}{\partial\theta}((1)(1)A_\theta) + \frac{\partial}{\partial z}((1)(r)A_z)\right]$$

$$\nabla.\vec{A} = \frac{1}{r}\left[\frac{\partial}{\partial r}(rA_r) + \frac{\partial A_\theta}{\partial\theta} + \frac{\partial}{\partial z}(rA_z)\right]$$

Example: Express the velocity $\vec{v}$ and acceleration $\vec{a}$ of a particle in cylindrical coordinates.

Solution:

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$\Rightarrow \vec{r} = r\cos\theta(\cos\theta\hat{e}_r - \sin\theta\hat{e}_\theta) + r\sin\theta(\sin\theta\hat{e}_r + \cos\theta\hat{e}_\theta) + z\hat{e}_z$$

$$= (r\cos^2\theta + r\sin^2\theta)\hat{e}_r + (r\cos\theta\sin\theta - r\cos\theta\sin\theta)\hat{e}_\theta + z\hat{e}_z$$

$$\Rightarrow \vec{r} = r\hat{e}_r + z\hat{e}_z$$

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{dr}{dt}\hat{e}_r + r\frac{d}{dt}(\hat{e}_r) + \frac{dz}{dt}\hat{e}_z + z\frac{d\hat{e}_z}{dt}$$

$$\vec{v} = \dot{r}\hat{e}_r + r(\dot{\theta}\hat{e}_\theta) + \dot{z}\hat{e}_z$$

Curl:

In orthogonal curvilinear coordinates, we have

$$\nabla\times\vec{A} = \frac{1}{h_1h_2h_3}\begin{vmatrix} h_1\hat{e}_1 & h_2\hat{e}_2 & h_3\hat{e}_3 \\ \frac{\partial}{\partial u_1} & \frac{\partial}{\partial u_2} & \frac{\partial}{\partial u_3} \\ A_1h_1 & A_2h_2 & A_3h_3 \end{vmatrix}$$

In cylindrical polar coordinates, we have

$$\nabla\times\vec{A} = \frac{1}{r}\begin{vmatrix} \hat{e}_r & r\hat{e}_\theta & \hat{e}_z \\ \frac{\partial}{\partial u_1} & \frac{\partial}{\partial u_2} & \frac{\partial}{\partial u_3} \\ A_r & rA_\theta & A_z \end{vmatrix}$$

$$\nabla\times\vec{A} = \frac{1}{r}\left[\left(\frac{\partial A_z}{\partial\theta} - \frac{\partial}{\partial z}(rA_\theta)\right)\hat{e}_r + \left(r\frac{\partial A_r}{\partial z} - r\frac{\partial A_z}{\partial r}\right)\hat{e}_\theta + \left(\frac{\partial}{\partial r}(rA_\theta) - \frac{\partial A_r}{\partial z}\right)\hat{e}_z\right]$$

Now consider:

$$\vec{v} = \dot{r}\hat{e}_r + r\dot{\theta}\hat{e}_\theta + \dot{z}\hat{e}_z$$

$$\Rightarrow \vec{a} = \frac{d\vec{v}}{dt} = \frac{d}{dt}(\dot{r}\hat{e}_r + r\dot{\theta}\hat{e}_\theta + \dot{z}\hat{e}_z)$$

$$\vec{a} = \ddot{r}\hat{e}_r + \dot{r}\frac{d}{dt}(\hat{e}_r) + \dot{r}\dot{\theta}\hat{e}_\theta + r\ddot{\theta}\hat{e}_\theta + r\dot{\theta}\frac{d}{dt}(\hat{e}_\theta) + \ddot{z}\hat{e}_z$$

$$= \ddot{r}\hat{e}_r + \dot{r}\dot{\theta}\hat{e}_\theta + \dot{r}\dot{\theta}\hat{e}_\theta + r\ddot{\theta}\hat{e}_\theta + r\dot{\theta}(-\dot{\theta}\hat{e}_r) + \ddot{z}\hat{e}_z$$

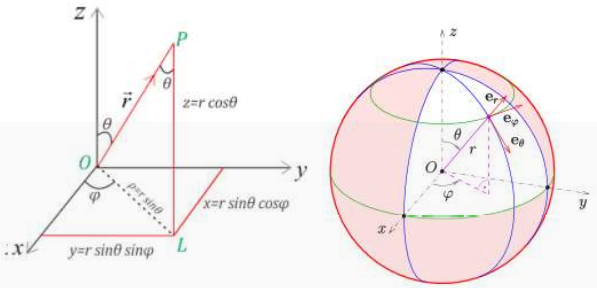
$$\vec{a} = (\ddot{r} - r\dot{\theta}^2)\hat{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{e}_\theta + \ddot{z}\hat{e}_z$$

Spherical Coordinate System

Spherical Polar Coordinates

Let $P(x,y,z)$ be any point whose projection on the $xy-plane$ is $L(x,y,0)$.

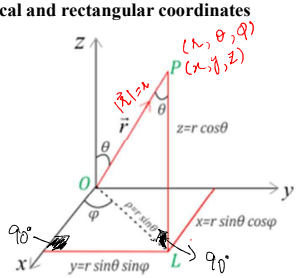
Then the spherical polar coordinates of P are (r,θ,ϕ) in which $\vec{r}=\overrightarrow{OP}$.



Transformation equations between spherical and rectangular coordinates

The transformation equations expressing the rectangular coordinates in terms of spherical polar coordinates are

$$x = r\sin\theta\cos\phi$$
$$y = r\sin\theta\sin\phi$$
$$z = r\cos\theta$$



squaring and adding above equations

$$x^2+y^2+z^2 = r^2\sin^2\theta\cos^2\phi + r^2\sin^2\theta\sin^2\phi + r^2\cos^2\theta$$
$$= r^2\sin^2\theta(\cos^2\phi+\sin^2\phi) + r^2\cos^2\theta$$
$$= r^2\sin^2\theta + r^2\cos^2\theta = r^2(\sin^2\theta+\cos^2\theta) = r^2$$

$$\Rightarrow \quad \boxed{r=\sqrt{x^2+y^2+z^2}}$$

squaring and adding first two equations, we get

$$x^2+y^2 = r^2\sin^2\theta\cos^2\phi + r^2\sin^2\theta\sin^2\phi$$
$$= r^2\sin^2\theta(\cos^2\phi+\sin^2\phi) = r^2\sin^2\theta$$

$$or \qquad r\sin\theta = \sqrt{x^2+y^2}$$

Divide this equation, by third equation, we *have*

$$\frac{r\sin\theta}{r\cos\theta} = \frac{\sqrt{x^2+y^2}}{z} \qquad or \qquad \tan\theta = \frac{\sqrt{x^2+y^2}}{z}$$

$$\Rightarrow \quad \boxed{\theta = \tan^{-1}\frac{\sqrt{x^2+y^2}}{z}}$$

Now Dividing first two equations

$$\frac{r\sin\theta\sin\phi}{r\sin\theta\cos\phi} = \frac{y}{x} \qquad or \qquad \tan\phi = \frac{y}{x}$$

$$\Rightarrow \quad \boxed{\phi = \tan^{-1}\frac{y}{x}}$$

Where $\quad r \geq 0,\ 0 \leq \theta \leq \pi,\ 0 \leq \phi \leq 2\pi$

Unit vectors :

The position vector of point $P(x,y,z)$ in spherical polar coordinates is given by

$$\vec{r} = x\hat{i} + y\,j + z\,k$$
$$\vec{r} = r\sin\theta\cos\phi\,\hat{i} + r\sin\theta\sin\phi\,j + r\cos\theta\,k$$

The tangent vectors in the direction of r, θ *and* ϕ respectively are given by

$$\vec{r} = x\hat{i} + y\,j + z\,k$$
$$\vec{r} = r\sin\theta\cos\phi\,\hat{i} + r\sin\theta\sin\phi\,j + r\cos\theta\,k$$
$$\frac{\partial\vec{r}}{\partial r} = \sin\theta\cos\phi\,\hat{i} + \sin\theta\sin\phi\,j + \cos\theta\,k$$
$$\frac{\partial\vec{r}}{\partial\theta} = r\cos\theta\cos\phi\,\hat{i} + r\cos\theta\sin\phi\,j - r\sin\theta\,k$$
$$\frac{\partial\vec{r}}{\partial\phi} = -r\sin\theta\sin\phi\,\hat{i} + r\sin\theta\cos\phi\,j$$

$$\hat{e}_r = \frac{\frac{\partial\vec{r}}{\partial r}}{\left|\frac{\partial\vec{r}}{\partial r}\right|} = \frac{\sin\theta\cos\phi\,\hat{i} + \sin\theta\sin\phi\,j + \cos\theta\,k}{\sqrt{\sin^2\theta\cos^2\phi + \sin^2\theta\sin^2\phi + \cos^2\theta}}$$

$$\hat{e}_r = \sin\theta\cos\phi\,\hat{i} + \sin\theta\sin\phi\,j + \cos\theta\,k$$

$$\hat{e}_\theta = \frac{\frac{\partial\vec{r}}{\partial\theta}}{\left|\frac{\partial\vec{r}}{\partial\theta}\right|} = \frac{r\cos\theta\cos\phi\,\hat{i} + r\cos\theta\sin\phi\,j - r\sin\theta\,k}{\sqrt{r^2\cos^2\theta\cos^2\phi + r^2\cos^2\theta\sin^2\phi + r^2\sin^2\theta}}$$

$$\hat{e}_\theta = \cos\theta\cos\phi\,\hat{i} + \cos\theta\sin\phi\,j - \sin\theta\,k$$

$$\hat{e}_\phi = \frac{\frac{\partial\vec{r}}{\partial\phi}}{\left|\frac{\partial\vec{r}}{\partial\phi}\right|} = \frac{-r\sin\theta\sin\phi\,\hat{i} + r\sin\theta\cos\phi\,j}{\sqrt{r^2\sin^2\theta\sin^2\phi + r^2\sin^2\theta\cos^2\phi}}$$

$$\hat{e}_\phi = -\sin\phi\,\hat{i} + \cos\phi\,j$$

In matrix notation

$$\begin{bmatrix}\hat{e}_r\\ \hat{e}_\theta\\ \hat{e}_\phi\end{bmatrix} = \begin{bmatrix}\sin\theta\cos\phi & \sin\theta\sin\phi & \cos\theta\\ \cos\theta\cos\phi & \cos\theta\sin\phi & -\sin\theta\\ -\sin\phi & \cos\phi & 0\end{bmatrix}\begin{bmatrix}\hat{i}\\ j\\ k\end{bmatrix}$$

Scale factors:

The scales factors for the spherical coordinate system are given by.

$$h_1 = h_r = \left|\frac{\partial\vec{r}}{\partial r}\right| = \sqrt{\sin^2\theta\cos^2\phi + \sin^2\theta\sin^2\phi + \cos^2\theta}$$
$$= \sqrt{\sin^2\theta(\cos^2\phi+\sin^2\phi) + \cos^2\theta}$$
$$= \sqrt{\sin^2\theta + \cos^2\theta} = 1$$
$$\boxed{h_1 = h_r = 1}$$

$$h_2 = h_\theta = \sqrt{r^2\cos^2\theta\cos^2\phi + r^2\cos^2\theta\sin^2\phi + r^2\sin^2\theta}$$
$$= \sqrt{r^2\cos^2\theta(\cos^2\phi+\sin^2\phi) + r^2\sin^2\theta}$$
$$= \sqrt{r^2\cos^2\theta + r^2\sin^2\theta} = r$$
$$\boxed{h_2 = h_\theta = r}$$

$$h_3 = h_\phi = \sqrt{r^2\sin^2\theta\sin^2\phi + r^2\sin^2\theta\cos^2\phi}$$
$$= \sqrt{r^2\sin^2\theta(\sin^2\phi+\cos^2\phi)}$$
$$= \sqrt{r^2\sin^2\theta} = r\sin\theta$$
$$\boxed{h_3 = h_\phi = r\sin\theta}$$

Arc length element:

In orthogonal curvilinear coordinates, the elements of arc length is det *er* min *ed by*

$$(ds)^2 = h_1^2(du_1)^2 + h_2^2(du_2)^2 + h_3^2(du_3)^2$$

In cylindrical polar coorrdinates, these becomes

$$(ds)^2 = (1)^2(dr)^2 + (r)^2(d\theta)^2 + (r\sin\theta)^2(d\phi)^2$$
$$= (dr)^2 + r^2(d\theta)^2 + r^2\sin^2\theta(d\phi)^2$$

Area element

we know that the elements of area in an orthogonal curvilinear coordinates are :

$$dA_1 = h_2h_3du_2du_3, \qquad dA_2 = h_1h_3du_1du_3 \qquad dA_3 = h_1h_2du_1du_2$$

In cylindrical polar coordinates, these becomes

$$dA_1 = (r)(r\sin\theta)d\theta d\phi = r^2\sin\theta d\theta d\phi$$
$$dA_2 = (1)(r\sin\theta)drd\phi = r\sin\theta drd\phi$$
$$dA_3 = (1)(r)drd\theta = rdrd\theta$$

Volume element

$$dv = h_1h_2h_3du_1du_2du_3$$

In spherical polar coordinate, this becomes

$$dv = (1)(r)(r\sin\theta)drd\theta d\phi = r^2\sin\theta drd\theta dz$$

Expression for Jacobian in spherical polar coordinates:

We know that the Jacobian in orthogonal curvilinear coordinates u_1, u_2, u_3 is given by

$$J\left(\frac{x,y,z}{u_1,u_2,u_3}\right) = h_1h_2h_3$$

In spherical polar coordinates $\quad h_1 = 1,\ h_2 = r,\ h_3 = r\sin\theta \quad$ *and* $\quad u_1 = r,\ u_2 = \theta,\ u_3 = \phi$

Therefore $\quad J\left(\frac{x,y,z}{u_1,u_2,u_3}\right) = (1)(r)(r\sin\theta) = r^2\sin\theta$

Spherical Coordinate System

Expression for gradient, divergence, curl and Laplcian for spherical polar coordinates:

we know that for spherical polar coordinates (r,θ,ϕ)

$u_1 = r, \quad u_2 = \theta, \quad u_3 = \phi; \quad \hat{e}_1 = \hat{e}_r, \quad \hat{e}_2 = \hat{e}_\theta, \quad \hat{e}_3 = \hat{e}_\phi$

$h_1 = h_r = 1, \quad h_2 = h_\theta = r, \quad h_3 = h_\phi = r\sin\theta$

$A_1 = A_r \; A_2 = A_\theta \quad A_3 = A_\phi$

Gradient:

In orthogonal curvilinear coordinates, we have

$$\nabla\psi = \frac{1}{h_1}\frac{\partial\psi}{\partial u_1}\hat{e}_1 + \frac{1}{h_2}\frac{\partial\psi}{\partial u_2}\hat{e}_2 + \frac{1}{h_3}\frac{\partial\psi}{\partial u_3}\hat{e}_3$$

In spherical polar coordinates it becomes

$$\nabla\psi = \frac{1}{1}\frac{\partial\psi}{\partial r}\hat{e}_r + \frac{1}{r}\frac{\partial\psi}{\partial\theta}\hat{e}_\theta + \frac{1}{r\sin\theta}\frac{\partial\psi}{\partial\phi}\hat{e}_\phi$$

$$\boxed{\nabla\psi = \frac{\partial\psi}{\partial r}\hat{e}_r + \frac{1}{r}\frac{\partial\psi}{\partial\theta}\hat{e}_\theta + \frac{1}{r\sin\theta}\frac{\partial\psi}{\partial\phi}\hat{e}_\phi}$$

Laplacian:

In orthogonal curvilinear coordinates, we have

$$\nabla^2\psi = \frac{1}{h_1h_2h_3}\left[\frac{\partial}{\partial u_1}\left(\frac{h_2h_3}{h_1}\frac{\partial\psi}{\partial u_1}\right) + \frac{\partial}{\partial u_2}\left(\frac{h_3h_1}{h_2}\frac{\partial\psi}{\partial u_2}\right) + \frac{\partial}{\partial u_3}\left(\frac{h_1h_2}{h_3}\frac{\partial\psi}{\partial u_3}\right)\right]$$

In spherical polar coordinates, this becomes

$$\nabla^2\psi = \frac{1}{(1)(r)(r\sin\theta)}\left[\frac{\partial}{\partial r}\left(\frac{(r)(r\sin\theta)}{(1)}\frac{\partial\psi}{\partial r}\right) + \frac{\partial}{\partial\theta}\left(\frac{(r\sin\theta)(1)}{r}\frac{\partial\psi}{\partial\theta}\right) + \frac{\partial}{\partial\phi}\left(\frac{(1)(r)}{(r\sin\theta)}\frac{\partial\psi}{\partial\phi}\right)\right]$$

$$\nabla^2\psi = \frac{1}{(r^2\sin\theta)}\left[(r\sin\theta)\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi}{\partial r}\right) + \frac{\partial}{\partial\theta}(\sin\theta\frac{\partial\psi}{\partial\theta}) + \frac{1}{(\sin\theta)}\frac{\partial^2\psi}{\partial\phi^2}\right]$$

$$\boxed{\nabla^2\psi = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta\frac{\partial\psi}{\partial\theta}) + \frac{1}{r^2\sin\theta}\frac{\partial^2\psi}{\partial\phi^2}}$$

Divergence:

In orthogonal curvilinear coordinates, we have

$$\nabla.\vec{A} = \frac{1}{h_1h_2h_3}\left[\frac{\partial}{\partial u_1}(A_1h_2h_3) + \frac{\partial}{\partial u_2}(A_2h_3h_1) + \frac{\partial}{\partial u_3}(A_3h_1h_2)\right]$$

In spherical polar coordinates this becomes

$$\nabla.\vec{A} = \frac{1}{(1)(r)(r\sin\theta)}\left[\frac{\partial}{\partial r}(r)(r\sin\theta)A_r + \frac{\partial}{\partial\theta}((r\sin\theta)(1)A_\theta) + \frac{\partial}{\partial\phi}(1)(r)(A_\phi)\right]$$

$$\nabla.\vec{A} = \frac{1}{(r^2\sin\theta)}\left[\frac{\partial}{\partial r}(r^2\sin\theta)A_r + \frac{\partial}{\partial\theta}(r\sin\theta A_\theta) + \frac{\partial}{\partial\phi}(rA_\phi)\right]$$

$$\boxed{\nabla.\vec{A} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2A_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(r\sin\theta A_\theta) + \frac{1}{r\sin\theta}\frac{\partial}{\partial\phi}(rA_\phi)}$$

Curl:

In orthogonal curvilinear coordinates, we have

$$\nabla\times\vec{A} = \frac{1}{h_1h_2h_3}\begin{vmatrix} h_1\hat{e}_1 & h_2\hat{e}_2 & h_3\hat{e}_3 \\ \frac{\partial}{\partial u_1} & \frac{\partial}{\partial u_2} & \frac{\partial}{\partial u_3} \\ A_1h_1 & A_2h_2 & A_3h_3 \end{vmatrix}$$

In spherical polar coordinates, this becomes

$$\nabla\times\vec{A} = \frac{1}{(1)(r)(r\sin\theta)}\begin{vmatrix} \hat{e}_r & r\hat{e}_\theta & r\sin\theta\hat{e}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial\theta} & \frac{\partial}{\partial\phi} \\ A_r & rA_\theta & r\sin\theta A_\phi \end{vmatrix}$$

$$\boxed{\nabla\times\vec{A} = \frac{1}{(r^2\sin\theta)}\begin{bmatrix} \hat{e}_r\{\frac{\partial}{\partial\theta}(r\sin\theta A_\phi) - \frac{\partial}{\partial\phi}(rA_\theta)\} - r\hat{e}_\theta\{\frac{\partial}{\partial r}(r\sin\theta A_\phi) - \frac{\partial}{\partial\phi}(A_r)\} \\ +\{\frac{\partial}{\partial r}(rA_\theta) - \frac{\partial A_r}{\partial\phi}\}r\sin\theta\hat{e}_\phi \end{bmatrix}}$$

Transformation equations from cylindrical to spherical system:

If point P has cylindrical polar coordinates $(\rho, \boldsymbol{\varphi}, \boldsymbol{z})$ then

$\rho = \sqrt{x^2+y^2}$ $\varphi = \tan^{-1}\frac{y}{x}$

$z = z$

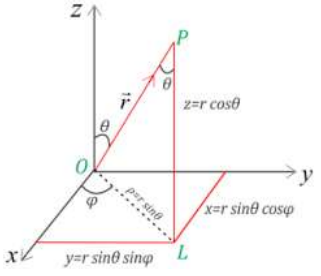
If spherical polar coordinates of P are (r,θ,φ) then

$r = \sqrt{x^2+y^2+z^2}$ $\theta = \tan^{-1}\frac{\sqrt{x^2+y^2}}{z}$

$\varphi = \tan^{-1}\frac{y}{x}$

Then from the figure and above equations we have

$r = \sqrt{\rho^2+z^2}$ $\theta = \tan^{-1}\frac{\rho}{z}$ $\varphi = \varphi$



Example: Prove that the spherical coordinate system is orthogonal.

Solution: Consider the basis vectors:

$\hat{e}_r = \sin\theta\cos\phi\hat{i} + \sin\theta\sin\phi\hat{j} + \cos\theta\hat{k}$

$\hat{e}_\theta = \cos\theta\cos\phi\hat{i} + \cos\theta\sin\phi\hat{j} - \sin\theta\hat{k}$

$\hat{e}_\phi = -\sin\phi\hat{i} + \cos\phi\hat{j}$

$\hat{e}_r.\hat{e}_\theta = \sin\theta\cos\theta\cos^2\phi + \sin\theta\cos\theta\sin^2\phi - \sin\theta\cos\theta$

$= \sin\theta\cos\theta(\cos^2\phi + \sin^2\phi) - \sin\theta\cos\theta$

$= \sin\theta\cos\theta - \sin\theta\cos\theta$

$\hat{e}_r.\hat{e}_\theta = 0 = \hat{e}_\theta.\hat{e}_r$

Now,

$\hat{e}_\theta.\hat{e}_\phi = -\sin\phi\cos\phi\cos\theta + \sin\phi\cos\phi\cos\theta$

$\Rightarrow \hat{e}_\theta.\hat{e}_\phi = 0 = \hat{e}_\phi.\hat{e}_\theta$

$\hat{e}_\phi.\hat{e}_r = -\sin\theta\sin\phi\cos\phi + \sin\theta\sin\phi\cos\phi$

$\Rightarrow \hat{e}_\phi.\hat{e}_r = 0 = \hat{e}_r.\hat{e}_\phi$

Implies that the spherical polar coordinate system is orthogonal.

Example: Express the unit vectors $\hat{i},\hat{j},\hat{k}$ in terms of $\hat{e}_r, \hat{e}_\theta, \hat{e}_\phi$.

Solution:

We have the expressions of $\hat{e}_r, \hat{e}_\theta, \hat{e}_\phi$ in terms of $\hat{i},\hat{j},\hat{k}$. We consider those equations and solve them simultaneously to evaluate $\hat{i},\hat{j},\hat{k}$ in terms of $\hat{e}_r, \hat{e}_\theta, \hat{e}_\phi$.

We have:

$$\hat{e}_r = \sin\theta\cos\phi\,\hat{i} + \sin\theta\sin\phi\,\hat{j} + \cos\theta\,\hat{k}$$

$$\hat{e}_\theta = \cos\theta\cos\phi\,\hat{i} + \cos\theta\sin\phi\,\hat{j} - \sin\theta\,\hat{k}$$

$$\hat{e}_\phi = -\sin\phi\,\hat{i} + \cos\phi\,\hat{j}$$

$$\sin\theta\,\hat{e}_r + \cos\theta\,\hat{e}_\theta = \sin^2\theta\,(\cos\phi\,\hat{i} + \sin\phi\,\hat{j}) + \cos^2\theta\,(\cos\phi\,\hat{i} + \sin\phi\,\hat{j})$$

$$\sin\theta\,\hat{e}_r + \cos\theta\,\hat{e}_\theta = \cos\phi\,\hat{i} + \sin\phi\,\hat{j}$$

also $\hat{e}_\phi = -\sin\phi\,\hat{i} + \cos\phi\,\hat{j}$

From last two equations, we have

$$\hat{j} = \sin\phi\sin\theta\,\hat{e}_r + \sin\phi\cos\theta\,\hat{e}_\theta + \cos\phi\,\hat{e}_\phi$$

and

$$\hat{i} = \cos\phi\sin\theta\,\hat{e}_r + \cos\phi\cos\theta\,\hat{e}_\theta - \sin\phi\,\hat{e}_\phi$$

and similarly, from the expressions of $\hat{e}_r$ and $\hat{e}_\theta$:

$$\hat{k} = \cos\theta\,\hat{e}_r - \sin\theta\,\hat{e}_\theta$$

Example: Find $\frac{d}{dt}(\hat{e}_r)$, $\frac{d}{dt}(\hat{e}_\theta)$ and $\frac{d}{dt}(\hat{e}_\phi)$.

Solution:

As $\hat{e}_r = \sin\theta\cos\phi\,\hat{i} + \sin\theta\sin\phi\,\hat{j} + \cos\theta\,\hat{k}$

$$\frac{d}{dt}(\hat{e}_r) = \frac{d}{dt}(\sin\theta\cos\phi)\,\hat{i} + \frac{d}{dt}(\sin\theta\sin\phi)\,\hat{j} + \frac{d}{dt}(\cos\theta)\,\hat{k}$$

$$= [\cos\theta\,\dot{\theta}\cos\phi - \sin\theta\sin\phi\,\dot{\phi}]\,\hat{i} + [\cos\theta\,\dot{\theta}\sin\phi + \sin\theta\cos\phi\,\dot{\phi}]\,\hat{j} + (-\sin\theta\,\dot{\theta})\,\hat{k}$$

Separating the terms of $\dot{\theta}$ and $\dot{\phi}$, we have:

$$\frac{d}{dt}(\hat{e}_r) = [\cos\theta\cos\phi\,\hat{i} + \cos\theta\sin\phi\,\hat{j} - \sin\theta\,\hat{k}]\,\dot{\theta} + \sin\theta[-\sin\phi\,\hat{i} + \cos\phi\,\hat{j}]\,\dot{\phi}$$

$$\Rightarrow \frac{d}{dt}(\hat{e}_r) = \hat{e}_\theta\,\dot{\theta} + \sin\theta\,\hat{e}_\phi\,\dot{\phi}$$

Similarly:

$$\frac{d}{dt}(\hat{e}_\theta) = -\hat{e}_r\,\dot{\theta} + \cos\theta\,\hat{e}_\phi\,\dot{\phi}$$

and

$$\frac{d}{dt}(\hat{e}_\phi) = -(\cos\phi\,\hat{i} + \sin\phi\,\hat{j})\,\dot{\phi}$$

here we replace the expressions of $\hat{i}$ and $\hat{j}$.

$$\frac{d}{dt}(\hat{e}_\phi) = -\dot{\phi}[\cos\phi\{\sin\theta\cos\phi\,\hat{e}_r + \cos\theta\cos\phi\,\hat{e}_\theta - \sin\phi\,\hat{e}_\phi\}] + \sin\phi\{\sin\theta\sin\phi\,\hat{e}_r + \cos\theta\sin\phi\,\hat{e}_\theta + \cos\phi\,\hat{e}_\phi\}]$$

$$= -\dot{\phi}[\sin\theta\cos^2\phi\,\hat{e}_r + \cos\theta\cos^2\phi\,\hat{e}_\theta - \sin\phi\cos\phi\,\hat{e}_\phi + \sin\theta\sin^2\phi\,\hat{e}_r + \cos\theta\sin^2\phi\,\hat{e}_\theta + \sin\phi\cos\phi\,\hat{e}_\phi]$$

$$= -(\sin\theta\,\hat{e}_r + \cos\theta\,\hat{e}_\theta)\,\dot{\phi}$$

Spherical Coordinate System

Example: Express the velocity and acceleration of a particle in spherical coordinates.

Solution: Consider:

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

where $x = r\sin\theta\cos\phi$

$y = r\sin\theta\sin\phi$

$z = r\cos\theta$

Now express $\hat{i}$, $\hat{j}$ and $\hat{k}$ in terms of $\hat{e}_r$, $\hat{e}_\theta$ and $\hat{e}_\phi$.

$$\hat{j} = \sin\phi\sin\theta\,\hat{e}_r + \sin\phi\cos\theta\,\hat{e}_\theta + \cos\phi\,\hat{e}_\phi$$

$$\hat{i} = \cos\phi\sin\theta\,\hat{e}_r + \cos\phi\cos\theta\,\hat{e}_\theta - \sin\phi\,\hat{e}_\phi$$

and

$$\hat{k} = \cos\theta\,\hat{e}_r - \sin\theta\,\hat{e}_\theta$$

Now:

$$\vec{r} = r\sin\theta\cos\phi\{\cos\phi\sin\theta\,\hat{e}_r + \cos\phi\cos\theta\,\hat{e}_\theta - \sin\phi\,\hat{e}_\phi\} + r\sin\theta\sin\phi\{\sin\phi\sin\theta\,\hat{e}_r + \sin\phi\cos\theta\,\hat{e}_\theta + \cos\phi\,\hat{e}_\phi\} + r\cos\theta\{\cos\theta\,\hat{e}_r - \sin\theta\,\hat{e}_\theta\}$$

after simplifications:

$\vec{r} = r\hat{e}_r$

and

$\vec{V} = \frac{d}{dt}(\vec{r}) = \frac{d}{dt}(r\hat{e}_r) = \dot{r}\hat{e}_r + r\frac{d}{dt}(\hat{e}_r)$

$= \dot{r}\hat{e}_r + r(\dot{\theta}\hat{e}_\theta + \sin\theta\hat{e}_\phi\dot{\phi})$

$\Rightarrow \vec{V} = \dot{r}\hat{e}_r + r\dot{\theta}\hat{e}_\theta + r\sin\theta\dot{\phi}\hat{e}_\phi$

Now;

$\vec{a} = \frac{d}{dt}(\vec{V}) = \frac{d}{dt}(\dot{r}\hat{e}_r + r\dot{\theta}\hat{e}_\theta + r\sin\theta\dot{\phi}\hat{e}_\phi)$

$\vec{a} = \ddot{r}\hat{e}_r + \dot{r}(\dot{\theta}\hat{e}_\theta + \sin\theta\dot{\phi}\hat{e}_\phi) + \dot{r}\dot{\theta}\hat{e}_\theta + r\ddot{\theta}\hat{e}_\theta + r\dot{\theta}(-\dot{\theta}\hat{e}_r + \cos\theta\hat{e}_\phi\dot{\phi}) + \dot{r}\sin\theta\dot{\phi}\hat{e}_\phi + r\ddot{\phi}\sin\theta\hat{e}_\phi + r\cos\theta\dot{\theta}\dot{\phi}\hat{e}_\phi + r\sin\theta\dot{\phi}(-\sin\theta\dot{\phi}\hat{e}_r - \cos\theta\dot{\phi}\hat{e}_\theta)$

$\vec{a} = (\ddot{r} - r\dot{\theta}^2 - r\sin^2\theta\dot{\phi}^2)\hat{e}_r + (2\dot{r}\dot{\theta} + r\ddot{\theta} - r\sin\theta\cos\theta\dot{\phi}^2)\hat{e}_\theta + (2\dot{r}\dot{\phi}\sin\theta + 2r\dot{\theta}\dot{\phi}\cos\theta + r\ddot{\phi}\sin\theta)\hat{e}_\phi$

Example; Represent the following vector in spherical coordinates.
$\vec{A} = 2y\hat{i} - z\hat{j} + 3x\hat{k}$.

Solution: Since

$\hat{i} = \sin\theta\cos\phi\hat{e}_r + \cos\theta\cos\phi\hat{e}_\theta - \sin\phi\hat{e}_\phi$

$\hat{j} = \sin\theta\sin\phi\hat{e}_r + \cos\theta\sin\phi\hat{e}_\theta + \cos\phi\hat{e}_\phi$

$\hat{k} = \cos\theta\hat{e}_r - \sin\theta\hat{e}_\theta$

and

$x = r\sin\theta\cos\phi$
$y = r\sin\theta\sin\phi$
$z = r\cos\theta$

Implies:

$\vec{A} = 2y\hat{i} - z\hat{j} + 3x\hat{k}$, becomes:

$\vec{A} = A_r\hat{e}_r + A_\theta\hat{e}_\theta + A_\phi\hat{e}_\phi$, where

$A_r = 2r\sin^2\theta\sin\phi\cos\phi - r\sin\theta\cos\theta\sin\phi + 3r\sin\theta\cos\theta\cos\phi$

$A_\theta = 2r\sin\theta\cos\theta\sin\phi\cos\phi - r\cos^2\theta\sin\phi - 3r\sin^2\theta\cos\phi$

$A_\phi = -2r\sin\theta\sin^2\phi - r\cos\theta\cos\phi$

Example: Express the following in spherical coordinates.

a) the sphere $x^2 + y^2 + z^2 = 9$

using transformation equations:

$(r\sin\theta\cos\phi)^2 + (r\sin\theta\sin\phi)^2 + (r\cos\theta)^2 = 9$

$r^2\sin^2\theta(\cos^2\phi + \sin^2\phi) + r^2\cos^2\theta = 9$

$\Rightarrow r^2(\sin^2\theta + \cos^2\theta) = 9$

$\Rightarrow r^2 = 9$ or $r = 3$

b) the paraboloid: $z = x^2 + y^2$

$\Rightarrow r\cos\theta = r^2\sin^2\theta\cos^2\phi + r^2\sin^2\theta\sin^2\phi$

$r\cos\theta = r^2\sin^2\theta$

$\Rightarrow \cos\theta = r\sin^2\theta$

c) the plane $y = x$

$\Rightarrow r\sin\theta\sin\phi = r\sin\theta\cos\phi$

$\Rightarrow \tan\phi = 1 \Rightarrow \phi = \tan^{-1}(1)$

$\Rightarrow \phi = \pi/4$ and $5\pi/4$ b/c $0 \le \phi \le 2\pi$

d) the cone $z^2 = 3(x^2 + y^2)$

$\Rightarrow r^2\cos^2\theta = 3r^2\sin^2\theta(\cos^2\phi + \sin^2\phi)$

$\Rightarrow \tan^2\theta = 1/3$

$\Rightarrow \tan\theta = 1/\sqrt{3}$

$\Rightarrow \theta = \tan^{-1}(1/\sqrt{3})$

$\Rightarrow \theta = \pi/6$ b/c $0 \le \theta \le \pi$

e) the plane $z = 0$

$\Rightarrow r\cos\theta = 0$

$r \neq 0 \Rightarrow \cos\theta = 0$

$\Rightarrow \theta = \cos^{-1}(0)$

$\Rightarrow \theta = \pi/2$

Example: Express the heat equation $\frac{\partial u}{\partial t} = k\nabla^2 u$ in spherical coordinates if u is independent of

a) ϕ

b) ϕ and θ

c) r and t

d) ϕ, θ and t

a) Consider $\frac{\partial u}{\partial t} = k\nabla^2 u$

first consider ∇^2 in Spherical coordinates

$$\nabla^2 u = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial u}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial u}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 u}{\partial\phi^2}$$

when u is independent of ϕ, we have

$$\frac{\partial u}{\partial t} = k\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial u}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial u}{\partial\theta}\right)\right]$$

b) Similarly when u is independent of θ and ϕ:

$$\frac{\partial u}{\partial t} = \frac{k}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial u}{\partial r}\right)$$

c) independent of r and t gives:

$$\sin\theta\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial u}{\partial\theta}\right) + \frac{\partial^2 u}{\partial\phi^2} = 0$$

d) $\frac{d}{dr}\left(r^2\frac{du}{dr}\right) = 0$ OR $r^2\frac{du}{dr} = \text{constant}$

Contravariant Vectors

Important points:

- On transforming a vector the components change and the basis vectors change however, the vector itself does not change its direction and magnitude remains same.
- This actually defines a tensor which does not change with a change of coordinate transformation. i.e., it remains invariant under a change of coordinate systems.

The components of the vector change under a coordinate transformation. These components may change in one of the two ways

- Contravariant
- Covariant

A vector whose components transform in a contravariant fashion is a contravariant vector.

Similarly, a vector whose components transform in a covariant fashion is a covariant vector.

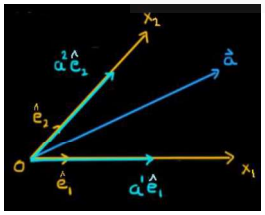
Contravariant vectors:

- A contravariant vector or tangent vector is often abbreviated simply as vector, such as a direction vector or velocity vector.
- The contravariant vector has components that contra-vary with a change of basis to compensate.
- That is, the matrix that transforms the vector components must be the inverse of the matrix that transforms the basis vectors.
- Examples of vectors with contravariant components include the position of an object relative to an observer, or any derivative of position with respect to time, including velocity and acceleration.
- In Einstein notation, contravariant components are denoted with upper indices.

Contravariant Vectors:

Consider a coordinate system with $x_1 - axis\ and\ x_2 - axis$ and $\hat{e}_1$ and $\hat{e}_2$ are unit vectors along these axes. Further consider a vector $\vec{a}$ in this coordinate system such that in component form it can be written as

$$\vec{a} = a^1\hat{e}_1 + a^2\hat{e}_2 \rightarrow (1)$$



Now we will observe change in above vector when the basis vector $\hat{e}_2$ becomes $2\hat{e}_2$. It means we are transforming the vector $\vec{a}$ from the x_1x_2 coordinate system to a new one named as $x'_1x'_2$ system such that the new bases vectors $\hat{e}'_1$ and $\hat{e}'_2$ are related with $\hat{e}_1$ and $\hat{e}_2$ as: $\hat{e}'_1 = \hat{e}_1$ and $\hat{e}'_2 = 2\hat{e}_2$.

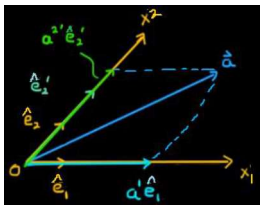
In this new system:

$$\vec{a} = a^1\hat{e}_1 + a'^2\hat{e}'_2 \rightarrow (2)$$

Now comparing (1) and (2), we have

$a^2\hat{e}_2 = a^{2\prime}\hat{e}'_2$ using $\hat{e}'_2 = 2\hat{e}_2$, we obtain:

$a^2\hat{e}_2 = 2a'^2\hat{e}_2$ implies $a^2 = 2a'^2$ or $a'^2 = \frac{a^2}{2}$



Covariant Vectors

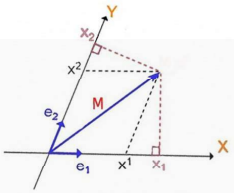
Covariant Vectors:

- In contrast, co-vectors also called dual vectors typically have units of the inverse of distance.
- An example of a co-vector is the gradient, which has units of a spatial derivative, or distance inverse. The components of co-vectors change in the same way as changes to scale of the reference axes and consequently are called covariant.
- A covariant vector often abbreviated as co-vector has components that **co-vary** with a change of basis.
- That is, the components must be transformed by the same matrix as the change of basis matrix.
- In Einstein notation, covariant components are denoted with lower indices.

Relation between contravariant and covariant components of a vector:

Note that:

- Contravariant components of a vector represent its parallel projections onto coordinate axes.
- Covariant components of a vector represent its perpendicular projection.



(x_1, x_2) covariant components.
(x^1, x^2) contravariant components.

Consider the vector $\vec{a}$ in contravariant component form as

$\vec{a} = a^1\hat{e}_1 + a^2\hat{e}_2 \rightarrow (1)$

Since its covariant components are perpendicular projections of $\vec{a}$ onto the coordinate axes so,

$a_1 = \vec{a}.\ \hat{e}_1 \rightarrow (2)$,

$a_2 = \vec{a}.\ \hat{e}_2 \rightarrow (3)$

Now replacing the expression of vector $\vec{a}$ from Eq.1 into (2) and (3)

We have:

$a_1 = (a^1\hat{e}_1 + a^2\hat{e}_2).\ \hat{e}_1$ and $a_2 = (a^1\hat{e}_1 + a^2\hat{e}_2).\ \hat{e}_2$

Implies

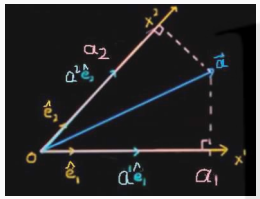
$a_1 = a^1(\hat{e}_1.\ \hat{e}_1) + a^2(\hat{e}_2.\ \hat{e}_1) \rightarrow (2')$

$a_2 = a^1(\hat{e}_1.\ \hat{e}_2) + a^2(\hat{e}_2.\ \hat{e}_2) \rightarrow (3')$

Covariant vector components transformation:

Now similar to contravariant case, consider a coordinate system with $x_1 - axis\ and\ x_2 - axis$ and $\hat{e}_1$ and $\hat{e}_2$ are unit vectors along these axes. Further consider a vector $\vec{a}$ in this coordinate system such that in component form it can be written as

$\vec{a} = a^1\hat{e}_1 + a^2\hat{e}_2 \rightarrow (1)$



To obtain a relation for covariant components, consider

$a'_1 = \vec{a}.\ \hat{e}_1 = a^1(\hat{e}_1.\ \hat{e}_1) + a^2(\hat{e}'_2.\ \hat{e}_1)$

$a'_2 = \vec{a}.\ \hat{e}'_2 = a^1(\hat{e}_1.\ \hat{e}'_2) + a^2(\hat{e}'_2.\ \hat{e}'_2)$

Now using $\hat{e}'_2 = 2\hat{e}_2\ and\ a^2 = \frac{a^2}{2}$, we get

$a'_1 = a^1(\hat{e}_1.\ \hat{e}_1) + a^2(\hat{e}_2.\ \hat{e}_1) = a_1$ and

$a'_2 = 2(a^1(\hat{e}_1.\ \hat{e}_2) + a^2(\hat{e}_2.\ \hat{e}_2)) = 2a_2$

Implies that a covariant component transforms in the same manner as the basis vectors.

Note:

If we restrict ourselves to Cartesian co-ordinate systems having orthonormal basis vectors then we do not have to make a distinction between co- and contravariant indices.

First order tensors

Tensor of rank one:

Tensors of order one are actually vectors. Here we study the transformation rules for contravariant and covariant first order tensors.

Transformation Law for Contravariant tensor:

A vector field $\vec{A}$ is a contravariant tensor of order one provided its components A^i and $\tilde{A}^i$ relative to the coordinate systems x^i and $\tilde{x}^i$ obey the transformation

$\tilde{A}^i = A^r \frac{\partial \tilde{x}^i}{\partial x^r}\ ;\ 1 \le i \le n$

$A_i = a_{ij}\ \tilde{A}_j$

Question: Show that the tangent vector of curve is a contravariant vector.

Solution: Let C be curve whose parametric representation in $x^i -$ system is $x^i = x^i(t);\ a \le t \le b$.

The tangent vector $T = T^i$ is defined by the usual differentiation:

$$T^i = \frac{dx^i}{dt}$$

Under a change of coordinates to $\tilde{x}^i$ system, the same curve is given by $\tilde{x}^i = \tilde{x}^i(t);\ a \le t \le b$

And the tangent for this system is

$$\tilde{T}^i = \frac{d\tilde{x}^i}{dt}$$

Then by chain rule:

$\frac{d\tilde{x}^i}{dt} = \frac{\partial \tilde{x}^i}{\partial x^r}\frac{dx^r}{dt}$ or $\tilde{T}^i = T^r\frac{\partial \tilde{x}^i}{\partial x^r}$

Implying that tangent vector $\vec{T}$ is a contravariant vector.

Transformation Law for Covariant tensor:

A vector field $\vec{A}$ is a covariant tensor of order one provided its components A_i and $\tilde{A}_i$ relative to the coordinate systems x^i and $\tilde{x}^i$ obey the transformation

$\tilde{A}_i = A_r\frac{\partial x^r}{\partial \tilde{x}^i}\ ;\ 1 \le i \le n$

Example: Suppose that T^i is a contravariant vector on R^2 and that $T^i = (x^2, x^1)$ in the x^i *system*. Calculate $\tilde{T}^i$ in the $\tilde{x}^i$ −system, under the change of coordinates.

$$\tilde{x}^1 = (x^2)^2$$
$$\tilde{x}^2 = x^1 x^2$$

By definition of contravariance,

$$\tilde{T}^i = T^r \frac{\partial \tilde{x}^i}{\partial x^r} = T^1 \frac{\partial \tilde{x}^i}{\partial x^1} + T^2 \frac{\partial \tilde{x}^i}{\partial x^2} = x^2 \frac{\partial \tilde{x}^i}{\partial x^1} + x^1 \frac{\partial \tilde{x}^i}{\partial x^2}$$

For $i = 1$: $\tilde{T}^1 = x^2 \frac{\partial \tilde{x}^1}{\partial x^1} + x^1 \frac{\partial \tilde{x}^1}{\partial x^2} = 2x^1 x^2$

For $i = 2$: $\tilde{T}^2 = x^2 \frac{\partial \tilde{x}^2}{\partial x^1} + x^1 \frac{\partial \tilde{x}^2}{\partial x^2} = (x^2)^2 + (x^1)^2$

Which in terms of tilted coordinates become:

$\tilde{T}^1 = 2\tilde{x}^2$, $\tilde{T}^2 = \tilde{x}^1 + \frac{(\tilde{x}^2)^2}{\tilde{x}^1}$

Example: Calculate $\tilde{T}^i$ in the $\tilde{x}^i$ −system if $T^i = (x^2, x^1 + 2x^2)$ is a contravariant vector.

Solution: Given $T^i = (x^2, x^1 + 2x^2)$

we have to find $\tilde{T}^i$.

using the transformation law:

$$\tilde{T}^i = \frac{\partial \tilde{x}^i}{\partial x^j} T^j$$

$$\Rightarrow \tilde{T}^1 = \frac{\partial \tilde{x}^1}{\partial x^1} T^1 + \frac{\partial \tilde{x}^1}{\partial x^2} T^2$$

and $\tilde{T}^2 = \frac{\partial \tilde{x}^2}{\partial x^1} T^1 + \frac{\partial \tilde{x}^2}{\partial x^2} T^2$

using $\tilde{x}^1 = (x^2)^2$ and $\tilde{x}^2 = x^1 x^2$

$$\Rightarrow \frac{\partial \tilde{x}^1}{\partial x^1} = 0, \frac{\partial \tilde{x}^1}{\partial x^2} = 2x^2 \quad \Big| \quad \frac{\partial \tilde{x}^2}{\partial x^1} = x^2 \text{ and } \frac{\partial \tilde{x}^2}{\partial x^2} = x^1$$

$$\Rightarrow \tilde{T}^1 = (0)(x^2) + 2x^2(x^1 + 2x^2) = 2x^1x^2 + 4(x^2)^2$$

$$\tilde{T}^2 = x^2(x^2) + (x^1)(x^1 + 2x^2) = (x^2)^2 + (x^1)^2 + 2x^1x^2$$

$$\Rightarrow \tilde{T}^1 = 2\tilde{x}^2 + 4\tilde{x}^1$$

and $\tilde{T}^2 = \tilde{x}^1 + 2\tilde{x}^2 + \frac{(\tilde{x}^2)^2}{x^1}$

$$\begin{cases} \sqrt{\tilde{x}^1} = x^2 \\ \Rightarrow \tilde{x}^2 = x^1 \sqrt{\tilde{x}^1} \\ \Rightarrow x^1 = \frac{\tilde{x}^2}{\sqrt{\tilde{x}^1}} \end{cases}$$

Question: Show that there is no distinction between contravariant and covariant vectors when use the transformation of the type: $\tilde{x}^i = a^i_m x^m + b^i$, where a's and b's are constants such that: $a^i_\lambda a^i_m = \delta^\lambda_m$

Solution: consider, $\tilde{x}^i = a^i_m x^m + b^i$

$$\Rightarrow a^i_m x^m = \tilde{x}^i - b^i$$

multiplying by a^i_λ on both sides:

$$a^i_\lambda a^i_m x^m = a^i_\lambda \tilde{x}^i - a^i_\lambda b^i$$

$$\Rightarrow \delta^\lambda_m x^m = a^i_\lambda \tilde{x}^i - b^i a^i_\lambda$$

$$\Rightarrow x^\lambda = a^i_\lambda \tilde{x}^i - b^i a^i_\lambda$$

$$\Rightarrow \frac{\partial x^\lambda}{\partial \tilde{x}^i} = a^i_\lambda$$

Now consider: $\tilde{x}^i = a^i_s x^s + b^i$

$$\Rightarrow \frac{\partial \tilde{x}^i}{\partial x^s} = a^i_s$$

Hence the contravariant transformation law is: $\tilde{A}^i = \frac{\partial \tilde{x}^i}{\partial x^s} A^s = a^i_s A^s$

and from covariant transformation law: $\tilde{A}_i = \frac{\partial x^s}{\partial \tilde{x}^i} A_s = a^i_s A_s$

$\Rightarrow$ There is no distinction between covariant and contravariant vectors under the given transformation.

Second Order Tensors

Higher order tensors:

Tensors of arbitrary order may be defined. We begin with the three types of second order tensors.

- Contravariant tensor
- Covariant tensor
- Mixed tensor

Let $V = [V^{ij}]$ denote a matrix field and $[V^{ij}]$ is an $n \times n$ matrix of scalar fields. These are defined over the same region in R^n. As before, there are two systems of coordinates x^i and $\tilde{x}^i$ with $V = [V^{ij}]$ in x^i-system and $V = [\tilde{V}^{ij}]$ in the tilted coordinate system. Where the coordinates from one system to another can be transformed via some transformation.

Contravariant tensor:

The matrix field V is a contravariant tensor of order two if its components T^{ij} in x^i-system and $\tilde{T}^{ij}$ in $\tilde{x}^i$-system obey the law of transformation

$$\bar{T}^{ij} = T^{rs} \frac{\partial \tilde{x}^i}{\partial x^r} \frac{\partial \tilde{x}^j}{\partial x^s}$$

$T^{ij} = T^{rs} \frac{\partial \tilde{x}^i}{\partial x^r} \frac{\partial \tilde{x}^j}{\partial x^s}$

Covariant tensor:

The matrix field V is a covariant tensor of order two if its components T_{ij} in x^i-system and $\tilde{T}_{ij}$ in $\tilde{x}^i$-system obey the law of transformation

$$\bar{T}_{ij} = T_{rs} \frac{\partial x^r}{\partial \tilde{x}^i} \frac{\partial x^s}{\partial \tilde{x}^j}$$

Mixed tensor:

The matrix field V is a mixed tensor of order two, contravariant of order one and covariant of order one, if its components T_j^i in x^i-system and $\tilde{T}_{ij}^{i}$ in $\tilde{x}^i$-system obey the law of transformation

$$\tilde{T}_j^i = T_s^r \frac{\partial \tilde{x}^i}{\partial x^r} \frac{\partial x^s}{\partial \tilde{x}^j}$$

Im

Question: Suppose that the components of a contravariant tensor T of order 2 in a coordinate system x^i of R^2 are $T^{11}=1$, $T^{12}=1$, and $T^{21}=-1$ and $T^{22}=2$

a) Find the components $\tilde{T}^{ij}$ of T in the $\tilde{x}^i$-system, Connected to the x^i-system via $\tilde{x}^1 = (x^1)^2 \neq 0$, $\tilde{x}^2 = x^1 x^2$

b) Compute the values of the point which corresponds to $x^1=1$, $x^2=-2$.

Solution: (a) using the transformation

$$\tilde{T}^{ij} = T^{rs} \frac{\partial \tilde{x}^i}{\partial x^r} \frac{\partial \tilde{x}^j}{\partial x^s} \; ; \; i,j,r,s \text{ have values } 1,2$$

$$\Rightarrow \tilde{T}^{11} = T^{rs} \frac{\partial \tilde{x}^1}{\partial x^r} \frac{\partial \tilde{x}^1}{\partial x^s} \quad (\because r \text{ and } s \text{ are dummy indices})$$

$$= T^{1s} \frac{\partial \tilde{x}^1}{\partial x^1} \frac{\partial \tilde{x}^1}{\partial x^s} + T^{2s} \frac{\partial \tilde{x}^1}{\partial x^2} \frac{\partial \tilde{x}^1}{\partial x^s}$$

$$= T^{11} \frac{\partial \tilde{x}^1}{\partial x^1} \frac{\partial \tilde{x}^1}{\partial x^1} + T^{12} \frac{\partial \tilde{x}^1}{\partial x^1} \frac{\partial \tilde{x}^1}{\partial x^2}$$

$$+ T^{21} \frac{\partial \tilde{x}^1}{\partial x^2} \frac{\partial \tilde{x}^1}{\partial x^1} + T^{22} \frac{\partial \tilde{x}^1}{\partial x^2} \frac{\partial \tilde{x}^1}{\partial x^2}$$

$\tilde{x}^1 = (x^1)^2$, $\tilde{x}^2 = x^1 x^2$

$\Rightarrow \frac{\partial \tilde{x}^1}{\partial x^1} = 2x^1$ and $\frac{\partial \tilde{x}^1}{\partial x^2} = 0$

$\frac{\partial \tilde{x}^2}{\partial x^1} = x^2$ and $\frac{\partial \tilde{x}^2}{\partial x^2} = x^1$

$\Rightarrow \tilde{T}^{11} = (1)(2x^1)(2x^1) + 0 + 0 + 0$

$\tilde{T}^{11} = 4(x^1)^2$

Similarly:

$\tilde{T}_{12} = 2x^1 x^2 + 2(x^1)^2$

$\tilde{T}_{21} = 2x^1 x^2 - 2(x^1)^2$

$\tilde{T}_{22} = 2(x^1)^2 + (x^2)^2$

(b) At the point $(1,-2)$

$\tilde{T}^{11} = 4(1)^2 = 4$

$\tilde{T}^{12} = 2(1)(-2) + 2(1)^2 = -2$

$\tilde{T}^{21} = 2(1)(-2) - 2(1)^2 = -6$

$\tilde{T}^{22} = 2(1)^2 + (-2)^2 = 6$

Question: Show that if S^i and T^i are Contravariant vectors on R^n, the matrix $[U^{ij}] = [S^i T^j]_{nn}$, defined in this manner for all coordinates systems represents a contravariant tensor of order 2.

Solution: Given S^i and T^i are contravariant vectors in R^n, we have

$$\tilde{S}^i = S^r \frac{\partial \tilde{x}^i}{\partial x^r} \text{ and } \tilde{T}^j = T^s \frac{\partial \tilde{x}^j}{\partial x^s}$$

By multiplying both we have

$$\tilde{S}^i \tilde{T}^j = S^r T^s \frac{\partial \tilde{x}^i}{\partial x^r} \frac{\partial \tilde{x}^j}{\partial x^s}$$

where $S^i T^j = U^{ij}$

$$\Rightarrow \tilde{U}^{ij} = U^{rs} \frac{\partial \tilde{x}^i}{\partial x^r} \frac{\partial \tilde{x}^j}{\partial x^s} \Rightarrow U^{ij} \text{ is a contravariant tensor of rank 2.}$$

Question: Let T_{ij} be a covariant tensor of order 2. Find the components $\tilde{T}_{ij}$ in $\tilde{x}_i$-system if $T_{11} = x^2$, $T_{12} = T_{21} = 0$, $T_{22} = x^1$ are the components in x^i-system. Also the transformations b/w x^i and $\tilde{x}^i$ system are:

$\tilde{x}^1 = (x^1)^2$

$\tilde{x}^2 = x^1 x^2$

Solution: First consider

$\tilde{x}^1 = (x^1)^2$ and $\tilde{x}^2 = x^1 x^2$

$\Rightarrow x^1 = \sqrt{\tilde{x}^1}$, $x^2 = \frac{\tilde{x}^2}{x^1} = \frac{\tilde{x}^2}{\sqrt{\tilde{x}^1}}$

$\Rightarrow x^1 = \sqrt{\tilde{x}^1}$ and $x^2 = \frac{\tilde{x}^2}{\sqrt{\tilde{x}^1}}$

Now using the transformation equation:

$$\tilde{T}_{ij} = T_{rs}\frac{\partial x^r}{\partial \tilde{x}^i}\frac{\partial x^s}{\partial \tilde{x}^j}$$

$$\Rightarrow \tilde{T}_{11} = T_{rs}\frac{\partial x^r}{\partial \tilde{x}^1}\frac{\partial x^s}{\partial \tilde{x}^1} = T_{1s}\frac{\partial x^1}{\partial \tilde{x}^1}\frac{\partial x^s}{\partial \tilde{x}^1} + T_{2s}\frac{\partial x^2}{\partial \tilde{x}^1}\frac{\partial x^s}{\partial \tilde{x}^1}$$

$$\Rightarrow \tilde{T}_{11} = T_{11}\frac{\partial x^1}{\partial \tilde{x}^1}\frac{\partial x^1}{\partial \tilde{x}^1} + T_{12}\frac{\partial x^1}{\partial \tilde{x}^1}\frac{\partial x^2}{\partial \tilde{x}^1} + T_{21}\frac{\partial x^2}{\partial \tilde{x}^1}\frac{\partial x^1}{\partial \tilde{x}^1} + T_{22}\frac{\partial x^2}{\partial \tilde{x}^1}\frac{\partial x^2}{\partial \tilde{x}^1}$$

now: $\frac{\partial x^1}{\partial \tilde{x}^1} = \frac{1}{2\sqrt{\tilde{x}_1}}$, $\frac{\partial x^2}{\partial \tilde{x}^1} = \frac{-\tilde{x}^2}{2(\tilde{x}^1)^{3/2}}$

$$\Rightarrow \tilde{T}_{11} = x^2\frac{1}{2\sqrt{\tilde{x}_1}}\cdot\frac{1}{2\sqrt{\tilde{x}_1}} + 0 + 0 + x^1\left(\frac{-\tilde{x}^2}{2(\tilde{x}^1)^{3/2}}\right)\left(\frac{-\tilde{x}^2}{2(\tilde{x}^1)^{3/2}}\right)$$

$$\Rightarrow \tilde{T}_{11} = \frac{1}{4}\frac{x^2}{\tilde{x}^1} + x^1\frac{(\tilde{x}^2)^2}{4(\tilde{x}^1)^3}$$

$$= \frac{1}{4}\frac{x^2}{(x^1)^2} + x^1\frac{(x^1x^2)^2}{4(x^1)^6}$$

$$= \frac{1}{4}\frac{x^2}{(x^1)^2} + \frac{(x^2)^2}{4(x^1)^3} = \frac{1}{4(x^1)^3}\left[x^1x^2 + (x^2)^2\right]$$

Similarly:

$$\tilde{T}_{12} = \frac{-x^2}{2(x^1)^2}$$

$$\tilde{T}_{21} = \frac{-x^2}{2(x^1)^2}$$

$$\tilde{T}_{22} = \frac{1}{x^1}$$

Question: Display the transformation law for a third order tensor that is contravariant of order 2 and covariant of order 1.

Take $p=2$ and $q=1$ in the definition of general tensors.

$$\Rightarrow \tilde{T}^{ij}_{k} = T^{rs}_{t}\frac{\partial \tilde{x}^i}{\partial x^r}\frac{\partial \tilde{x}^j}{\partial x^s}\frac{\partial x^t}{\partial \tilde{x}^k}$$

Question: T^{ij}_{klm} is a mixed tensor of order five. Prove that T^{ij}_{kij} is a covariant tensor of order one.

Solution:

$$\tilde{T}^{ij}_{klm} = T^{rs}_{tuv}\frac{\partial \tilde{x}^i}{\partial x^r}\frac{\partial \tilde{x}^j}{\partial x^s}\frac{\partial x^t}{\partial \tilde{x}^k}\frac{\partial x^u}{\partial \tilde{x}^l}\frac{\partial x^v}{\partial \tilde{x}^m}$$

set $l=i$ and $m=j$:

$$\tilde{T}^{ij}_{kij} = T^{rs}_{tuv}\frac{\partial \tilde{x}^i}{\partial x^r}\frac{\partial \tilde{x}^j}{\partial x^s}\frac{\partial x^t}{\partial \tilde{x}^k}\frac{\partial x^u}{\partial \tilde{x}^i}\frac{\partial x^v}{\partial \tilde{x}^j}$$

$$= T^{rs}_{tuv}\left(\frac{\partial \tilde{x}^i}{\partial x^r}\frac{\partial x^u}{\partial \tilde{x}^i}\right)\left(\frac{\partial \tilde{x}^j}{\partial x^s}\frac{\partial x^v}{\partial \tilde{x}^j}\right)\frac{\partial x^t}{\partial \tilde{x}^k}$$

$$\tilde{T}^{ij}_{kij} = T^{rs}_{tuv}\,\delta^u_r\,\delta^v_s\frac{\partial x^t}{\partial \tilde{x}^k}$$

T_k ← $T^{rs}_{tuv} \to T_t$ when $r=s$

$\frac{\partial x^u}{\partial x^r} = \delta^u_r$

OR

$$\tilde{T}_k = T_t\frac{\partial x^t}{\partial \tilde{x}^k}$$

Question: Show that kronecker delta is a mixed tensor of rank two.

Solution: x^i – System, $\tilde{x}^i$ – System

$$\tilde{\delta}^i_j = 1 \text{ if } i=j, \quad = 0 \text{ if } i\neq j$$

so, $\tilde{\delta}^i_j = \frac{\partial \tilde{x}^i}{\partial \tilde{x}^j} = \frac{\partial \tilde{x}^i}{\partial x^k}\cdot\frac{\partial x^l}{\partial \tilde{x}^j}\frac{\partial x^k}{\partial x^l}$

$\tilde{\delta}^i_j = \frac{\partial \tilde{x}^i}{\partial x^k}\cdot\frac{\partial x^l}{\partial \tilde{x}^j}\delta^k_l$ $\Rightarrow \delta^i_j$ is a mixed tensor of rank 2.

Question: Compute the formulas for the tensor components $\tilde{T}^{ij}$ in polar coordinates if the components T^{ij} in rectangular coordinates are given.

Solution: here $x^1 = x$, $x^2 = y$, $\tilde{x}^1 = r$, $\tilde{x}^2 = \theta$

also, $x = r\cos\theta$, $y = r\sin\theta$, $r = \sqrt{x^2+y^2}$, $\theta = \tan^{-1}y/x$

using the transformation:

$$\tilde{T}^{ij} = T^{uv}\frac{\partial \tilde{x}^i}{\partial x^u}\frac{\partial \tilde{x}^j}{\partial x^v}$$

So, we require

$$\frac{\partial \tilde{x}^1}{\partial x^1} = \frac{\partial r}{\partial x} = \frac{x}{\sqrt{x^2+y^2}} = \frac{r\cos\theta}{r} = \cos\theta$$

$$\frac{\partial \tilde{x}^2}{\partial x^1} = \frac{\partial \theta}{\partial x} = \frac{x^2(-y/x^2)}{x^2+y^2} = \frac{-y}{x^2+y^2} = \frac{-r\sin\theta}{r^2} = -\frac{\sin\theta}{r}$$

$$\frac{\partial \tilde{x}^1}{\partial x^2} = \frac{\partial r}{\partial y} = \frac{y}{\sqrt{x^2+y^2}} = \frac{r\sin\theta}{r} = \sin\theta$$

$$\frac{\partial \tilde{x}^2}{\partial x^2} = \frac{\partial \theta}{\partial y} = \frac{x^2(1/x)}{x^2+y^2} = \frac{x}{x^2+y^2} = \frac{r\cos\theta}{r^2} = \frac{\cos\theta}{r}$$

Tensor of arbitrary order:

Vectors and matrix fields are inadequate for higher order tensors. The generalized vector field V is a tensor of order $m = p+q$, contravariant of order p and covariant of order q, if its components $T^{i_1i_2i_3\ldots i_p}_{j_1j_2j_3\ldots j_q}$ in x^i-system and $\tilde{T}^{i_1i_2i_3\ldots i_p}_{j_1j_2j_3\ldots j_q}$ in $\tilde{x}^i$- system obey the law of transformation

$$\tilde{T}^{i_1i_2i_3\ldots i_p}_{j_1j_2j_3\ldots j_q} = T^{i_1i_2i_3\ldots i_p}_{j_1j_2j_3\ldots j_q}\frac{\partial \tilde{x}^{i_1}}{\partial x^{r_1}}\frac{\partial \tilde{x}^{i_2}}{\partial x^{r_2}}\cdots\frac{\partial \tilde{x}^{i_p}}{\partial x^{r_p}}\frac{\partial x^{s_1}}{\partial \tilde{x}^{j_1}}\frac{\partial x^{s_1}}{\partial \tilde{x}^{j_1}}\cdots\frac{\partial x^{s_1}}{\partial \tilde{x}^{j_1}}$$

Implies:

$$\bar{T}^{11} = T^{1\nu}\frac{\partial \bar{x}^1}{\partial x^\nu}\frac{\partial \bar{x}^1}{\partial x^\nu} + T^{2\nu}\frac{\partial \bar{x}^1}{\partial x^2}\cdot\frac{\partial \bar{x}^1}{\partial x^\nu}$$

$$= T^{11}\frac{\partial \bar{x}^1}{\partial x^1}\frac{\partial \bar{x}^1}{\partial x^1} + T^{12}\frac{\partial \bar{x}^1}{\partial x^1}\frac{\partial \bar{x}^1}{\partial x^2} + T^{21}\frac{\partial \bar{x}^1}{\partial x^2}\cdot\frac{\partial \bar{x}^1}{\partial x^1} + T^{22}\frac{\partial \bar{x}^1}{\partial x^2}\frac{\partial \bar{x}^1}{\partial x^2}$$

$$= T^{11}(\cos\theta)(\cos\theta) + T^{12}(\cos\theta)(\sin\theta) + T^{21}(\sin\theta)(\cos\theta) + T^{22}(\sin\theta)(\sin\theta)$$

$$\bar{T}^{11} = \cos^2\theta T^{11} + \frac{1}{2}\sin 2\theta T^{12} + \frac{1}{2}\sin 2\theta T^{21} + \sin^2\theta T^{22}$$

Implies:

$$\bar{T}^{11} = \cos^2\theta T^{11} + \frac{1}{2}\sin 2\theta T^{21} + \frac{1}{2}\sin 2\theta T^{12} + \sin^2\theta T^{22}$$

$$\bar{T}^{12} = \frac{-1}{2r}\sin 2\theta T^{11} - \frac{\sin^2\theta}{r}T^{21} + \frac{\cos^2\theta}{r}T^{12} + \frac{1}{2r}\sin 2\theta T^{22}$$

$$\bar{T}^{21} = \frac{-1}{2r}\sin 2\theta T^{11} + \frac{1}{r}\cos^2\theta T^{21} - \frac{1}{r}\sin^2\theta T^{12} + \frac{1}{2r}\sin 2\theta T^{22}$$

$$\bar{T}^{22} = \frac{\sin^2\theta}{r^2}T^{11} - \frac{\sin 2\theta}{2r^2}T^{21} - \frac{\sin 2\theta}{2r^2}T^{12} + \frac{\cos^2\theta}{r^2}T^{22}$$

Question: Compute the formulas for the tensor components $\bar{T}_{ij}$ in cylindrical coordinates if the components T_{ij} in rectangular coordinates are given.

Solution: $x^1 = x$, $x^2 = y$, $x^3 = z$ and $\bar{x}^1 = r$, $\bar{x}^2 = \theta$, $\bar{x}^3 = z$

also, $x = r\cos\theta$, $y = r\sin\theta$, $r = \sqrt{x^2+y^2}$, $\theta = \tan^{-1}\frac{y}{x}$ and $z = z$

using the transformation:

$$\bar{T}_{ij} = T_{\mu\nu}\frac{\partial x^\mu}{\partial \bar{x}^i}\frac{\partial x^\nu}{\partial \bar{x}^j}$$

$$\frac{\partial x^1}{\partial \bar{x}^1} = \frac{\partial x}{\partial r} = \cos\theta \qquad \frac{\partial x^2}{\partial \bar{x}^1} = \frac{\partial y}{\partial r} = \sin\theta$$

$$\frac{\partial x^1}{\partial \bar{x}^2} = \frac{\partial x}{\partial \theta} = -r\sin\theta \qquad \frac{\partial x^2}{\partial \bar{x}^2} = \frac{\partial y}{\partial \theta} = r\cos\theta$$

$$\frac{\partial x^1}{\partial \bar{x}^3} = \frac{\partial x}{\partial z} = 0 \qquad \frac{\partial x^2}{\partial \bar{x}^3} = \frac{\partial y}{\partial z} = 0$$

$$\frac{\partial x^3}{\partial \bar{x}^1} = \frac{\partial z}{\partial r} = 0$$

$$\frac{\partial x^3}{\partial \bar{x}^2} = \frac{\partial z}{\partial \theta} = 0$$

$$\frac{\partial x^3}{\partial \bar{x}^3} = \frac{\partial z}{\partial z} = 1$$

Implies:

$$\bar{T}_{11} = T_{\mu\nu}\frac{\partial x^\mu}{\partial \bar{x}^1}\cdot\frac{\partial x^\nu}{\partial \bar{x}^1}$$

$$= T_{1\nu}\frac{\partial x^1}{\partial \bar{x}^1}\frac{\partial x^\nu}{\partial \bar{x}^1} + T_{2\nu}\frac{\partial x^2}{\partial \bar{x}^1}\cdot\frac{\partial x^\nu}{\partial \bar{x}^1} + T_{3\nu}\frac{\partial x^3}{\partial \bar{x}^1}\frac{\partial x^\nu}{\partial \bar{x}^1}$$

$$= T_{11}\frac{\partial x^1}{\partial \bar{x}^1}\frac{\partial x^1}{\partial \bar{x}^1} + T_{12}\frac{\partial x^1}{\partial \bar{x}^1}\frac{\partial x^2}{\partial \bar{x}^1} + T_{13}\frac{\partial x^1}{\partial \bar{x}^1}\frac{\partial x^3}{\partial \bar{x}^1}\to 0$$

$$+ T_{21}\frac{\partial x^2}{\partial \bar{x}^1}\frac{\partial x^1}{\partial \bar{x}^1} + T_{22}\frac{\partial x^2}{\partial \bar{x}^1}\frac{\partial x^2}{\partial \bar{x}^1} + T_{23}\frac{\partial x^2}{\partial \bar{x}^1}\frac{\partial x^3}{\partial \bar{x}^1}\to 0$$

$$+ T_{31}\frac{\partial x^3}{\partial \bar{x}^1}\frac{\partial x^1}{\partial \bar{x}^1} + T_{32}\frac{\partial x^3}{\partial \bar{x}^1}\frac{\partial x^2}{\partial \bar{x}^1} + T_{33}\frac{\partial x^3}{\partial \bar{x}^1}\frac{\partial x^3}{\partial \bar{x}^1}\to 0$$

$$\Rightarrow \bar{T}_{11} = \cos^2\theta T_{11} + T_{12}(\cos\theta\sin\theta) + T_{21}(\cos\theta\sin\theta) + \sin^2\theta T_{22}$$

$$\Rightarrow \bar{T}_{11} = \cos^2\theta T_{11} + \sin^2\theta T_{22} + \sin\theta\cos\theta(T_{12}+T_{21})$$

Alternative Formulas (For tensors of order 2)

In terms of Jacobian matrix:

where $J = \frac{\partial(\bar{x}^1,\bar{x}^2)}{\partial(x^1,x^2)}$, $J^{-1} = \frac{\partial(x^1,x^2)}{\partial(\bar{x}^1,\bar{x}^2)}$

The transformation law for contravariant tensor is: $\bar{T} = JTJ^t$

Similarly, for a covariant tensor:

$$\bar{T} = (J^{-1})^t T J^{-1}$$

and for mixed tensor:

$$\bar{T} = JTJ^{-1}$$

Question: Given the components of tensor T^{ij} in x^i-System: $T^{11}=1$, $T^{12}=1$, $T^{21}=-1$ and $T^{22}=2$. Find the components of T^{ij} in $\bar{x}^i$-system given: $\bar{x}^1 = (x^1)^2$, $\bar{x}^2 = x^1x^2$

Solution: Using $T = [T^{ij}]$

and $\bar{T} = JTJ^t$

First we find J matrix.

$$J = \frac{\partial(\bar{x}^1,\bar{x}^2)}{\partial(x^1,x^2)} = \begin{bmatrix} \partial\bar{x}^1/\partial x^1 & \partial\bar{x}^1/\partial x^2 \\ \partial\bar{x}^2/\partial x^1 & \partial\bar{x}^2/\partial x^2 \end{bmatrix}$$

$\bar{x}^1 = (x^1)^2$, $\bar{x}^2 = x^1x^2$

$$\Rightarrow J = \begin{bmatrix} 2x^1 & 0 \\ x^2 & x^1 \end{bmatrix} \text{ and } J^t = \begin{bmatrix} 2x^1 & x^2 \\ 0 & x^1 \end{bmatrix}$$

Now $\bar{T} = JTJ^t$

gives

$$\bar{T} = \begin{bmatrix} 2x^1 & 0 \\ x^2 & x^1 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 2x^1 & x^2 \\ 0 & x^1 \end{bmatrix}$$

$$\Rightarrow \bar{T} = \begin{bmatrix} 2x^1 & 2x^1 \\ x^2 - x^1 & x^2 + 2x^1 \end{bmatrix}\begin{bmatrix} 2x^1 & x^2 \\ 0 & x^1 \end{bmatrix}$$

$$= \begin{bmatrix} 4(x^1)^2 & 2x^1x^2 + 2(x^1)^2 \\ 2x^1x^2 - 2(x^1)^2 & (x^2)^2 - x^1x^2 + x^1x^2 + 2(x^1)^2 \end{bmatrix}$$

$$\bar{T} = \begin{bmatrix} 4(x^1)^2 & 2x^1x^2 + 2(x^1)^2 \\ 2x^1x^2 - 2(x^1)^2 & (x^2)^2 + 2(x^1)^2 \end{bmatrix}$$

Question: Compute the formulas for the tensor components $\bar{T}^{ij}$ in polar coordinates if the components T^{ij} in rectangular coordinates are given.

Solution: let $T = [T^{ij}]$ and $\bar{T} = J\,T\,J^t$ where $J = \dfrac{\partial(r,\theta)}{\partial(x,y)}$ and $J^{-1} = \dfrac{\partial(x,y)}{\partial(r,\theta)}$

As $x = r\cos\theta$ and $y = r\sin\theta$

$r = \sqrt{x^2+y^2}$, $\theta = \tan^{-1} y/x$

$$J^{-1} = \begin{bmatrix} \partial x/\partial r & \partial x/\partial\theta \\ \partial y/\partial r & \partial y/\partial\theta \end{bmatrix} = \begin{bmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{bmatrix}$$

$$\Rightarrow J = \begin{bmatrix} \cos\theta & \sin\theta \\ -\frac{\sin\theta}{r} & \frac{\cos\theta}{r} \end{bmatrix}$$

and $J^t = \begin{bmatrix} \cos\theta & -\sin\theta/r \\ \sin\theta & \cos\theta/r \end{bmatrix}$

Now

$$\bar{T} = J\,T\,J^t$$

$$\Rightarrow \bar{T} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\frac{\sin\theta}{r} & \frac{\cos\theta}{r} \end{bmatrix}\begin{bmatrix} T^{11} & T^{12} \\ T^{21} & T^{22} \end{bmatrix}\begin{bmatrix} \cos\theta & -\frac{\sin\theta}{r} \\ \sin\theta & \frac{\cos\theta}{r} \end{bmatrix}$$

$$= \begin{bmatrix} \cos\theta T^{11} + \sin\theta T^{21} & \cos\theta T^{12} + \sin\theta T^{22} \\ -\frac{\sin\theta}{r}T^{11} + \frac{\cos\theta}{r}T^{21} & -\frac{\sin\theta}{r}T^{12} + \frac{\cos\theta}{r}T^{22} \end{bmatrix}\begin{bmatrix} \cos\theta & -\frac{\sin\theta}{r} \\ \sin\theta & \frac{\cos\theta}{r} \end{bmatrix}$$

Implies:

$$\bar{T}^{11} = \cos^2\theta T^{11} + \frac{1}{2}\sin 2\theta T^{21} + \frac{1}{2}\sin 2\theta T^{12} + \sin^2\theta T^{22}$$

$$\bar{T}^{12} = -\frac{1}{2r}\sin 2\theta T^{11} - \frac{\sin^2\theta}{r}T^{21} + \frac{\cos^2\theta}{r}T^{12} + \frac{1}{2r}\sin 2\theta T^{22}$$

$$\bar{T}^{21} = -\frac{1}{2r}\sin 2\theta T^{11} + \frac{1}{r}\cos^2\theta T^{21} - \frac{1}{r}\sin^2\theta T^{12} + \frac{1}{2r}\sin 2\theta T^{22}$$

$$\bar{T}^{22} = \frac{\sin^2\theta}{r^2}T^{11} - \frac{\sin 2\theta}{2r^2}T^{21} - \frac{\sin 2\theta}{2r^2}T^{12} + \frac{\cos^2\theta}{r^2}T^{22}$$

✓

Invariant:

A function $\varphi(x^1, x^2, x^3, \ldots, x^n)$ is called scalar or an invariant if its original value does not change upon transformation of coordinates from x^i-system to $\bar{x}^i$-system. *i.e.*,

$$\varphi(x^1, x^2, x^3, \ldots, x^n) = \bar{\varphi}(\bar{x}^1, \bar{x}^2, \bar{x}^3, \ldots, \bar{x}^n)$$

- Scalar is a tensor of rank zero.
- Objects, functions, equations or formulas that are independent of the coordinate system used to express them are invariants and are of fundamental significance.
- An invariant of a vector (rank-1 tensor) under coordinate transformations is its magnitude, i.e. length (the direction is also invariant but it is not scalar!).
- The main three independent scalar invariants of a rank-2 tensor under change of basis are:

$$\text{If } A = [A_{ij]},\ \text{then}$$
$$I = tr(A) = A_{ii}$$
$$II = tr(A^2) = A_{ij}A_{ji}$$
$$III = tr(A^3) = A_{ij}A_{jk}A_{ki}$$

$$I_1 = I = A_{ii}$$
$$I_2 = \frac{1}{2}(I^2 - II)$$

$$I_3 = \det(A) = \frac{1}{3!}(I^3 - 3I.\ II + 2III)$$

Question: Let A^i and A_i are contravariant and covariant components of a vector $\vec{A}$ then $A^i A_i$ is an invariant.

Solution: Using the transformation rules for A^i and A_i:

$$\bar{A}^i = \frac{\partial \bar{x}^i}{\partial x^k}A^k \quad \text{and} \quad \bar{A}_i = \frac{\partial x^j}{\partial \bar{x}^i}A_j \qquad \bar{A}^i\bar{A}_i = A^iA_i$$

$$\Rightarrow \bar{A}^i\bar{A}_i = \frac{\partial \bar{x}^i}{\partial x^k}\cdot\frac{\partial x^j}{\partial \bar{x}^i}A^kA_j = \delta^j_k A^kA_j$$

$$\bar{A}^i\bar{A}_i = A^iA_j = A^iA_i$$

$$\Rightarrow \bar{A}^i\bar{A}_i = A^iA_i$$

mid

Question: Prove that if T^i_j is a mixed tensor of order two T^i_i is an invariant.

Solution:

$$\bar{T}^i_j = \frac{\partial \bar{x}^i}{\partial x^k}\frac{\partial x^s}{\partial \bar{x}^j}T^k_s \qquad \bar{T}^i_i = T^i_i$$

$$\Rightarrow \bar{T}^i_i = \frac{\partial \bar{x}^i}{\partial x^k}\frac{\partial x^s}{\partial \bar{x}^i}T^k_s$$

$$\Rightarrow \bar{T}^i_i = \delta^s_k T^k_s = T^k_k$$

$$\Rightarrow \bar{T}^i_i = T^i_i \Rightarrow T^i_i \text{ is invariant}$$

Question: If S^i_{jk} and T^i are tensors of the type and order indicated, $S^i_{jk}T^j$ is an invariant.

Solution:

$$\bar{T}^i = \frac{\partial \bar{x}^i}{\partial x^l}T^l$$

$$\bar{S}^i_{jk} = \frac{\partial \bar{x}^i}{\partial x^m}\frac{\partial x^n}{\partial \bar{x}^j}\frac{\partial x^q}{\partial \bar{x}^k}S^m_{nq}$$

Now,

$$\tilde{S}^{ip}_{jk}\tilde{T}^{l} = \frac{\partial \tilde{x}^i}{\partial x^m}\frac{\partial x^n}{\partial \tilde{x}^j}\frac{\partial x^q}{\partial \tilde{x}^k}\cdot\frac{\partial \tilde{x}^p}{\partial x^l}S^{m}_{nq}T^{l}$$

$$\Rightarrow \tilde{S}^{ir}_{jl}\tilde{T}^{j} = \frac{\partial \tilde{x}^i}{\partial x^m}\frac{\partial x^n}{\partial \tilde{x}^j}\frac{\partial x^q}{\partial \tilde{x}^r}\cdot\frac{\partial \tilde{x}^r}{\partial x^l}\cdot S^{m}_{nq}T^{l}$$

$$\tilde{S}^{ir}_{jl}\tilde{T}^{j} = \delta^{q}_{m}\delta^{n}_{l}S^{m}_{nq}T^{l}$$

for $q=m$ and $n=l$

$$\Rightarrow \tilde{S}^{ir}_{jr}\tilde{T}^{j} = S^{m}_{nm}T^{n}$$

$m \to r$, $n \to j$

$$\Rightarrow \tilde{S}^{ir}_{jr}\tilde{T}^{j} = S^{r}_{j}T^{j} \Rightarrow \text{It is invariant.}$$

Addition and subtraction of tensors

The sum or difference of two tensors which have same number of covariant and contravariant indices is again a tensor of the rank and type as the given tensors.

Since the transformation law for tensor of any arbitrary order is linear in the tensor components, it is clear that if $T_1, T_2, \ldots, T_n$ are tensors of the same type and order and if $\alpha_1, \alpha_2, \ldots, \alpha_n$ are invariant scalars, then

$$\alpha_1 T_1 + \alpha_2 T_2 + \cdots + \alpha_n T_n$$

Is a tensor of the same type and order as T_i.

Question: Show that δ^i_j is an invariant.

Solution:

$$\tilde{\delta}^i_j = \frac{\partial \tilde{x}^i}{\partial x^k}\left(\frac{\partial x^l}{\partial \tilde{x}^j}\delta^k_l\right)$$

$$= \frac{\partial \tilde{x}^i}{\partial x^k}\cdot\frac{\partial x^k}{\partial \tilde{x}^j}$$

$$= \frac{\partial \tilde{x}^i}{\partial \tilde{x}^j} = \delta^i_j$$

$\Rightarrow$ Kronecker delta has same components in every coordinate system.

Question: Show that if S^i and T^i are components of contravariant vectors then $\lambda S^i + \mu T^i$ is a contravariant vector, where λ and μ are scalars.

Solution: We have to find the transformation law for $\lambda S^i + \mu T^i$, where

$$\tilde{S}^i = \frac{\partial \tilde{x}^i}{\partial x^m}S^m \qquad \tilde{T}^i = \frac{\partial \tilde{x}^i}{\partial x^n}T^n$$

also $\tilde{\lambda} = \lambda$ and $\tilde{\mu} = \mu$

So,

$$\tilde{\lambda}\tilde{S}^i + \tilde{\mu}\tilde{T}^i = \lambda\left(\frac{\partial \tilde{x}^i}{\partial x^m}S^m\right) + \mu\left(\frac{\partial \tilde{x}^i}{\partial x^n}T^n\right)$$

$$= \lambda\left(\frac{\partial \tilde{x}^i}{\partial x^m}S^m\right) + \mu\left(\frac{\partial \tilde{x}^i}{\partial x^m}T^m\right)$$

$$= \left(\lambda S^m + \mu T^m\right)\frac{\partial \tilde{x}^i}{\partial x^m}$$

Question: If A^{ij}_k and B^{lm}_n are tensors then their sum and difference are tensors of the same rank and type.

Solution:

$$\tilde{A}^{ij}_k = \frac{\partial \tilde{x}^i}{\partial x^p}\frac{\partial \tilde{x}^j}{\partial x^q}\frac{\partial x^r}{\partial \tilde{x}^k}A^{pq}_r$$

$$\tilde{B}^{ij}_k = \frac{\partial \tilde{x}^i}{\partial x^p}\frac{\partial \tilde{x}^j}{\partial x^q}\frac{\partial x^r}{\partial \tilde{x}^k}B^{pq}_r$$

$$\tilde{A}^{ij}_k \pm \tilde{B}^{ij}_k = \frac{\partial \tilde{x}^i}{\partial x^p}\frac{\partial \tilde{x}^j}{\partial x^q}\frac{\partial x^r}{\partial \tilde{x}^k}\left(A^{pq}_r \pm B^{pq}_r\right)$$

If $\tilde{A}^{ij}_k \pm \tilde{B}^{ij}_k = \tilde{C}^{ij}_k$

and $A^{pq}_r \pm B^{pq}_r = C^{pq}_r$

$$\Rightarrow \tilde{C}^{ij}_k = \frac{\partial \tilde{x}^i}{\partial x^p}\frac{\partial \tilde{x}^j}{\partial x^q}\frac{\partial x^r}{\partial \tilde{x}^k}C^{pq}_r$$

Question: Show that $\frac{\partial \phi}{\partial \tilde{x}^i}$ is a covariant vector, where ϕ is a scalar function.

Solution: Consider there are two coordinate systems: x^i-system and $\tilde{x}^i$-system.

ϕ is a scalar function such that:

$$\phi(\tilde{x}^1, \tilde{x}^2, \ldots, \tilde{x}^n) = \phi(x^1, x^2, \ldots, x^n)$$

$$\frac{\partial \phi}{\partial \tilde{x}^i} = \frac{\partial \phi}{\partial x^j}\frac{\partial x^j}{\partial \tilde{x}^i}$$

$$\Rightarrow \frac{\partial \phi}{\partial \tilde{x}^i} = \frac{\partial x^j}{\partial \tilde{x}^i}\frac{\partial \phi}{\partial x^j}; \quad i = 1, 2, \ldots, n \quad j = 1, 2, \ldots, n$$

Outer product

The outer product of the tensors

$$S = S^{i_1 i_2 \cdots i_p}_{j_1 j_2 \cdots j_q} \quad \text{and} \quad T = T^{k_1 k_2 \cdots k_r}_{l_1 l_2 \cdots l_s}$$

Is the tensor

$$ST = S^{i_1 i_2 \cdots i_p}_{j_1 j_2 \cdots j_q}\cdot T^{k_1 k_2 \cdots k_r}_{l_1 l_2 \cdots l_s}$$

Which is of order

$$p + q + r + s$$

i.e., the sum of orders of S and T. contravariant of order $p+r$ and covariant of order $q+s$.

Question: Find the outer product of two tensors S^i_j and T_k.

Solution: Let $S^i_j T_k = P^i_{jk}$

$$\tilde{P}^i_{jk} = S^r_s \frac{\partial \tilde{x}^i}{\partial x^r} \frac{\partial x^s}{\partial \tilde{x}^j} \cdot T_\mu \frac{\partial x^\mu}{\partial \tilde{x}^k}$$

$$\tilde{P}^i_{jk} = P^r_{s\mu} \frac{\partial \tilde{x}^i}{\partial x^r} \frac{\partial x^s}{\partial \tilde{x}^j} \frac{\partial x^\mu}{\partial \tilde{x}^k}$$

Question: From the tensors $S = S^{ij}$ and $T = T_{klm}$ form the inner product $U^j_{km} = S^{uj} T_{kum}$ and show that it is contravariant of order 1 and covariant of order 2.

Solution: Consider,

$$U^j_{km} = S^{uj} T_{kum}$$

$$\Rightarrow \tilde{U}^j_{km} = \tilde{S}^{uj} \tilde{T}_{kum}$$

Question: Show that the product of A^i and B_j is a mixed tensor of rank two.

Solution:

$$\tilde{A}^i = \frac{\partial \tilde{x}^i}{\partial x^k} A^k$$

and $$\tilde{B}_j = \frac{\partial x^l}{\partial \tilde{x}^j} B_l$$

$$\Rightarrow \tilde{A}^i \tilde{B}_j = \frac{\partial \tilde{x}^i}{\partial x^k} \frac{\partial x^l}{\partial \tilde{x}^j} A^k B_l$$

$$\Rightarrow \tilde{P}^i_j = \frac{\partial \tilde{x}^i}{\partial x^k} \frac{\partial x^l}{\partial \tilde{x}^j} P^k_l$$

$$\Rightarrow \tilde{U}^j_{km} = S^{pr} \frac{\partial \tilde{x}^u}{\partial x^p} \frac{\partial \tilde{x}^j}{\partial x^v} T_{sqt} \frac{\partial x^s}{\partial \tilde{x}^k} \frac{\partial x^q}{\partial \tilde{x}^u} \frac{\partial x^t}{\partial \tilde{x}^m}$$

$$= S^{pr} T_{sqt} \frac{\partial \tilde{x}^j}{\partial x^r} \frac{\partial x^s}{\partial \tilde{x}^k} \frac{\partial x^t}{\partial \tilde{x}^m} \frac{\partial x^q}{\partial x^p}$$

$$\tilde{U}^j_{km} = S^{pr} T_{sqt} \frac{\partial \tilde{x}^j}{\partial x^r} \frac{\partial x^s}{\partial \tilde{x}^k} \frac{\partial x^t}{\partial \tilde{x}^m} \delta^q_p$$

$$\Rightarrow \tilde{U}^j_{km} = U^r_{st} \frac{\partial \tilde{x}^j}{\partial x^r} \frac{\partial x^s}{\partial \tilde{x}^k} \frac{\partial x^t}{\partial \tilde{x}^m}$$

Question: Prove that if g_{ij} is a covariant tensor of order two and v^i and v^i are contravariant vectors, then the double inner product $g_{ij} v^i v^j$ is an invariant.

Solution: Consider the transformation laws

$$\tilde{g}_{ij} = g_{rs} \frac{\partial x^r}{\partial \tilde{x}^i} \frac{\partial x^s}{\partial \tilde{x}^j}$$

$$\tilde{v}^i = v^t \frac{\partial \tilde{x}^i}{\partial x^t}, \quad v^j = v^\mu \frac{\partial \tilde{x}^j}{\partial x^\mu}$$

Now take the product:

$$\tilde{g}_{ij} \tilde{v}^i \tilde{v}^j = g_{rs} v^t v^\mu \frac{\partial x^r}{\partial \tilde{x}^i} \frac{\partial x^s}{\partial \tilde{x}^j} \frac{\partial \tilde{x}^i}{\partial x^t} \frac{\partial \tilde{x}^j}{\partial x^\mu}$$

$$= g_{rs} v^t v^\mu \delta^r_t \delta^s_\mu$$

$$\tilde{g}_{ij} \tilde{v}^i \tilde{v}^j = g_{rs} v^r v^s$$

$$= g_{ij} v^i v^j$$

$\Rightarrow g_{ij} v^i v^j$ is invariant.

Inner product

To take an inner product of two tensors, one equates an upper index of one tensor to a lower index of the other, and then apply sum over the repeated dummy index. In effect, the contravariant and covariant behaviors cancel out, which lowers the total order of the tensor product.

$$ST = S^{i_1 \cdots u \cdots i_p}_{j_1 j_2 \cdots j_q} T^{k_1 k_2 \cdots k_r}_{l_1 \cdots u \cdots l_s}$$

It can be observed that there will exist $ps + rq$ inner products, in general, all of these will be distinct. Each will be a tensor of order

$$m = p + q + r + s - 2$$

Question: If A^i and B_i are two tensors then $A^i B_i$ is scalar or invariant.

Solution:

$$\tilde{A}^i = \frac{\partial \tilde{x}^i}{\partial x^p} A^p \quad \text{and} \quad \tilde{B}_i = \frac{\partial x^q}{\partial \tilde{x}^i} B_q$$

$$\Rightarrow \tilde{A}^i \tilde{B}_i = \frac{\partial \tilde{x}^i}{\partial x^p} \frac{\partial x^q}{\partial \tilde{x}^i} A^p B_q$$

$$= \delta^q_p A^p B_q$$

$\Rightarrow \tilde{A}^i \tilde{B}_i = A^p B_p = A^i B_i \Rightarrow$ It is invariant.

Contraction

This is another order reducing operation like the inner product but is applied to single tensors. In contraction of a tensor we equate its one contravariant index and one covariant index. For example, in the tensor

$$S = S^{i_1 i_2 \cdots i_\alpha \cdots i_p}_{j_1 j_2 \cdots j_\beta \cdots j_q}$$

we equate $i_\alpha = \mu = j_\beta$, then resulting tensor is

$$S' = S^{i_1 i_2 \cdots \mu \cdots i_p}_{j_1 j_2 \cdots \mu \cdots j_q}$$

this is called a contraction of S with indices i_α and j_β.

$\Rightarrow S'$ is contravariant of order $p-1$ and covariant of order $q-1$.

Question: How many tensors may be created by repeated contraction of the tensor $T = T^{ij}_{kl}$?

Solution: Single contraction produces the following four mixed tensors:

1) for $i=k=v$: T^{vj}_{vl}

2) for $j=k=v$: T^{iv}_{vl}

3) for $i=l=v$: T^{vj}_{kv}

4) for $j=l=v$: T^{iv}_{kv}

$$\tilde{T}^{i}_{ik} = T^{r}_{ts}\frac{\partial \tilde{x}^{i}}{\partial x^{r}}\frac{\partial x^{t}}{\partial \tilde{x}^{i}}\frac{\partial x^{s}}{\partial \tilde{x}^{k}}$$

$$\tilde{T}^{i}_{ik} = T^{r}_{ts}\frac{\partial x^{s}}{\partial \tilde{x}^{k}}\delta^{t}_{r} = T^{r}_{rs}\frac{\partial x^{s}}{\partial \tilde{x}^{k}}$$

Similarly contracting: $i=k$

$$\Rightarrow \tilde{T}^{i}_{ji} = T^{r}_{ts}\frac{\partial \tilde{x}^{i}}{\partial x^{r}}\frac{\partial x^{t}}{\partial \tilde{x}^{j}}\frac{\partial x^{s}}{\partial \tilde{x}^{i}}$$

$$\tilde{T}^{i}_{ji} = T^{r}_{ts}\frac{\partial x^{t}}{\partial x^{j}}\delta^{s}_{r} = T^{r}_{tr}\frac{\partial x^{t}}{\partial x^{j}}$$

$\Rightarrow$ Both contractions yield a covariant vector.

a) Here $S = S^{ij}_{k}$ and $T = T^{l}_{m}$

consider outer product:

$ST = S^{ij}_{k}T^{l}_{m} = U^{ijl}_{km} = U$ (say)

then contravariant vectors can be obtained as the double contractions of $U = U^{ijl}_{km}$:

U^{uvk}_{uv}, U^{uvK}_{vu}, U^{ujv}_{uv}, U^{ujv}_{vu},

U^{iuv}_{uv}, U^{iuv}_{vu}

and double contraction produces the two zero-order tensors:

5) In first contractions 1 and 4 equating the indices we have:

T^{uv}_{uv}

6) and from contractions 2 and 3 we have T^{uv}_{vu}

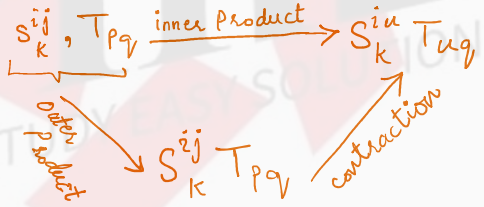
$\Rightarrow$ there are Six distinct tensors are formed by contractions.

Combined operations

We can form new tensors from the old ones in a variety of ways by performing different tensor operations discussed above.

One might form the outer product of two tensors, then take an inner product of this with a third tensor, or contract on one or more pair of indices either before or after taking a product.

It is noteworthy to mention that an inner product of two tensors may be characterized as a contraction of their outer product.



b) The vector

$U^{uvk}_{uv} = S^{uv}_{u}T^{k}_{v}$

may be obtained by first taking the inner product $S^{iv}_{l}T^{k}_{v}$ and then contracting on $i=u=l$.

Similar can be shown for the other vectors of part (a).

Question: Show that any contraction of the tensor $T = T^{i}_{jk}$ results in a covariant vector.

Solution: Consider the transformation law:

$$\tilde{T}^{i}_{jk} = T^{r}_{ts}\frac{\partial \tilde{x}^{i}}{\partial x^{r}}\frac{\partial x^{t}}{\partial \tilde{x}^{j}}\frac{\partial x^{s}}{\partial \tilde{x}^{k}}$$

Contracting the indices: $i=j$

Question: Suppose that $S = S^{ij}_{k}$ and $T = T^{i}_{j}$ are tensors from which a contravariant vector V^{i} is to be constructed using a combination of outer/inner products and contractions.

a) Show that there are Six distinct possibilities for V^{i}.

b) Show that each possible V^{i} is obtainable as a contraction of an inner product ST.

Quotient Law

By this law, we can test the given quantity is a tensor or not. Suppose given quantity be A and we do not know that A is a tensor or not. To test A, we take inner product of A with an arbitrary tensor, if this inner product is a tensor then A is also a tensor.

Statement

If the inner product of a set of functions with an arbitrary tensor is a tensor then these set of functions are components of a tensor.

Question: Show that the expression $A(i,j,k)B^k$ is a covariant tensor of rank three if $A(i,j,k)B^k$ is a covariant tensor of rank two and B^k is contravariant vector.

Solution: Given $A(i,j,k)B^k$ is a covariant tensor of rank two then,

$$\tilde{A}(i,j,k)\tilde{B}^k = \frac{\partial x^l}{\partial \tilde{x}^i}\frac{\partial x^q}{\partial \tilde{x}^j} A(p,q,r) B^r$$

Since B^k is contravariant vector

$$\tilde{B}^k = \frac{\partial \tilde{x}^k}{\partial x^r} B^r \quad \text{or} \quad B^r = \frac{\partial x^r}{\partial \tilde{x}^k}\tilde{B}^k$$

$$\Rightarrow \tilde{A}(i,j,k)\tilde{B}^k = \frac{\partial x^l}{\partial \tilde{x}^i}\frac{\partial x^q}{\partial \tilde{x}^j} A(p,q,r)\frac{\partial x^r}{\partial \tilde{x}^k}\tilde{B}^k$$

$$\Rightarrow \tilde{A}(i,j,k) = \frac{\partial x^l}{\partial \tilde{x}^i}\frac{\partial x^q}{\partial \tilde{x}^j}\frac{\partial x^r}{\partial \tilde{x}^k} A(p,q,r)$$

$\Rightarrow A(i,j,k)$ is a covariant tensor of rank three.

Similarly: $B^q = \frac{\partial x^q}{\partial \tilde{x}^j}\tilde{B}^j$

$$C_r = \frac{\partial \tilde{x}^k}{\partial x^r}\tilde{C}_k$$

$\Rightarrow$

$$\tilde{A}(i,j,k)\tilde{A}^i\tilde{B}^j\tilde{C}_k = A(p,q,r)\frac{\partial x^l}{\partial \tilde{x}^i}\frac{\partial x^q}{\partial \tilde{x}^j}\frac{\partial x^k}{\partial x^r}\tilde{A}^i\tilde{B}^j\tilde{C}_k$$

$$\Rightarrow \tilde{A}(i,j,k) = A(p,q,r)\frac{\partial x^l}{\partial \tilde{x}^i}\frac{\partial x^q}{\partial \tilde{x}^j}\frac{\partial \tilde{x}^k}{\partial x^r}$$

$\Rightarrow A(i,j,k)$ is a tensor of type $(1,2)$.

Question: Prove that if T^i_{jkl} is a tensor in x^i-system and $T^i_{jkl} = 3T^i_{ljk}$. Then $\tilde{T}^i_{jkl} = 3\tilde{T}^i_{ljk}$ in all coordinate systems.

Solution:

Here we have to prove that

$$\tilde{T}^i_{jkl} = 3\tilde{T}^i_{ljk} \text{ in } \tilde{x}^i\text{-system.}$$

Tensor Equations

Much of the importance of tensors in mathematical physics and engineering resides in the fact that if a tensor equation or identity is true in one coordinate system, then it is true in all coordinate systems.

Example: Suppose that in x^i-System the covariant tensor $T = T_{ij}$ vanishes. The components of T in any other system say, $\tilde{x}^i$-System are: if $T_{ij} = 0$

$$\tilde{T}_{ij} = T_{rs}\frac{\partial x^r}{\partial \tilde{x}^i}\frac{\partial x^s}{\partial \tilde{x}^j} = 0 \Rightarrow \tilde{T}_{ij} = 0$$

Therefore $T = 0$ in every coordinate system.

$\tilde{T}^i_{jkl} = 3\tilde{T}^i_{ljk}$ means: $\tilde{T}^i_{jkl} - 3\tilde{T}^i_{ljk} = 0$

So consider:

$$\tilde{T}^i_{jkl} - 3\tilde{T}^i_{ljk} = T^p_{rst}\frac{\partial \tilde{x}^i}{\partial x^p}\frac{\partial x^r}{\partial \tilde{x}^j}\frac{\partial x^s}{\partial \tilde{x}^k}\frac{\partial x^t}{\partial \tilde{x}^l} - 3T^p_{rst}\frac{\partial \tilde{x}^i}{\partial x^p}\frac{\partial x^r}{\partial \tilde{x}^l}\frac{\partial x^s}{\partial \tilde{x}^j}\frac{\partial x^t}{\partial \tilde{x}^k}$$

$$= T^p_{rst}\frac{\partial \tilde{x}^i}{\partial x^p}\frac{\partial x^r}{\partial \tilde{x}^i}\frac{\partial x^s}{\partial \tilde{x}^k}\frac{\partial x^t}{\partial \tilde{x}^l}\begin{cases} r \to t \\ s \to r \\ t \to s \end{cases} - 3T^p_{tsr}\frac{\partial \tilde{x}^i}{\partial x^p}\frac{\partial x^t}{\partial \tilde{x}^l}\frac{\partial x^r}{\partial \tilde{x}^j}\frac{\partial x^s}{\partial \tilde{x}^k}$$

$$\Rightarrow \tilde{T}^i_{jkl} - 3\tilde{T}^i_{ljk} = (T^p_{rst} - 3T^p_{trs})\frac{\partial \tilde{x}^i}{\partial x^p}\frac{\partial x^r}{\partial \tilde{x}^j}\frac{\partial x^s}{\partial \tilde{x}^k}\frac{\partial x^t}{\partial \tilde{x}^l}$$

but $T^i_{jkl} = 3T^i_{ljk}$ in x^i-system

$$\Rightarrow \tilde{T}^i_{jkl} - 3\tilde{T}^i_{ljk} = 0$$

$\Rightarrow \tilde{T}^i_{jkl} = 3\tilde{T}^i_{ljk}$ in every coordinate system.

Question: If $A(i,j,k)A^iB^jC_k$ is a scalar for arbitrary vectors $A^iB^jC_k$ then show that $A(i,j,k)$ is a tensor of type $(1,2)$.

Solution: As $A(i,j,k)A^iB^jC_k$ is a scalar: then

$$\tilde{A}(i,j,k)\tilde{A}^i\tilde{B}^j\tilde{C}_k = A(p,q,r)A^pB^qC_r$$

Also A^i, B^j and C_k are vectors. Then

$$\tilde{A}_i = \frac{\partial \tilde{x}^i}{\partial x^p}A^p \quad \text{or} \quad A^p = \frac{\partial x^p}{\partial \tilde{x}^i}\tilde{A}_i$$

Example: Consider an equation: $T_{ij} = R_{ijk}U^k - AW^{kl}_i M_{jk} I_l = 0 \Rightarrow \tilde{T}_{ij} = 0$

$$R_{ijk}U^k = AW^{kl}_i M_{jk} I_l$$

which connect six entities that may or may not be tensors. Then if it can be shown that (i) $T_{ij} = R_{ijk}U^k - AW^{kl}_i M_{jk} I_l$ is a tensor and (ii) there exist a coordinate system in which all T_{ij} are zero. Then the given tensor equation is valid in every coordinate system.

Question: Prove that if a mixed tensor T^i_j can be expressed as the outer product of contravariant and covariant vectors U^i and V_j in one coordinate system then T^i_j is the outer product of those vectors in general.

Solution: As $T^i_j = U^i V_j$ in x^i-System (Say) and we have to show that

$$\tilde{T}^i_j = \tilde{U}^i \tilde{V}_j \text{ in any } \tilde{x}^i\text{-System.}$$

Consider:

$$\tilde{T}^i_j - \tilde{U}^i \tilde{V}_j = T^r_s \frac{\partial \tilde{x}^i}{\partial x^r}\frac{\partial x^s}{\partial \tilde{x}^j} - U^r \frac{\partial \tilde{x}^i}{\partial x^r} V_s \frac{\partial x^s}{\partial \tilde{x}^j}$$

$$= \left(T^r_s - U^r V_s\right)\left(\frac{\partial \tilde{x}^i}{\partial x^r}\frac{\partial x^s}{\partial \tilde{x}^j}\right)$$

$$\Rightarrow \tilde{T}^i_j - \tilde{U}^i \tilde{V}_j = 0 \Rightarrow \tilde{T}^i_j = \tilde{U}^i \tilde{V}_j \text{ as required.}$$

Question: A Symmetric tensor of rank two has only $\frac{1}{2}n(n+1)$ different components in an n-dimensional space.

Solution: Let A^{ij} be a Symmetric tensor of rank two, i.e., $A^{ij} = A^{ji}$. In matrix form:

$$A = [A^{ij}] = \begin{bmatrix} A^{11} & A^{12} & A^{13} & \cdots & A^{1n} \\ A^{21} & A^{22} & A^{23} & \cdots & A^{2n} \\ \vdots & \vdots & \vdots & & \vdots \\ A^{n1} & A^{n2} & A^{n3} & \cdots & A^{nn} \end{bmatrix}$$

$\Rightarrow A^{ij}$ will have n^2-components. Out of these n^2-Components n are diagonal, i.e., $A^{11}, A^{22}, \ldots, A^{nn}$. In remaining $n^2 - n$ components: $A^{12} = A^{21}$, $A^{23} = A^{32}$, and so on. due to Symmetry. So, the distinct components are $\frac{1}{2}(n^2 - n)$.

Hence the total number of different components are $= n + \frac{1}{2}(n^2 - n) = \frac{2n - n^2 - n}{2}$

$$= \frac{1}{2}(n + n^2) = \frac{1}{2}n(n+1)$$

Question: A skew symmetric tensor of second order has only $\frac{1}{2}n(n-1)$ different non-zero components.

Solution: Let A^{ij} be a skew Symmetric tensor. Then $A^{ij} = -A^{ji}$ when $i = j$ (i.e., for diagonal elements)

$$A^{ii} = -A^{ii} \Rightarrow 2A^{ii} = 0 \Rightarrow A^{ii} = 0$$

$\Rightarrow$ diagonal elements of a second order skew symmetric tensor are Zero.

Hence

$$A = [A^{ij}] = \begin{bmatrix} 0 & A^{12} & A^{13} & \cdots & A^{1n} \\ A^{21} & 0 & A^{23} & \cdots & A^{2n} \\ \vdots & \vdots & \vdots & & \vdots \\ A^{n1} & A^{n2} & A^{n3} & \cdots & A^{nn} = 0 \end{bmatrix}$$

Now A^{ij} has n^2 components out of which n-diagonal components are Zero. So we are left with $n^2 - n$ components. Also $A^{12} = -A^{21}$, $A^{13} = -A^{31}$ and so on.

$\Rightarrow$ The different or distinct components of A^{ij} are

$$\frac{1}{2}(n^2 - n) \quad \text{or} \quad \frac{1}{2}n(n-1)$$

Symmetric Tensors

A tensor is said to symmetric with respect to two contravariant or two covariant indices if its components remain unchanged on an interchange of the two indices.

For example: (1) The tensor A^{ij} is symmetric if: $A^{ij} = A^{ji}$

(2) The tensor A^{ijk}_{lm} is symmetric w.r.t indices i and j if: $A^{ijk}_{lm} = A^{jik}_{lm}$

and w.r.t indices l and m if:

$$A^{ijk}_{lm} = A^{ijk}_{ml}$$

Skew symmetric tensor

A tensor is said to be skew symmetric with respect to two contravariant or two covariant indices if its components change sign on interchange of the two indices.

For example: (1) The tensor A^{ij} is skew-Symmetric if $A^{ij} = -A^{ji}$

(2) The tensor A^{ijk}_{lm} is skew-Symmetric w.r.t the indices j and k if:

$$A^{ijk}_{lm} = -A^{ikj}_{lm}$$

Question: A contravariant or covariant tensor of rank two say A_{ij} can always be written as the sum of a symmetric and skew symmetric tensor.

Solution: consider a covariant tensor

$$A_{ij} = \frac{1}{2}(A_{ij} + A_{ji}) + \frac{1}{2}(A_{ij} - A_{ji})$$

$$\Rightarrow A_{ij} = S_{ij} + T_{ij}$$

$$\Rightarrow \phi = b_{jk} A^j A^k$$

where $b_{jk} = \frac{1}{2}(a_{jk} + a_{kj})$

and $b_{kj} = \frac{1}{2}(a_{kj} + a_{jk}) = b_{jk}$

$\Rightarrow b_{jk}$ is symmetric.

where $S_{ij} = \frac{1}{2}(A_{ij} + A_{ji})$

and $T_{ij} = \frac{1}{2}(A_{ij} - A_{ji})$

Now, $S_{ji} = \frac{1}{2}(A_{ji} + A_{ij}) = S_{ij}$

$\Rightarrow S_{ij}$ is symmetric tensor.

and $T_{ji} = \frac{1}{2}(A_{ji} - A_{ij}) = -\frac{1}{2}(A_{ij} - A_{ji})$

$\Rightarrow T_{ji} = -T_{ij}$

$\Rightarrow T_{ij}$ is skew symmetric tensor.

Question: If T_i be the component of a covariant vector show that $\left(\frac{\partial T_i}{\partial x^j} - \frac{\partial T_j}{\partial x^i}\right)$ are components of a skew symmetric tensor of rank 2.

Solution: As T_i is a covariant tensor then

$$\bar{T}_i = \frac{\partial x^k}{\partial \bar{x}^i} T_k$$

differentiating it w.r.t $\bar{x}^j$ partially,

$$\frac{\partial \bar{T}_i}{\partial \bar{x}^j} = \frac{\partial}{\partial \bar{x}^j}\left(\frac{\partial x^k}{\partial \bar{x}^i} T_k\right)$$

$$= \frac{\partial^2 x^k}{\partial \bar{x}^j \partial \bar{x}^i} T_k + \frac{\partial x^k}{\partial \bar{x}^i}\frac{\partial T_k}{\partial \bar{x}^j}$$

$$\Rightarrow \frac{\partial \bar{T}_i}{\partial \bar{x}^j} = \frac{\partial^2 x^k}{\partial \bar{x}^j \partial \bar{x}^i} T_k + \frac{\partial x^k}{\partial \bar{x}^i}\frac{\partial x^l}{\partial \bar{x}^j}\frac{\partial T_k}{\partial x^l} \rightarrow (1)$$

and Similarly:

$$\frac{\partial \bar{T}_j}{\partial \bar{x}^i} = \frac{\partial^2 x^k}{\partial \bar{x}^i \partial \bar{x}^j} T_k + \frac{\partial x^k}{\partial \bar{x}^i}\frac{\partial x^l}{\partial \bar{x}^j}\frac{\partial T_k}{\partial x^l}$$

interchanging the dummy indices k and l in second term:

$$\frac{\partial \bar{T}_j}{\partial \bar{x}^i} = \frac{\partial^2 x^k}{\partial \bar{x}^i \partial \bar{x}^j} T_k + \frac{\partial x^k}{\partial \bar{x}^i}\frac{\partial x^l}{\partial \bar{x}^j}\frac{\partial T_k}{\partial \bar{x}^l} \rightarrow (2)$$

now Substracting (1) and (2), we have

$$\frac{\partial \bar{T}_i}{\partial \bar{x}^j} - \frac{\partial \bar{T}_j}{\partial \bar{x}^i} = \frac{\partial x^k}{\partial \bar{x}^i}\frac{\partial x^l}{\partial \bar{x}^j}\left(\frac{\partial T_k}{\partial x^l} - \frac{\partial T_l}{\partial x^k}\right)$$

$\Rightarrow \frac{\partial T_i}{\partial x^j} - \frac{\partial T_j}{\partial x^i}$ is a second rank tensor.

Now consider:

$$A_{ij} = \frac{\partial T_i}{\partial x^j} - \frac{\partial T_j}{\partial x^i}$$

$$\Rightarrow A_{ji} = \frac{\partial T_j}{\partial x^i} - \frac{\partial T_i}{\partial x^j}$$

$$= -\left(\frac{\partial T_i}{\partial x^j} - \frac{\partial T_j}{\partial x^i}\right) = -A_{ij}$$

or $A_{ij} = -A_{ji}$

$\Rightarrow A_{ij}$ is Skew Symmetric tensor of rank two.

Question: If $\phi = a_{jk} A^j A^k$. Show that we can always write $\phi = b_{jk} A^j A^k$ where b_{jk} is Symmetric.

Solution: As $\phi = a_{jk} A^j A^k$

Interchange the indices j and k

$$\phi = a_{kj} A^k A^j$$

and adding both expressions

$$2\phi = (a_{jk} + a_{kj}) A^j A^k$$

$$\Rightarrow \phi = \frac{1}{2}(a_{jk} + a_{kj}) A^j A^k$$

Question: If the tensors a_{ij} and g_{ij} are symmetric and u^i, v^i are components of contravariant vectors satisfying the equations

$$(a_{ij} - k g_{ij}) u^i = 0, \quad i, j = 1, 2, \ldots, n$$

$$(a_{ij} - k' g_{ij}) v^i = 0, \quad k \neq k'$$

Prove that $g_{ij} u^i v^j = 0$

$$a_{ij} u^i v^j = 0$$

Solution: consider the given equations:

$$(a_{ij} - k g_{ij}) u^i = 0 \longrightarrow (1)$$

$$(a_{ij} - k' g_{ij}) v^i = 0 \longrightarrow (2)$$

Multiplying (1) by v^j and (2) by u^j:

$$(a_{ij} - k g_{ij}) u^i v^j = 0 \longrightarrow (1)'$$

$$(a_{ij} - k' g_{ij}) v^i u^j = 0 \longrightarrow (2)'$$

Subtracting (1)' and (2)':

$$a_{ij} u^i v^j - a_{ij} v^i u^j - k g_{ij} u^i v^j + k' g_{ij} u^j v^i = 0$$

Interchanging i and j in the second and fourth terms:

$$a_{ij} u^i v^j - a_{ji} u^i v^j - k g_{ij} u^i v^j + k' g_{ji} u^i v^j = 0$$

$$\Rightarrow (a_{ij} - a_{ji}) u^i v^j - (k g_{ij} - k' g_{ji}) u^i v^j = 0$$

also, $a_{ij} = a_{ji}$, $g_{ij} = g_{ji}$ and $k \neq k'$

$$\Rightarrow 0 - k g_{ij} u^i v^j + k' g_{ij} u^i v^j = 0$$

$$\Rightarrow (k' - k) g_{ij} u^i v^j = 0 \qquad k' - k \neq 0$$

$$\Rightarrow g_{ij} u^i v^j = 0$$

Now multiplying (1) by v^j:

$$a_{ij} v^j u^i - k g_{ij} u^i v^j = 0$$

$$\Rightarrow a_{ij} u^i v^j = 0$$

Question: If a_{ij} is symmetric tensor and b_i is a vector and

$$a_{ij} b_k + a_{jk} b_i + a_{ki} b_j = 0$$

then prove that $a_{ij} = 0$ or $b_k = 0$.

Solution: Consider:

$$a_{ij} b_k + a_{jk} b_i + a_{ki} b_j = 0$$

$$\Rightarrow \bar{a}_{ij} \bar{b}_k + \bar{a}_{jk} \bar{b}_i + \bar{a}_{ki} \bar{b}_j = 0$$

using transformation laws we have:

$$a_{pq} \frac{\partial x^p}{\partial \bar{x}^i} \frac{\partial x^q}{\partial \bar{x}^j} b_r \frac{\partial x^r}{\partial \bar{x}^k} + a_{pq} \frac{\partial x^p}{\partial \bar{x}^j} \frac{\partial x^q}{\partial \bar{x}^k} b_r \frac{\partial x^r}{\partial \bar{x}^i} + a_{pq} \frac{\partial x^p}{\partial \bar{x}^k} \frac{\partial x^q}{\partial \bar{x}^i} b_r \frac{\partial x^r}{\partial \bar{x}^j} = 0$$

$$\Rightarrow a_{pq} b_r \left[\frac{\partial x^p}{\partial \bar{x}^i} \frac{\partial x^q}{\partial \bar{x}^j} \frac{\partial x^r}{\partial \bar{x}^k} + \frac{\partial x^p}{\partial \bar{x}^j} \frac{\partial x^q}{\partial \bar{x}^k} \frac{\partial x^r}{\partial \bar{x}^i} + \frac{\partial x^p}{\partial \bar{x}^k} \frac{\partial x^q}{\partial \bar{x}^i} \frac{\partial x^r}{\partial \bar{x}^j} \right] = 0$$

$$\Rightarrow a_{pq} b_r = 0 \Rightarrow a_{pq} = 0 \text{ or } b_r = 0$$

$$\Rightarrow a_{ij} = 0 \text{ or } b_k = 0$$

Question: If A_{ij} is a skew symmetric tensor then $(B^i_j B^k_l + B^i_l B^k_j) A_{ik} = 0$

Solution: As A_{ij} is skew symmetric

$$\Rightarrow A_{ij} = -A_{ji}$$

Consider,

$$T = (B^i_j B^k_l + B^i_l B^k_j) A_{ik}$$

$$= B^i_j B^k_l A_{ik} + B^i_l B^k_j A_{ik}$$

$$= B^i_j B^k_l A_{ik} + B^k_l B^i_j A_{ki}$$

$$= B^i_j B^k_l A_{ik} - B^k_l B^i_j A_{ik} = 0 \text{ proved.}$$

Question: If A_{ij} is a skew symmetric tensor prove that:

$$(\delta^i_j \delta^k_l + \delta^i_l \delta^k_j) A_{ik} = 0$$

Solution: Given A_{ij} is a skew symmetric tensor then $A_{ij} = -A_{ji}$. Consider

$$(\delta^i_j \delta^k_l + \delta^i_l \delta^k_j) A_{ik} = \delta^i_j \delta^k_l A_{ik} + \delta^i_l \delta^k_j A_{ik}$$

$$= \delta^i_j A_{il} + \delta^i_l A_{ij}$$

$$= A_{jl} + A_{lj}$$

$$= A_{jl} - A_{jl} = 0$$

Question: If $P_{ij} \neq 0$ are components of a tensor and if the equation $f P_{ij} + g P_{ji} = 0$ holds where f and g are non-zero scalars then prove that either $f = g$ and P_{ij} is skew symmetric or $f = -g$ and P_{ij} is symmetric.

Solution: As $f P_{ij} + g P_{ji} = 0$

$$\Rightarrow f P_{ij} = -g P_{ji}$$

$$\Rightarrow f^2 P_{ij} = -fg P_{ji} = -g(f P_{ji})$$

$$= g(g P_{ij})$$

$$f^2 P_{ij} = g^2 P_{ij}$$

$$\Rightarrow (f^2 - g^2) P_{ij} = 0$$

$\Rightarrow P_{ij}=0$ or $f^2-g^2=0$

but $P_{ij}\neq 0$

$\Rightarrow f^2-g^2=0 \Rightarrow f=\pm g$

Case-I: when $f=g$

then $f P_{ij} = -g P_{ji}$

$\Rightarrow f P_{ij} = -f P_{ji}$

$\Rightarrow P_{ij} = -P_{ji}$

$\Rightarrow P_{ij}$ is skew symmetric.

Case II: when $f=-g$

$\Rightarrow f P_{ij} = f P_{ji}$

$\Rightarrow P_{ij} = P_{ji}$

$\Rightarrow P_{ij}$ is Symmetric.

Conjugate symmetric tensor:

Consider a covariant symmetric tensor A_{ij} of rank two. Let d denote the determinant of this tensor i.e., $d=|A_{ij}|$ and $d\neq 0$. Then define A^{ij} by

$$A^{ij}=\frac{\text{Cofactor of } A_{ij} \text{ in the determinant } |A_{ij}|}{d}$$

'A^{ij}' is a contravariant symmetric tensor of rank two which is called conjugate OR reciprocal tensor of A_{ij}.

Question: Find the conjugate or reciprocal of the tensor

$$[A_{ij}]=\begin{bmatrix}1 & x & y\\ x & 1 & 0\\ y & 0 & -1\end{bmatrix}$$

Solution: The conjugate of A_{ij} is:

$$A^{ij}=\frac{\text{Cofactor of } A_{ij} \text{ in } |A_{ij}|}{d}$$

where $d=|A_{ij}|$

First find $|A_{ij}|=d$.

$$d=\begin{vmatrix}1 & x & y\\ x & 1 & 0\\ y & 0 & -1\end{vmatrix}$$

Expanding the determinant by first row:

$$d=1\begin{vmatrix}1 & 0\\ 0 & -1\end{vmatrix}-x\begin{vmatrix}x & 0\\ y & -1\end{vmatrix}+y\begin{vmatrix}x & 1\\ y & 0\end{vmatrix}$$

$$=1(-1)-x(-x)+y(-y)$$

$$d=-1+x^2+y^2$$

$$d=x^2+y^2-1$$

So

$$A^{11}=\frac{\text{cof. of } A_{11}}{d}=(-1)^2\frac{\begin{vmatrix}1 & 0\\ 0 & -1\end{vmatrix}}{d}=\frac{-1}{d}$$

$$A^{12}=\frac{\text{cof. of } A_{12}}{d}=(-1)^3\frac{\begin{vmatrix}x & 0\\ y & -1\end{vmatrix}}{d}=\frac{x}{d}$$

$$A^{13}=\frac{\text{cof. of } A_{13}}{d}=(-1)^4\frac{\begin{vmatrix}x & 1\\ y & 0\end{vmatrix}}{d}=\frac{-y}{d}$$

$$A^{21}=\frac{\text{cof. of } A_{21}}{d}=(-1)^3\frac{\begin{vmatrix}x & y\\ 0 & -1\end{vmatrix}}{d}=\frac{x}{d}$$

$$A^{22}=\frac{\text{cof. of } A_{22}}{d}=(-1)^4\frac{\begin{vmatrix}1 & y\\ y & -1\end{vmatrix}}{d}=\frac{-1-y^2}{d}$$

$$A^{23}=\frac{\text{cof. of } A_{23}}{d}=(-1)^5\frac{\begin{vmatrix}1 & x\\ y & 0\end{vmatrix}}{d}=\frac{xy}{d}$$

$$A^{31}=\frac{\text{cof. of } A_{31}}{d}=(-1)^4\frac{\begin{vmatrix}x & y\\ 1 & 0\end{vmatrix}}{d}=\frac{-y}{d}$$

$$A^{32}=\frac{\text{cof. of } A_{32}}{d}=(-1)^5\frac{\begin{vmatrix}1 & y\\ x & 0\end{vmatrix}}{d}=\frac{xy}{d}$$

$$A^{33}=\frac{\text{cof. of } A_{33}}{d}=(-1)^6\frac{\begin{vmatrix}1 & x\\ x & 1\end{vmatrix}}{d}=\frac{1-x^2}{d}$$

$$\Rightarrow [A^{ij}]=\begin{bmatrix}\frac{-1}{d} & \frac{x}{d} & \frac{-y}{d}\\ \frac{x}{d} & \frac{-1-y^2}{d} & \frac{xy}{d}\\ \frac{-y}{d} & \frac{xy}{d} & \frac{1-x^2}{d}\end{bmatrix}\begin{bmatrix}1 & x & y\\ x & 1 & 0\\ y & 0 & -1\end{bmatrix}$$

$$\downarrow$$

$$[A_{ij}]$$

You can check that

$$[A_{ij}][A^{ij}]=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}=I_3$$

Also if A^{ij} is a symmetric tensor.

Note: If B_{ij} is the cofactor of the element A_{ij} in the determinant $d=|A_{ij}|\neq 0$ and A^{ij} defined as:

$$A^{ij}=\frac{B_{ij}}{d}$$

Then:

$$A_{ij}A^{kj}=\delta_i^k$$

Cartesian Tensors:

A cartesian tensor is a tensor in a three dimensional Euclidean space. Unlike general tensors there is no distinction between covariant and contravariant indices for Cartesian tensors. However, tensors in non Euclidean spaces do require this distinction.

In other words, tensors defined in a geometry obeying the coordinate transformation from non tilde system to tilde system as

$\tilde{x}^i = a^i_j x^j$ with $|a^i_j| \neq 0$

and the transformation matrix a^i_j is orthogonal are called Cartesian tensors. This transformation should also be distance preserving.

Transformation laws for Cartesian tensors:

$\star J^{-1} = J^t$, $b^i_j = a^j_i$

If $J = \left[\frac{\partial \tilde{x}^i}{\partial x^j}\right]_{nn}$ and $J^{-1} = \left[\frac{\partial x^i}{\partial \tilde{x}^j}\right]$

Or $J = [a^i_j]_{nn}$ and $J^{-1} = [b^i_j]_{nn}$

the transformation laws for Cartesian tensors are,

Contra: $\tilde{T}^i = a^i_r T^r$

Cov: $\tilde{T}_i = a^i_r T_r$

} Rank one tensors

For rank two tensors:

$\tilde{T}^{ij} = a^i_r a^j_s T^{rs}$

$\tilde{T}_{ij} = a^i_r a^j_s T_{rs}$

$\tilde{T}^i_j = a^i_r a^j_s T^r_s$

Here contravariant and covariant behaviors do not distinguishable so, Consequently Cartesian tensors can be written as:

$\tilde{x}_i = a_{ij} x_j$ or $x_i = a_{ji} \tilde{x}_j$

coordinate transformations

$\tilde{T}_i = a_{ir} T_i$, $\tilde{T}_{ij} = a_{ir} a_{js} T_{rs}$

Note that $J J^t = I$ implies

$|J J^t| = |I|$

$\Rightarrow |J||J^t| = |J| = 1$

$\Rightarrow |J||J| = 1$

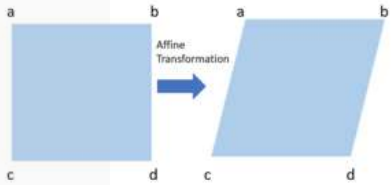
$\Rightarrow |J|^2 = 1$

$\Rightarrow |J| = \pm 1$

when $|J| = 1$, then the corresponding tensors are called direct tensors.

Affine Tensors

The adjective "affine" indicates everything that is related to the geometry of affine spaces. A coordinate system for the n-dimensional affine space is determined by any basis of n vectors, which are not necessarily orthonormal. Therefore the resulting axes are not mutually perpendicular.



Consider a transformation of the form that takes the points from a rectangular coordinate system into an affine coordinate system. Tensors defined on the class of all such affine coordinate systems are called affine tensors.

As $J = \left[\frac{\partial \tilde{x}^i}{\partial x^j}\right]_{nn} = [a^i_j]_{nn}$ and

$J^{-1} = \left[\frac{\partial x^i}{\partial \tilde{x}^j}\right]_{nn} = [b^i_j]_{nn}$

The transformation laws for affine tensors are:

$\tilde{T}^{ij} = a^i_r a^j_s T^{rs}$ → contravariant

$\tilde{T}_{ij} = b^r_i b^s_j T_{rs}$ → covariant

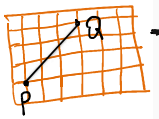
$\tilde{T}^i_j = a^i_r b^s_j T^r_s$ → Mixed

Metric Tensor: ✓

This is a rank two tensor and is also called the fundamental tensor. The main purpose of the metric tensor is to generalize the concept of distance to general curvilinear coordinate frames and maintain the invariance of distance in different coordinate systems.

Distance between 2-points:

Consider a rubber sheet



→ unstretched sheet

$ds^2 = dx^2 + dy^2$

(distance formula)

If this sheet is stretched in both x and y direction 2-times, then

$ds^2 = (2dx^2 + 2dy^2)$

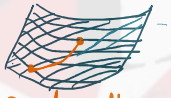
Explanation:

Every point in coordinate system has its own metric tensor. We do not care about this usually because our Euclidean coordinate system has same metric tensor everywhere.

In advance Physics and Mathematics, we have to deal with non-Euclidean Spaces with varing metric tensor at every point.

To understand this, first we discuss distance between two points in Euclidean Space.

But if we stretch it in some random direction. We cannot write the distance b/w two points using distance formula. As the rubber sheet may be like this



→ stretched in random direction.

and the corresponding coordinate system may be non-Euclidean.

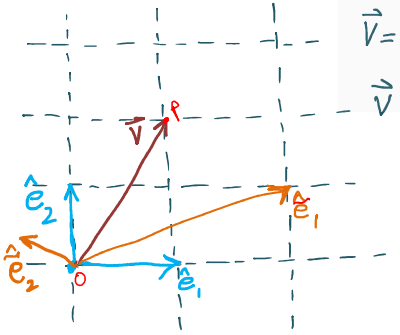
In this case the distance will be

$ds^2 = g_{11}dx^2 + (g_{12}+g_{21})dx\,dy + g_{22}dy^2$

Hence whole stretching information is captured by these coefficients $(g_{11}, g_{12}, g_{21}, g_{22})$. Furthermore, this information is stored in the form of tensor called metric tensor.

Another Explanation:

Consider a vector $\vec{V}$. Such that



$\vec{V} = V^1\hat{e}_1 + V^2\hat{e}_2$

$\vec{V} = \tilde{V}^1\tilde{\hat{e}}_1 + \tilde{V}^2\tilde{\hat{e}}_2$

Now:

$$|\vec{V}| = \vec{V}\cdot\vec{V} = (V^1\hat{e}_1 + V^2\hat{e}_2)\cdot(V^1\hat{e}_1 + V^2\hat{e}_2)$$

$$= (V^1)^2(\hat{e}_1\cdot\hat{e}_1) + 2V^1V^2(\hat{e}_1\cdot\hat{e}_2) + (V^2)^2(\hat{e}_2\cdot\hat{e}_2)$$

As system is orthonormal

$\Rightarrow |\vec{V}| = (V^1)^2 + (V^2)^2$ ✓

corresponding Metric: $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

Similarly in tild system, we have

$$|\vec{V}| = (\tilde{V}^1)^2(\tilde{\hat{e}}_1\cdot\tilde{\hat{e}}_1) + 2\tilde{V}^1\tilde{V}^2(\tilde{\hat{e}}_1\cdot\tilde{\hat{e}}_2) + (\tilde{V}^2)^2(\tilde{\hat{e}}_2\cdot\tilde{\hat{e}}_2)$$

consider

$\tilde{\hat{e}}_1 = c_1\hat{e}_1 + c_2\hat{e}_2$

$\tilde{\hat{e}}_2 = c_3\hat{e}_1 + c_4\hat{e}_2$

Implies:

$\hat{e}_1 \cdot \hat{e}_1 = c_1^2 + c_2^2 \rightarrow g_{11}$

$\hat{e}_1 \cdot \hat{e}_2 = c_1 c_3 + c_2 c_4 \rightarrow g_{12}, g_{21}$

$\hat{e}_2 \cdot \hat{e}_2 = c_3^2 + c_4^2 \rightarrow g_{22}$

$\Rightarrow |\vec{V}| = (V')^2 (c_1^2 + c_2^2) + 2V'V^2 (c_1 c_3 + c_2 c_4) + (V^2)^2 (c_3^2 + c_4^2)$

Implies in tild system the metric tensor is:

$$[g_{ij}] = \begin{bmatrix} c_1^2 + c_2^2 & c_1 c_3 + c_2 c_4 \\ c_1 c_3 + c_2 c_4 & c_3^2 + c_4^2 \end{bmatrix}$$

$$|\vec{V}| = (\vec{V})[g_{ij}](\vec{V})^t = [V' \; V^2] \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix} \begin{bmatrix} V' \\ V^2 \end{bmatrix}$$

Hence:

$$[g_{ij}] = \begin{bmatrix} \hat{e}_1 \cdot \hat{e}_1 & \hat{e}_1 \cdot \hat{e}_2 \\ \hat{e}_2 \cdot \hat{e}_1 & \hat{e}_2 \cdot \hat{e}_2 \end{bmatrix}$$

$$[\tilde{g}_{ij}] = \begin{bmatrix} \hat{\tilde{e}}_1 \cdot \hat{\tilde{e}}_1 & \hat{\tilde{e}}_1 \cdot \hat{\tilde{e}}_2 \\ \hat{\tilde{e}}_2 \cdot \hat{\tilde{e}}_1 & \hat{\tilde{e}}_2 \cdot \hat{\tilde{e}}_2 \end{bmatrix}$$

$\Rightarrow g_{ij} = \hat{e}_i \cdot \hat{e}_j$ and $\tilde{g}_{ij} = \hat{\tilde{e}}_i \cdot \hat{\tilde{e}}_j$

As $\tilde{g}_{ij} = \hat{\tilde{e}}_i \cdot \hat{\tilde{e}}_j$

$$\tilde{g}_{ij} = \frac{\partial x^k}{\partial \tilde{x}^i} \hat{e}_k \cdot \frac{\partial x^l}{\partial \tilde{x}^j} \hat{e}_l = \frac{\partial x^k}{\partial \tilde{x}^i} \frac{\partial x^l}{\partial \tilde{x}^j} (\hat{e}_k \cdot \hat{e}_l)$$

$\Rightarrow g_{ij}$ is a covariant tensor of rank 2.

The metric tensor 'g_{ij}' is a function which tells how to compute the distance between any two points in a given space. Generally

$ds^2 = g_{11}(dx^1)^2 + g_{12} dx^1 dx^2 + g_{22}(dx^2)^2 + \cdots$

or $ds^2 = g_{ij} dx^i dx^j$

Examples:

1) For Euclidean space in a rectangular coordinate system

$g_{ij} = \delta_{ij}$

$\Rightarrow$ In 3D

$ds^2 = (dx^1)^2 + (dx^2)^2 + (dx^3)^2$

$ds^2 = g_{ij} dx^i dx^j$

2) In polar coordinates: $(x^1, x^2) = (r, \theta)$

$ds^2 = (dx^1)^2 + (x^1)^2 (dx^2)^2$

or $ds^2 = dr^2 + r^2 d\theta^2 \rightarrow \begin{bmatrix} 1 & 0 \\ 0 & r^2 \end{bmatrix}$

3) In cylindrical coordinates:

$(x^1, x^2, x^3) = (r, \theta, z)$

$ds^2 = dr^2 + r^2 d\theta^2 + dz^2 \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & (x^1)^2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$ds^2 = (dx^1)^2 + (x^1)^2 (dx^2)^2 + (dx^3)^2$

4) Spherical coordinates:

$(x^1, x^2, x^3) = (r, \theta, \phi)$

$ds^2 = dr^2 + r^2 d\theta^2 + r^2 \sin^2\theta \, d\phi^2$

$ds^2 = (dx^1)^2 + (x^1)^2 (dx^2)^2 + (x^1 \sin x^2)^2 (dx^3)^2$

Properties of Metric Tensor:

Assume that a matrix $g = [g_{ij}]$ exists satisfying in all (admissible) coordinate systems x^i:

1) g is of differentiability class C^2.
2) g is symmetric i.e., $g_{ij} = g_{ji}$
3) g is non-singular i.e., $|g_{ij}| \neq 0$.

4) $ds^2 = g_{ij}\, dx^i dx^j$ and hence the distance concept generated by g is invariant with respect to a change of coordinates.

$d\tilde{s}^2 = ds^2$

Example: Suppose that on R^3 a matrix field is given in x^i-system by:

$$[g_{ij}] = \begin{bmatrix} (x^1)^2 - 1 & 1 & 0 \\ 1 & (x^2)^2 & 0 \\ 0 & 0 & 64/9 \end{bmatrix}$$

where $[(x^1)^2 - 1](x^2)^2 \neq 1$

Show that this matrix field is a metric.

Solution:

To prove that $[g_{ij}]$ is a metric tensor, we show that it satisfies the properties of metric tensor.

(1) Since g_{ij} is a polynomial in x^1 and x^2 for each i and j. Implies it is of differentiability class C^2.

(2) $$[g_{ji}] = \begin{bmatrix} (x^1)^2 - 1 & 1 & 0 \\ 1 & (x^2)^2 & 0 \\ 0 & 0 & 64/9 \end{bmatrix} = [g_{ij}]$$

$\Rightarrow g = [g_{ij}]$ is symmetric tensor.

3) Also $|g_{ij}| = \frac{64}{9}\begin{vmatrix} (x^1)^2 - 1 & 1 \\ 1 & (x^2)^2 \end{vmatrix}$

$= \frac{64}{9}\{(x^2)^2[(x^1)^2 - 1] - 1\} \neq 0$

4) As $\tilde{g}_{ij} = g_{rs}\frac{\partial x^r}{\partial \tilde{x}^i}\frac{\partial x^s}{\partial \tilde{x}^j}$

let $dx^i = v^i$ and $dx^j = v^j$

then $ds^2 = g_{ij} v^i v^j$

$\tilde{v}^i = v^t \frac{\partial \tilde{x}^i}{\partial x^t}$ and $\tilde{v}^j = v^l \frac{\partial \tilde{x}^j}{\partial x^l}$

Now:

$$(d\tilde{s})^2 = \tilde{g}_{ij}\tilde{v}^i\tilde{v}^j = g_{rs} v^t v^l \frac{\partial x^r}{\partial \tilde{x}^i}\frac{\partial x^s}{\partial \tilde{x}^j}\frac{\partial \tilde{x}^i}{\partial x^t}\frac{\partial \tilde{x}^j}{\partial x^l}$$

$$= g_{rs} v^t v^l \delta^r_t \delta^s_l$$

$$= g_{rs} v^r v^s$$

$$= g_{ij} v^i v^j$$

$$(d\tilde{s})^2 = (ds)^2$$

$\Rightarrow ds^2$ is invariant.

Important Result:

If the Jacobian matrix of the transformation from a given coordinate system x^i to a rectangular coordinate system $\tilde{x}^i$ is

$$J = \frac{\partial \tilde{x}^i}{\partial x^j},$$

then the matrix $g = [g_{ij}]$ of the Euclidean metric tensor in the x^i-System is given by

$$g = J^t J$$

Note:

Only Euclidean metric admits a representation of the form $g = J^t J$.

Because if g is non-Euclidean, there exists no coordinate system $\tilde{x}^i$ in which

$$\tilde{g} = I$$

This can be checked as:

Consider $g_{ij} = \frac{\partial \tilde{x}^r}{\partial x^i}\frac{\partial \tilde{x}^s}{\partial x^j}\tilde{g}_{rs}$

Now just consider:

$$g_{11} = \frac{\partial \tilde{x}^r}{\partial x^1}\frac{\partial \tilde{x}^s}{\partial x^1}\tilde{g}_{rs}$$

$$\Rightarrow g_{11} = \frac{\partial \tilde{x}^1}{\partial x^1}\frac{\partial \tilde{x}^1}{\partial x^1} + \frac{\partial \tilde{x}^2}{\partial x^1}\frac{\partial \tilde{x}^2}{\partial x^1} + \frac{\partial \tilde{x}^3}{\partial x^1}\frac{\partial \tilde{x}^3}{\partial x^1}$$

if we consider $[\tilde{g}_{rs}] = [\delta_{rs}] = I$

Similarly from $g = J^t J$

$$g_{11} = \frac{\partial \tilde{x}^1}{\partial x^1}\frac{\partial \tilde{x}^1}{\partial x^1} + \frac{\partial \tilde{x}^2}{\partial x^1}\frac{\partial \tilde{x}^2}{\partial x^1} + \frac{\partial \tilde{x}^3}{\partial x^1}\frac{\partial \tilde{x}^3}{\partial x^1}$$

hence $g = [g_{ij}] = J^t J = \left[\frac{\partial \tilde{x}^k}{\partial x^i}\right]\left[\frac{\partial \tilde{x}^k}{\partial x^j}\right]$

if $[\tilde{g}_{ij}] = [\delta_{ij}] = I$

Question: Find metric tensor for cylindrical coordinates x^i, connected to rectangular coordinates $\tilde{x}^i$ through the relations:

$$\tilde{x}^1 = x^1\cos x^2, \quad \tilde{x}^2 = x^1 \sin x^2, \quad \tilde{x}^3 = x^3$$

Solution: First we find Jacobian matrix:

$$J = \left[\frac{\partial \tilde{x}^i}{\partial x^j}\right] = \begin{bmatrix} \cos x^2 & -x^1 \sin x^2 & 0 \\ \sin x^2 & x^1 \cos x^2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Implies:

$$g = J^t J = \begin{bmatrix} 1 & 1 \\ 0 & \frac{1}{x^2} \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 1 & \frac{1}{x^2} \end{bmatrix}$$

$$= \begin{bmatrix} 1+1 & \frac{1}{x^2} \\ \frac{1}{x^2} & \left(\frac{1}{x^2}\right)^2 \end{bmatrix} = \begin{bmatrix} 2 & \frac{1}{x^2} \\ \frac{1}{x^2} & \left(\frac{1}{x^2}\right)^2 \end{bmatrix}$$

hence the Euclidean metric for cylindrical coordinates is given by:

$$g = J^t J = \begin{bmatrix} \cos x^2 & \sin x^2 & 0 \\ -x^1 \sin x^2 & x^1 \cos x^2 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} \cos x^2 & -x^1 \sin x^2 & 0 \\ \sin x^2 & x^1 \cos x^2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & (x^1)^2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Metric Tensor and Arc Lengths:

Arc length of a curve $x^i = x^i(t)$ is generally defined as:

$$L = \int_a^b \left|\frac{d\vec{r}}{dt}\right| dt \qquad ds^2 = g_{ij}\, dx^i dx^j$$

Now $\vec{r} = x^1 \hat{e}_1 + x^2 \hat{e}_2$

$$\frac{d\vec{r}}{dt} = \frac{\partial \vec{r}}{\partial x^1}\frac{dx^1}{dt} + \frac{\partial \vec{r}}{\partial x^2}\frac{dx^2}{dt}$$

$$\left|\frac{d\vec{r}}{dt}\right|^2 = \frac{d\vec{r}}{dt}\cdot\frac{d\vec{r}}{dt}$$

$$= \left(\frac{dx^1}{dt}\right)^2\left(\frac{\partial \vec{r}}{\partial x^1}\cdot\frac{\partial \vec{r}}{\partial x^1}\right) + 2\frac{dx^1}{dt}\frac{dx^2}{dt}\left(\frac{\partial \vec{r}}{\partial x^1}\cdot\frac{\partial \vec{r}}{\partial x^2}\right)$$

$$+\left(\frac{dx^2}{dt}\right)^2\left(\frac{\partial \vec{r}}{\partial x^2}\cdot\frac{\partial \vec{r}}{\partial x^2}\right)$$

$$\Rightarrow \left|\frac{d\vec{r}}{dt}\right|^2 = \frac{dx^i}{dt}\frac{\partial \vec{r}}{\partial x^i}\cdot\frac{dx^j}{dt}\frac{\partial \vec{r}}{\partial x^j}$$

$$= \frac{dx^i}{dt}\frac{dx^j}{dt}\left(\frac{\partial \vec{r}}{\partial x^i}\cdot\frac{\partial \vec{r}}{\partial x^j}\right)$$

$$= \frac{dx^i}{dt}\frac{dx^j}{dt}(\hat{e}_i\cdot\hat{e}_j) \qquad \left(\because \hat{e}_i\cdot\hat{e}_j = g_{ij}\right)$$

$$\left|\frac{d\vec{r}}{dt}\right| = \sqrt{g_{ij}\frac{dx^i}{dt}\frac{dx^j}{dt}}$$

$$\Rightarrow L = \int_a^b \sqrt{g_{ij}\frac{dx^i}{dt}\frac{dx^j}{dt}}\, dt$$

Example: Suppose that on R^3 a matrix field is given in x^i-system by:

$$[g_{ij}] = \begin{bmatrix} (x^1)^2 - 1 & 1 & 0 \\ 1 & (x^2)^2 & 0 \\ 0 & 0 & 64/9 \end{bmatrix}$$

where $[(x^1)^2 - 1](x^2)^2 \neq 1$

For this metric, compute the arc-length parameter and length of the given curve.

Question: Find the components g_{ij} of the Euclidean metric tensor in the special coordinate system x^i, related to rectangular coordinates $\tilde{x}^i$ by:

$$x^1 = \tilde{x}^1, \quad x^2 = e^{\tilde{x}^2 - \tilde{x}^1}$$

$$J = \begin{bmatrix} \frac{\partial \tilde{x}^1}{\partial x^1} & \frac{\partial \tilde{x}^1}{\partial x^2} \\ \frac{\partial \tilde{x}^2}{\partial x^1} & \frac{\partial \tilde{x}^2}{\partial x^2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1/x^2 \end{bmatrix}$$

$\star$ $\tilde{x}^1 = x^1$ and $\tilde{x}^2 = x^1 + \ln x^2$

Solution:

Using this metric tensor we have to find the arc length and length of the curve

$$C: \begin{cases} x^1 = 2t - 1 \\ x^2 = 2t^2 \\ x^3 = t^3 \end{cases} \quad ; 0 \le t \le 1$$

$$dx^1 = 2dt, \quad dx^2 = 4t\,dt, \quad dx^3 = 3t^2\,dt$$

Now:

$$ds^2 = g_{ij}\, dx^i dx^j$$

$$= g_{11}dx^1dx^1 + g_{12}dx^1dx^2 + g_{21}dx^2dx^1 + g_{22}dx^2dx^2 + g_{33}dx^3dx^3$$

$$\Rightarrow ds^2 = [(x')^2 - 1]\,4dt^2 + 2(2dt)(4tdt) + (x^2)^2(4tdt)^2 + \frac{64}{9}(3t^2\,dt)^2$$

$$= \left\{4[(x')^2 - 1] + 16t + (x^2)^2(16t^2) + 9t^4 \cdot \frac{64}{9}\right\} dt^2$$

$$ds^2 = 4\left[(x')^2 - 1 + 4t + 4t^2(x^2)^2 + 16t^4\right] dt^2$$

$$\Rightarrow ds^2 = 4\left[(2t-1)^2 - 1 + 4t + 4t^2(4t^4) + 16t^4\right] dt^2$$

$$ds^2 = 4\left\{4t^2 + 16t^4 + 16t^6\right\} dt^2$$

$$ds^2 = (8t^3 + 4t)^2\,dt^2$$

$$\Rightarrow ds = (8t^3 + 4t)\,dt$$

Arc length:

$$\Rightarrow S = \int_0^t (8u^3 + 4u)\,du$$

$$= 2t^4 + 2t^2$$

and Length of the curve:

$$L = \int_0^1 (8t^3 + 4t)\,dt = 4$$

$$\mathcal{L} = \int_a^b \sqrt{g_{ij}\frac{dx^i}{dt}\frac{dx^j}{dt}}\,dt$$

$$L = \int_a^b \sqrt{g_{ij}\,dx^i dx^j}$$

$$= \int_a^b \sqrt{ds^2}$$

$$L = \int_a^b ds$$

Question: A curve is given in spherical coordinates x^i by:

$x^1 = t$, $x^2 = \sin^{-1}\frac{1}{t}$, $x^3 = \sqrt{t^2-1}$

Find the length of the arc for $1 \le t \le 2$.

Solution:

$$\mathcal{L} = \int_a^b \sqrt{g_{ij}\frac{dx^i}{dt}\frac{dx^j}{dt}}\,dt$$

where g_{ij} for spherical coordinates is:

$$[g_{ij}] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & (x^1)^2 & 0 \\ 0 & 0 & (x^1\sin x^2)^2 \end{bmatrix}$$

$$\mathcal{L} = \int_1^2 \left[g_{11}\left(\frac{dx^1}{dt}\right)^2 + g_{22}\left(\frac{dx^2}{dt}\right)^2 + g_{33}\left(\frac{dx^3}{dt}\right)^2\right]^{1/2} dt$$

where

$$\frac{dx^1}{dt} = 1, \quad \frac{dx^2}{dt} = \frac{-1/t^2}{\sqrt{1-(1/t)^2}}, \quad \frac{dx^3}{dt} = \frac{t}{\sqrt{t^2-1}}$$

$$\left(\frac{dx^1}{dt}\right)^2 = 1, \quad \left(\frac{dx^2}{dt}\right)^2 = \frac{1}{t^2(t^2-1)}, \quad \left(\frac{dx^3}{dt}\right)^2 = \frac{t^2}{t^2-1}$$

$$\Rightarrow \mathcal{L} = \int_1^2 \left[1 + \frac{t^2}{t^2(t^2-1)} + t^2\left(\frac{1}{t^2}\right)\frac{t^2}{t^2-1}\right]^{1/2} dt$$

$$= \int_1^2 \left[1 + \frac{1}{t^2-1} + \frac{t^2}{t^2-1}\right]^{1/2} dt$$

$$= \int_1^2 \left(\frac{2t^2}{t^2-1}\right)^{1/2} dt$$

$$\Rightarrow \mathcal{L} = \int_1^2 \frac{\sqrt{2}\,t}{\sqrt{t^2-1}}\,dt = \frac{\sqrt{2}}{2}\int_1^2 \frac{2t}{\sqrt{t^2-1}}\,dt$$

$$= \frac{2}{\sqrt{2}}\left|(t^2-1)^{1/2}\right|_1^2 = \sqrt{2}\,(\sqrt{3}) = \sqrt{6}$$

Question: Find the length of the curve defined by $x^1 = 1$, $x^2 = t$; $1 \le t \le 2$ if the metric is that of the hyperbolic plane ($x^2 > 0$):

$$g_{11} = g_{22} = \frac{1}{(x^2)^2}, \quad g_{12} = g_{21} = 0$$

Solution:

consider the arc length formula:

$$\mathcal{L} = \int_1^2 \sqrt{g_{ij}\frac{dx^i}{dt}\frac{dx^j}{dt}}\,dt$$

$$\Rightarrow \mathcal{L}_1 = \int_1^2 \left[g_{11}\left(\frac{dx^1}{dt}\right)^2 + g_{22}\left(\frac{dx^2}{dt}\right)^2\right]^{1/2} dt$$

$$= \int_1^2 \left[\frac{1}{t^2}(0) + \frac{1}{t^2}(1)^2\right]^{1/2} dt$$

$$= \int_1^2 t^{-1}\,dt = \ln t\Big|_1^2 = \ln 2 - \ln 1 \quad (\ln 1 = 0)$$

$$\mathcal{L} = \ln 2$$

Question: Using Euclidean metric for cylindrical coordinates, calculate the length of arc along the circular helix $x^1 = a\cos t$, $x^2 = a\sin t$, $x^3 = bt$ with a and b positive constants, from $t = 0$ to $t = c > 0$.

Solution: $\bar{x}^i \rightarrow$ for rectangular coordinates, $x^i \rightarrow$ for cylindrical coordinates

also

$\bar{x}^1 = x^1 \cos x^2$, $\bar{x}^2 = x^1 \sin x^2$, $\bar{x}^3 = x^3$

$$[g_{ij}] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & (x^1)^2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Given curve has parametric Eqs:

$\bar{x}^1 = a\cos t$, $\bar{x}^2 = a\sin t$, $\bar{x}^3 = bt$

$\Rightarrow$ $x^1 = a$, $x^2 = t$ and $x^3 = bt$ are the parametric Eqs of curve in cylindrical coordinates.

Now consider, $\mathcal{L} = \int_a^b \sqrt{g_{ij}\frac{dx^i}{dt}\frac{dx^j}{dt}}\,dt$

$$\Rightarrow \mathcal{L} = \int_0^c \left[g_{11}\frac{dx^1}{dt}\frac{dx^1}{dt} + g_{22}\frac{dx^2}{dt}\frac{dx^2}{dt} + g_{33}\frac{dx^3}{dt}\frac{dx^3}{dt}\right]^{1/2} dt$$

$$= \int_0^c \left[0 + (x^1)^2(1)(1) + (1)(b)(b)\right]^{1/2} dt$$

$$= \int_0^c (a^2+b^2)^{1/2}\,dt \rightarrow \int_0^c dt$$

$$= c\,(a^2+b^2)^{1/2}$$

Question: Using the metric

$$[g_{ij}] = \begin{bmatrix} 12 & 4 & 0 \\ 4 & 1 & 1 \\ 0 & 1 & (x^1)^2 \end{bmatrix}$$

calculate the length of the curve given by

$x^1 = 3-t$, $x^2 = 6t+3$, $x^3 = \ln t$

where $1 \le t \le e$.

Solution:

Consider

$$\mathcal{L} = \int_a^b \left[g_{ij}\frac{dx^i}{dt}\frac{dx^j}{dt}\right]^{1/2} dt$$

$$= \int_a^b \left[g_{11}\left(\frac{dx^1}{dt}\right)^2 + 2g_{12}\frac{dx^1}{dt}\frac{dx^2}{dt} + g_{22}\left(\frac{dx^2}{dt}\right)^2 + 2g_{23}\frac{dx^2}{dt}\frac{dx^3}{dt} + g_{33}\left(\frac{dx^3}{dt}\right)^2\right]^{1/2} dt$$

$$\Rightarrow \mathcal{L} = \int_1^e \left[12(-1)^2 + 2(4)(-1)(6) + (1)(6)^2 + 2(1)(6)\left(\tfrac{1}{t}\right) + (3-t)^2\left(\tfrac{1}{t^2}\right)\right]^{1/2} dt$$

$$= \int_1^e \left[12 + 36 - 48 + \frac{12}{t} + \frac{9+t^2-6t}{t^2}\right]^{1/2} dt$$

$$= \int_1^e \left(9 + \frac{t^2}{t^2} + 6t\right)^{1/2} dt$$

$$\Rightarrow \mathcal{L} = \int_1^e \left(\frac{3+t}{t}\right)^{2\times 1/2} dt$$

$$= \int_1^e \left(\frac{3}{t} + 1\right) dt = \left|3\ln t + t\right|_1^e$$

$$= 3\ln e^1 - 3\ln 1^0 + e - 1$$

$$= 3 + e - 1$$

$$\mathcal{L} = 2 + e$$

Conjugate Metric Tensor: ✓

The inverse of the fundamental tensor or metric tensor

$$[g^{ij}]_{nn} = [g_{ij}]^{-1}_{nn}$$

is called the conjugate metric tensor.

* $g_{ij} = \hat{e}_i \cdot \hat{e}_j$ } where $\hat{e}_i$ and $\hat{e}'_i$ are the two sets of basis vectors associated normal to the coordinate surfaces and tangent to the coordinate curves.

then $g^{ij} = \hat{e}'_i \cdot \hat{e}'_j$

The covariant and contravariant metric tensors are inverses of each other that is

$$[g_{ij}] = [g^{ij}]^{-1}$$

and $[g^{ij}] = [g_{ij}]^{-1}$

$$\Rightarrow g_{ik}g^{kj} = \delta_i^{\ j}$$

and $g^{ik}g_{kj} = \delta^i_{\ j}$

* It is common to call covariant form as metric tensor and contravariant form as inverse metric, associate metric, conjugate metric or reciprocal metric tensor.

$$g_{ij} = \begin{bmatrix} \hat{e}_1\cdot\hat{e}_1 & \hat{e}_1\cdot\hat{e}_2 & \cdots & \hat{e}_1\cdot\hat{e}_n \\ \hat{e}_2\cdot\hat{e}_1 & \hat{e}_2\cdot\hat{e}_2 & \cdots & \hat{e}_2\cdot\hat{e}_n \\ \vdots & \vdots & \ddots & \vdots \\ \hat{e}_n\cdot\hat{e}_1 & & \cdots & \hat{e}_n\cdot\hat{e}_n \end{bmatrix}$$

* For orthogonal coordinate system the metric tensor is diagonal, i.e.,

$$g_{ij} = g^{ij} = 0 \quad \text{for } i \neq j$$

Raising and Lowering Indices:

The contravariant metric tensor is used for raising indices of covariant tensors and the covariant metric tensor is used for lowering indices of contravariant tensors

Example:

Consider a mixed tensor $T^{ij}_{\ k}$ and convert it into a covariant and contravariant tensor.

Solution:

Contravariant form: $T^{ijk} = T^{ij}_{\ \ r}\, g^{rk}$

Covariant form: $T_{ijk} = g_{is}\, g_{jr}\, T^{sr}_{\ \ k}$

Question: Given that V^i is a contravariant vector in R^3, find its associated covariant vector V_i in cylindrical coordinates x^i under the Euclidean metric.

Solution: Here the metric is

$$[g_{ij}] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & (x^1)^2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

and $V_i = g_{is}\, V^s \rightarrow [V_i] = [g_{ij}][V^i]$

$\Rightarrow$ In matrix form:

$$\begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & (x^1)^2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} V^1 \\ V^2 \\ V^3 \end{bmatrix}$$

$$\begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} V^1 \\ (x^1)^2 V^2 \\ V^3 \end{bmatrix} \Rightarrow V_1 = V^1$$

$$V_2 = (x^1)^2 V^2 = r^2 V^2$$

$$V_3 = V^3$$

Question: In Spherical coordinates $X^\alpha = (1, r, 0)$ and $Y_\alpha = (0, -r^2, \cos^2\theta)$. Find X_α and Y^α.

Solution: Here the metric is:

$$[g_{ij}] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & r^2\sin^2\theta \end{bmatrix}$$

Now

$$X_\alpha = g_{\alpha\beta} X^\beta \quad [X_\alpha] = [g_{\alpha\beta}][X^\beta]$$

$$\Rightarrow X_\alpha = g_{\alpha 1} X^1 + g_{\alpha 2} X^2 + g_{\alpha 3} X^3$$

$$\Rightarrow X_1 = g_{11} X^1 = 1$$

$$X_2 = g_{22} X^2 = r^2 (r) = r^3$$

$$X_3 = g_{33} X^3 = r^2\sin^2\theta\,(0) = 0$$

$$\Rightarrow X_\alpha = (1, r^3, 0)$$

Now

$$Y^\alpha = g^{\alpha\beta} Y_\beta$$

where

$$g^{\alpha\beta} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/r^2 & 0 \\ 0 & 0 & 1/r^2\sin^2\theta \end{bmatrix}$$

$$\Rightarrow Y^1 = g^{11} Y_1 = (1)(0) = 0$$

$$Y^2 = g^{22} Y_2 = \frac{1}{r^2}(-r^2) = -1$$

$$Y^3 = g^{33} Y_3 = \frac{1}{r^2\sin^2\theta}(\cos^2\theta) = \frac{\cot^2\theta}{r^2}$$

Question: Given the two dimensional line element $ds^2 = r^2 dx^2 + 2\,dx\,dy - dy^2$

Write down g_{ij}, g^{ij} and then raise and lower indices of

$V_i = (1, -1)$

$W^i = (0, 1)$

Solution: From the given line element we have

$$g_{ij} = \begin{bmatrix} x^2 & 1 \\ 1 & -1 \end{bmatrix} \rightarrow |g_{ij}| = -(x^2+1)$$

and its inverse is $g^{ij} = \frac{1}{|g_{ij}|} \text{adj } g_{ij}$

$$g^{ij} = \frac{1}{-(x^2+1)} \begin{bmatrix} -1 & -1 \\ -1 & x^2 \end{bmatrix} = \frac{1}{x^2+1} \begin{bmatrix} 1 & 1 \\ 1 & -x^2 \end{bmatrix}$$

Now:

$$V^2 = g^{ij} V_j = g^{21} V_1 + g^{22} V_2$$

$$\Rightarrow V^1 = g^{11} V_1 + g^{12} V_2$$

$$V^1 = \frac{1}{1+x^2}(1) + \frac{1}{1+x^2}(-1) = 0$$

$$V^2 = g^{21} V_1 + g^{22} V_2$$

$$= \frac{1}{1+x^2}(1) - \frac{x^2}{1+x^2}(-1) = \frac{1+x^2}{1+x^2} = 1$$

$$\Rightarrow V^i = (0, 1)$$

$$W_i = g_{ij} W^j$$

$$\Rightarrow W_1 = g_{11} W^1 + g_{12} W^2$$

$$= x^2(0) + (1)(1) = 1$$

$$W_2 = g_{21} W^1 + g_{22} W^2$$

$$= (1)(0) + (-1)(1) = -1$$

$$\Rightarrow W_i = (1, -1)$$

Question: In curvilinear coordinates x^i, two contravariant vectors are:

$u^i = \left(\frac{-x^1}{x^2}, 1, 0\right)$, $v^i = \left(\frac{1}{x^2}, 0, 0\right)$

where $\bar{x}^1 = x^2$, $\bar{x}^2 = x^3$ and $x^3 = x^1 x^2$; $x^2 \neq 0$.

Find u_i and v_i.

Solution: First we find the metric tensor by $[g_{ij}] = J^t J$

$J = \frac{\partial \bar{x}^i \rightarrow \text{rectg}}{\partial x^i \rightarrow \text{curvi}}$

$$J = \begin{bmatrix} \partial\bar{x}^1/\partial x^1 & \partial\bar{x}^1/\partial x^2 & \partial\bar{x}^1/\partial x^3 \\ \partial\bar{x}^2/\partial x^1 & \partial\bar{x}^2/\partial x^2 & \partial\bar{x}^2/\partial x^3 \\ \partial\bar{x}^3/\partial x^1 & \partial\bar{x}^3/\partial x^2 & \partial\bar{x}^3/\partial x^3 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ x^2 & x^1 & 0 \end{bmatrix}$$

$$\Rightarrow [g_{ij}] = J^t J = \begin{bmatrix} (x^2)^2 & x^1 x^2 & 0 \\ x^1 x^2 & 1+(x^1)^2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Now Given

$u^i = (-x^1/x^2, 1, 0)$

$$u_i = g_{ij} u^j = g_{i1} u^1 + g_{i2} u^2 + g_{i3} u^3$$

$$\Rightarrow u_1 = (x^2)^2\left(\frac{-x^1}{x^2}\right) + x^1 x^2 (1) + (0)$$

$$= -x^1 x^2 + x^1 x^2 = 0$$

$$u_2 = x^1 x^2\left(\frac{-x^1}{x^2}\right) + (1+(x^1)^2)(1) + 0$$

$$= -(x^1)^2 + 1 + (x^1)^2 = 1$$

$$u_3 = 0 + 0 + 0 = 0$$

$$\Rightarrow u_i = (0, 1, 0)$$

Given: $v^i = \left(\frac{1}{x^2}, 0, 0\right)$

$$v_i = g_{ij} v^j$$

$$\Rightarrow v_1 = g_{11} v^1 + g_{12} \overset{0}{v^2} + g_{13} \overset{0}{v^3} = (x^2)^2\left(\frac{1}{x^2}\right) = x^2$$

$$v_2 = g_{21} v^1 + g_{22} \overset{0}{v^2} + g_{23} \overset{0}{v^3} = x^1 x^2 \left(\frac{1}{x^2}\right) = x^1$$

$$v_3 = g_{31} \overset{0}{v^1} + g_{32} \overset{0}{v^2} + g_{33} \overset{0}{v^3} = 0$$

$$\Rightarrow v_i = (x^2, x^1, 0).$$

Generalized inner product:

$ds^2 = g_{ij} dx^i dx^j$

Suppose that g_{ij} is a metric on R^n and u^i, v^i are two vectors in R^n, then the generalized inner product of u^i and v^i is defined as

$$I = u^i v_i = u_i v^i = g_{ij} u^i v^j$$

$$= g^{ij} u_i v_j$$

$$\mathcal{L} = \int_a^b \sqrt{g_{ij} \frac{dx^i}{dt} \frac{dx^j}{dt}}\, dt$$

Concept of Length and Angle:

The norm (or length) of an arbitrary vector v^i or v_i is the non-negative real number.

$$\|\vec{v}\| = d\sqrt{v^2} = d\sqrt{v_i v^i}$$

The angle θ between two non-null contravariant vectors $\vec{U}$ and $\vec{V}$ is defined by:

$$\cos\theta = \frac{\vec{U}\cdot\vec{V}}{\|\vec{U}\|\,\|\vec{V}\|} = \frac{g_{ij}\,U^i V^j}{\sqrt{g_{pq}U^pU^q}\,\sqrt{g_{rs}V^rV^s}}$$

Question: Show that the length of any contravariant vector v^i equals the length of its associated covariant vector v_i.

Solution: By definition:

$$\|v^i\| = \sqrt{g_{ij}v^iv^j} \quad \text{and} \quad \|v_i\| = \sqrt{g^{ij}v_iv_j}$$

Consider:

$$g_{ij}v^iv^j = g_{ij}(g^{ir}v_r)(g^{js}v_s) = g_{ji}g^{ir}g^{js}v_rv_s$$
$$= \delta^r_j g^{js}v_rv_s = g^{rs}v_rv_s$$

Question: Show that under the Euclidean metric for polar coordinates, the vectors

$\vec{u} = u^i = \left(\frac{3}{5}, \frac{4}{5x^1}\right)$ and $\vec{v} = v^i = \left(\frac{-4}{5}, \frac{3}{5x^1}\right)$

are orthonormal.

Solution:

To show that these vectors are orthonormal, we have to show that

$$\|\vec{u}\| = 1, \quad \|\vec{v}\| = 1$$

and $\vec{u}\cdot\vec{v} = 0$

here $[g_{ij}] = \begin{bmatrix} 1 & 0 \\ 0 & (x^1)^2 \end{bmatrix}$, $u^i = \left(\frac{3}{5}, \frac{4}{5x^1}\right)$

$$\|\vec{u}\| = u^iu_i = g_{ij}u^iu^j = g_{11}u^1u^1 + g_{22}u^2u^2$$
$$= (1)\left(\frac{3}{5}\right)^2 + (x^1)^2\left(\frac{4}{5x^1}\right)^2$$
$$= \frac{9}{25} + \frac{16}{25} = 1$$

Similarly:

$$\|\vec{v}\| = v_iv^i = g_{ij}v^iv^i$$
$$= g_{11}v^1v^1 + g_{22}v^2v^2$$
$$= (1)\left(\frac{-4}{5}\right)^2 + \left(\frac{3}{5x^1}\right)^2(x^1)$$
$$= \frac{16}{25} + \frac{9}{25} = 1$$

Now: $\vec{u}\cdot\vec{v} = g_{ij}u^iv^j$

$$\Rightarrow = g_{11}u^1v^1 + g_{22}u^2v^2$$
$$= (1)\left(\frac{3}{5}\right)\left(\frac{-4}{5}\right) + (x^1)^2\left(\frac{4}{5x^1}\right)\left(\frac{3}{5x^1}\right)$$
$$= -\frac{12}{25} + \frac{12}{25}$$
$$= 0$$

$\Rightarrow \vec{u}$ and $\vec{v}$ are orthonormal.

Question: Find the condition for two curves on a sphere a to be orthogonal, if the curves are represented in spherical coordinates by:

C_1: $= f(\theta)$ and C_2: $= g(\theta)$

Solution: The two curves can be parameterized in spherical coordinates (r, θ, ϕ):

$$C_1: \begin{cases} r = a \\ \theta = t \\ \phi = f(t) \end{cases} \quad \text{and} \quad C_2: \begin{cases} r = a \\ \theta = u \\ \phi = f(u) \end{cases}$$

Also let (a, θ_0, ϕ_0) is the point of intersection.

The tangents to the curves are:

For C_1: $u^i = \left(\frac{dr}{dt}, \frac{d\theta}{dt}, \frac{d\phi}{dt}\right) = (0, 1, f'(\theta))$

at (a, θ_0, ϕ_0): $\quad u^i = \frac{dx^i}{dt}$

$$u^i = (0, 1, f'(\theta_0))$$

For C_2: $v^i = \left(\frac{dr}{du}, \frac{d\theta}{du}, \frac{d\phi}{du}\right) = (0, 1, g'(\theta))$

and at (a, θ_0, ϕ_0) $v^i(a, \theta_0, \phi_0) = (0, 1, g'(\theta_0))$

These are orthogonal if and only if

$$g_{ij}\,u^iv^j = 0$$

where $[g_{ij}] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & a^2 & 0 \\ 0 & 0 & a^2\sin^2\theta_0 \end{bmatrix}$ at (a, θ_0, ϕ_0)

$$g_{ij}u^iv^j = 0$$
$$\Rightarrow g_{11}u^1v^1 + g_{22}u^2v^2 + g_{33}u^3v^3 = 0$$
$$(1)(0) + a^2(1) + a^2\sin^2\theta_0 f'(\theta_0)g'(\theta_0) = 0$$
$$0 + a^2 + a^2\sin^2\theta_0 f'(\theta_0)g'(\theta_0) = 0$$
$$\Rightarrow \csc^2\theta_0 + f'(\theta_0)g'(\theta_0) = 0$$
$$\Rightarrow f'(\theta_0)g'(\theta_0) = -\csc^2\theta_0$$

at any point of intersection (a, θ_0, ϕ_0).

Question: In curvilinear coordinates x^i, Show that the vectors

$$u^i = \left(-\frac{x^1}{x^2}, 1, 0\right) \text{ and } v^i = \left(\frac{1}{x^2}, 0, 0\right)$$

are orthonormal if x^i are related to rectangular coordinates $\tilde{x}^i$ by:

$$\tilde{x}^1 = x^2, \quad \tilde{x}^2 = x^3 \text{ and } \tilde{x}^3 = x^1 x^2$$

Solution: We are to prove $\|u^i\| = 1$, $\|v^i\| = 1$ and $u_i v^i = 0$

$\vec{u}\cdot\vec{v} = u_i v^i = u^i v_i = g_{ij} u^i v^j$

Here the metric is:

$$[g_{ij}] = J^t J = \begin{bmatrix} (x^2)^2 & x^1 x^2 & 0 \\ x^1 x^2 & 1+(x^1)^2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \|u^i\| = \sqrt{g_{ij} u^i u^j} = \left(g_{11}u^1u^1 + 2g_{12}u^1u^2 + g_{22}u^2u^2 + g_{33}\overset{0}{u^3}\overset{0}{u^3}\right)^{1/2}$$

$$= \left((x^2)^2\left(\frac{-x^1}{x^2}\right)^2 + 2x^1x^2\left(\frac{-x^1}{x^2}\right)(1) + (1+(x^1)^2)(1)^2\right)^{1/2}$$

$$= \left((x^2) - 2(x^1)^2 + 1 + (x^1)^2\right)^{1/2} = 1$$

Similarly:

$$\|\vec{V}\| = \left(g_{ij} v^i v^j\right)^{1/2}$$

$$= \left((x^2)^2 \frac{1}{(x^2)^2} + 2x^1x^2(0) + 0 + 0\right)^{1/2}$$

$$= 1$$

$$u_i v^i = g_{ij} u^j v^i = (x^2)^2\left(\frac{-x^1}{x^2}\right)\left(\frac{1}{x^2}\right) + x^1x^2(1)\frac{1}{x^2}$$

$$= -x^1 + x^1 = 0$$

$\Rightarrow u^i$ and v^i are orthonormal.

Question: Under the metric

$$[g_{ij}] = \begin{bmatrix} 1 & \cos 2x^2 \\ \cos 2x^2 & 1 \end{bmatrix}$$

Compute the norm of $\vec{V} = \frac{dx^i}{dt}$, along the curve $x^1 = -\sin 2t$, $x^2 = t$ and use it to find the arc length between $t = 0$ and $t = \pi/2$.

Solution:

$$v^i = \frac{dx^i}{dt} = \left(\frac{dx^1}{dt}, \frac{dx^2}{dt}\right) = (-2\cos 2t, 1)$$

now

$$\|v^i\| = \sqrt{g_{ij} v^i v^j}$$

$$= \left[(1)(4\cos^2 2t) + 2\cos 2t(-2\cos 2t)(1) + (1)\right]^{1/2}$$

$$= (1)^{1/2} = 1$$

$$\mathcal{L} = \int_a^b \left(g_{ij} v^i v^j\right)^{1/2} dt$$

$$\mathcal{L} = \int_0^{\pi/2} dt$$

$$= |t|_0^{\pi/2} = \pi/2$$

The del Operator: '∇' is a partial differential operator. In Cartesian coordinate system, it is defined by:

$$\nabla = \frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}$$

OR $\nabla = \frac{\partial}{\partial x_1}\hat{e}_1 + \frac{\partial}{\partial x_2}\hat{e}_2 + \frac{\partial}{\partial x_3}\hat{e}_3$

Implies in tensor notation, we can write it: $\nabla = \sum_i \frac{\partial}{\partial x_i}\hat{e}_i = \sum_i \nabla_i \hat{e}_i$

where $\nabla_i = \frac{\partial}{\partial x_i}$

Gradient:

The gradient of a differentiable scalar function $\phi(x_i)$ is a vector given by:

$$\nabla\phi = \sum_i \frac{\partial\phi}{\partial x_i}\hat{e}_i$$

and in component form:

$$[\nabla\phi]_i = \nabla_i\phi = \frac{\partial\phi}{\partial x_i} = \phi_{,i}$$

Divergence:

The divergence of a differentiable vector $\vec{A}$ is a scalar given by:

$$\nabla\cdot\vec{A} = \sum_i \frac{\partial A_i}{\partial x_i}$$

$$\nabla\cdot\vec{A} = \delta_{ij}\frac{\partial A_i}{\partial x_j} = \frac{\partial A_i}{\partial x_i} = \nabla_i A_i = \partial_i A_i = A_{i,i}$$

curl:

The curl of a differentiable vector $\vec{A}$ is a vector given by:

(In component form)

$$[\nabla \times \vec{A}]_i = \varepsilon_{ijk} \frac{\partial A_k}{\partial x_j} = \varepsilon_{ijk} \nabla_j A_k$$
$$= \varepsilon_{ijk} \partial_j A_k$$
$$= \varepsilon_{ijk} A_{k,j}$$

(i^{th} component)

Laplacian:

The Laplacian operator, also called the harmonic operator, acting on a differentiable scalar function $\phi(x_i)$ is given by:

$$\Delta\phi = \nabla^2\phi = \delta_{ij} \frac{\partial \phi}{\partial x_i \partial x_j} = \frac{\partial^2 \phi}{\partial x_i \partial x_i} = \nabla_{ii}\phi$$
$$= \partial_i \partial_i \phi$$
$$= \phi_{,ii}$$

Tensor Techniques to prove vector Identities:

Examples:

(1) $\nabla . \vec{r} = n$

Consider $\nabla . \vec{r} = \partial_i x_i$
$= \frac{\partial x_i}{\partial x_i}$
$= \delta_{ii}$
$= n$

(i is dummy, Σ is not mentioned due to Einstein Summation Convention)

$\vec{r}$ is position vector
$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$
$= x_1\hat{e}_1 + x_2\hat{e}_2 + x_3\hat{e}_3$
$= x_i \hat{e}_i$

2) $\nabla \times \vec{r} = \vec{0}$

Proof:

$[\nabla \times \vec{r}]_i = \varepsilon_{ijk} \partial_j x_k$ (definition of curl)
$= \varepsilon_{ijk} \delta_{kj}$ ($\because \partial_j x_k = \frac{\partial x_k}{\partial x_j}$)
$= \varepsilon_{ijj}$ (using def. of δ_{ij})
$= 0$ (by the def. of ε_{ijk})

3) $\nabla(\vec{a}.\vec{r}) = \vec{a}$, where $\vec{a}$ is any constant vector.

Proof: consider L.H.S, where

$\vec{a}.\vec{r} = a_j x_j$ and $[\nabla\phi]_i = \partial_i \phi$

$\Rightarrow [\nabla(\vec{a}.\vec{r})]_i = \partial_i (a_j x_j)$
$= a_j \partial_i x_j + x_j \underbrace{\partial_i a_j}_{0}$ (product rule)
$= a_j \partial_i x_j$
$= a_j \delta_{ij}$
$= a_i$
$\Rightarrow \nabla(\vec{a}.\vec{r}) = \vec{a}$

4) $\nabla . \nabla\phi = \nabla^2\phi$ $\quad \nabla . \vec{A} = \partial_i A_i$, $\nabla\phi$

Proof: consider,

$\nabla . \nabla\phi = \partial_i [\nabla\phi]_i$ $\quad (\because \nabla_i = \partial_i$, $[\nabla\phi]_i = \partial_i\phi)$
$= \partial_i (\partial_i \phi)$
$= \partial_i \partial_i \phi$
$= \partial_{ii}\phi$
$= \nabla^2\phi$

5) $\nabla . (\nabla \times \vec{A}) = 0$

Proof: $[\nabla \times \vec{A}]_i = \varepsilon_{ijk} \partial_j A_k$

$\nabla . \vec{B} = \partial_i B_i$

replace $\nabla \times \vec{A}$ in place of $\vec{B}$:

$\Rightarrow \nabla . (\nabla \times \vec{A}) = \partial_i (\varepsilon_{ijk} \partial_j A_k)$
$= \varepsilon_{ijk} \partial_i \partial_j A_k = \varepsilon_{ijk} \partial_j \partial_i A_k$
$= -\varepsilon_{jik} \partial_j \partial_i A_k$
$= -\varepsilon_{ijk} \partial_i \partial_j A_k$

by interchanging the dummy indices i and j. we have:

$\nabla . (\nabla \times \vec{A}) = \varepsilon_{ijk} \partial_i \partial_j A_k$
$= -\varepsilon_{ijk} \partial_i \partial_j A_k$

and for any quantity x

if $x = -x \Rightarrow 2x = 0 \Rightarrow x = 0$

$\Rightarrow \varepsilon_{ijk} \partial_i \partial_j A_k = 0$

$\Rightarrow \nabla . (\nabla \times \vec{A}) = 0$

6) $\nabla \times \nabla\phi = \vec{0}$

Proof: we consider i^{th} component of the cross product on L.H.S.

$(\nabla \times \nabla\phi)_i = \varepsilon_{ijk} \partial_j (\nabla\phi)_k$
$= \varepsilon_{ijk} \partial_j (\partial_k \phi)$
$= \varepsilon_{ijk} \partial_j \partial_k \phi$

$\Rightarrow (\nabla \times \nabla\phi)_i = \varepsilon_{ijk}\, \partial_k \partial_j \phi$

$= -\varepsilon_{ikj}\, \partial_k \partial_j \phi$

$= -\varepsilon_{ijk}\, \partial_j \partial_k \phi$

(interchanging dummy indices j and k)

$\Rightarrow (\nabla \times \nabla\phi)_i = \varepsilon_{ijk}\, \partial_j \partial_k \phi = -\varepsilon_{ijk}\, \partial_j \partial_k \phi$

$\Rightarrow (\nabla \times \nabla\phi)_i = 0 \Rightarrow \nabla \times \nabla\phi = \vec{0}$

$\Rightarrow (\nabla \times \nabla\phi)_i = \varepsilon_{ijk}\, \partial_k \partial_j \phi$

$= -\varepsilon_{ikj}\, \partial_k \partial_j \phi$

$= -\varepsilon_{ijk}\, \partial_j \partial_k \phi$

(interchanging dummy indices j and k)

$\Rightarrow (\nabla \times \nabla\phi)_i = \varepsilon_{ijk}\, \partial_j \partial_k \phi = -\varepsilon_{ijk}\, \partial_j \partial_k \phi$

$\Rightarrow (\nabla \times \nabla\phi)_i = 0 \Rightarrow \nabla \times \nabla\phi = \vec{0}$

$\Rightarrow \vec{A}.(\vec{B}\times\vec{C}) = \varepsilon_{kij}\, A_i B_j C_k$

$= \varepsilon_{kij}\, C_k A_i B_j$

$= C_k\, \varepsilon_{kij} A_i B_j$

$= \vec{C}.(\vec{A}\times\vec{B})$

Similarly, we can prove

$\vec{A}.(\vec{B}\times\vec{C}) = \vec{B}.(C\times\vec{A})$

8) $\nabla.(\vec{A}\times\vec{B}) = \vec{B}.(\nabla\times\vec{A}) - \vec{A}.(\nabla\times\vec{B})$

Proof: $\nabla.(\vec{A}\times\vec{B}) = \partial_i (\vec{A}\times\vec{B})_i$

$= \partial_i (\varepsilon_{ijk} A_j B_k)$

$= \varepsilon_{ijk}\, \partial_i (A_j B_k)$

$= \varepsilon_{ijk} (B_k \partial_i A_j + A_j \partial_i B_k)$

$\Rightarrow \nabla.(\vec{A}\times\vec{B}) = \varepsilon_{ijk} B_k \partial_i A_j + \varepsilon_{ijk} A_j \partial_i B_k$

$= \varepsilon_{kij} B_k \partial_i A_j - \varepsilon_{jik} A_j \partial_i B_k$

$= B_k (\varepsilon_{kij} \partial_i A_j) - A_j (\varepsilon_{jik} \partial_i B_k)$

$= B_k (\nabla\times\vec{A})_k - A_j (\nabla\times\vec{B})_j$

$= \vec{B}.(\nabla\times\vec{A}) - \vec{A}.(\nabla\times\vec{B})$

9) $\nabla\times(\nabla\times\vec{A}) = \nabla(\nabla.\vec{A}) - \nabla^2\vec{A}$

Proof: Consider L.H.S:

$[\nabla\times(\nabla\times\vec{A})]_i = \varepsilon_{ijk}\, \partial_j (\nabla\times\vec{A})_k$

$= \varepsilon_{ijk}\, \partial_j (\varepsilon_{klm}\, \partial_l A_m)$

$= \varepsilon_{ijk} \varepsilon_{klm}\, \partial_j \partial_l A_m$

$\Rightarrow [\nabla\times(\nabla\times\vec{A})]_i = \varepsilon_{ijk} \varepsilon_{klm}\, \partial_j \partial_l A_m$

using $\varepsilon_{ijk}\varepsilon_{klm} = \delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}$

$\Rightarrow [\nabla\times(\nabla\times\vec{A})]_i = (\delta_{il}\delta_{jm} - \delta_{im}\delta_{jl})\, \partial_j \partial_l A_m$

$= \delta_{il}\delta_{jm}\partial_j\partial_l A_m - \delta_{im}\delta_{jl}\partial_j\partial_l A_m$

$= \partial_m \partial_i A_m - \partial_l \partial_l A_i$

$\Rightarrow = \partial_m \partial_i A_m - \partial_l \partial_l A_i$

$= \partial_i (\partial_m A_m) - \partial_{ll} A_i$

$= (\nabla(\nabla.\vec{A}))_i - (\nabla^2 A)_i$

$\Rightarrow \nabla\times(\nabla\times\vec{A}) = \nabla(\nabla.\vec{A}) - \nabla^2\vec{A}$

7) $\vec{A}.(\vec{B}\times\vec{C}) = \vec{C}.(\vec{A}\times\vec{B}) = \vec{B}.(\vec{C}\times\vec{A})$

Proof: Consider, $\vec{D} = \vec{B}\times\vec{C}$, then

$D_i = (\vec{B}\times\vec{C})_i = \varepsilon_{ijk}\, B_j C_k$

and

$\vec{A}.(\vec{B}\times\vec{C}) = \vec{A}.\vec{D} = A_i D_i$

$= A_i\, \varepsilon_{ijk} B_j C_k$

$= \varepsilon_{ijk} A_i B_j C_k$

10) $\vec{A}\times(\nabla\times\vec{B}) = \nabla\vec{B}\cdot\vec{A} - \vec{A}\cdot\nabla B$

Proof: consider,

$$\left[\vec{A}\times(\nabla\times\vec{B})\right]_i = \varepsilon_{ijk} A_j (\nabla\times\vec{B})_k$$

$$= \varepsilon_{ijk} A_j \varepsilon_{klm} \partial_l B_m$$

$$= \varepsilon_{ijk} \varepsilon_{klm} A_j \partial_l B_m$$

$$\Rightarrow = (\delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}) A_j \partial_l B_m$$

$$= \delta_{il}\delta_{jm} A_j \partial_l B_m - \delta_{im}\delta_{jl} A_j \partial_l B_m$$

$$= A_m \partial_i B_m - A_l \partial_l B_i$$

$$= (\partial_i B_m) A_m - A_l (\partial_l B_i)$$

$$= (\nabla\vec{B}\cdot\vec{A})_i - (\vec{A}\cdot\nabla\vec{B})_i$$

Since i is arbitrary index so the identity is true for all components.

EASY STUDY EASY SOLUTION